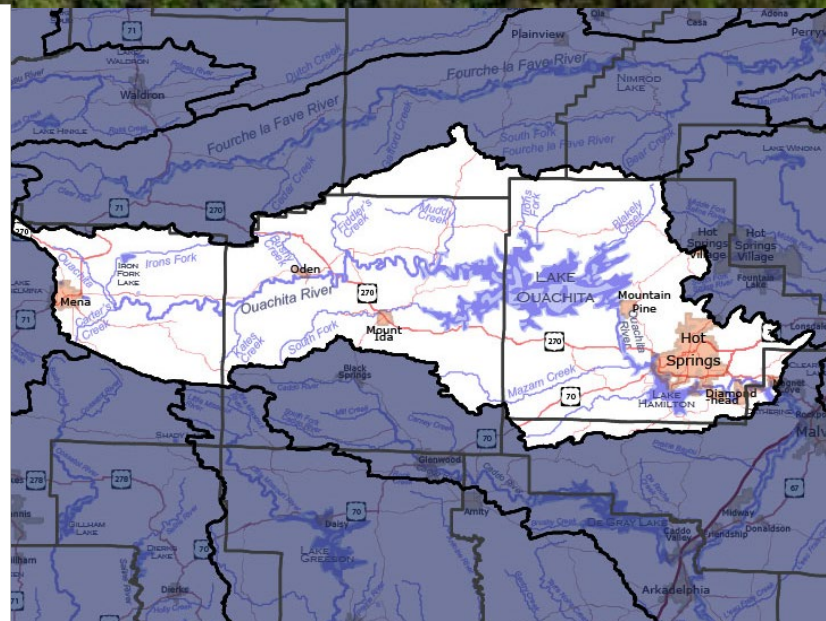
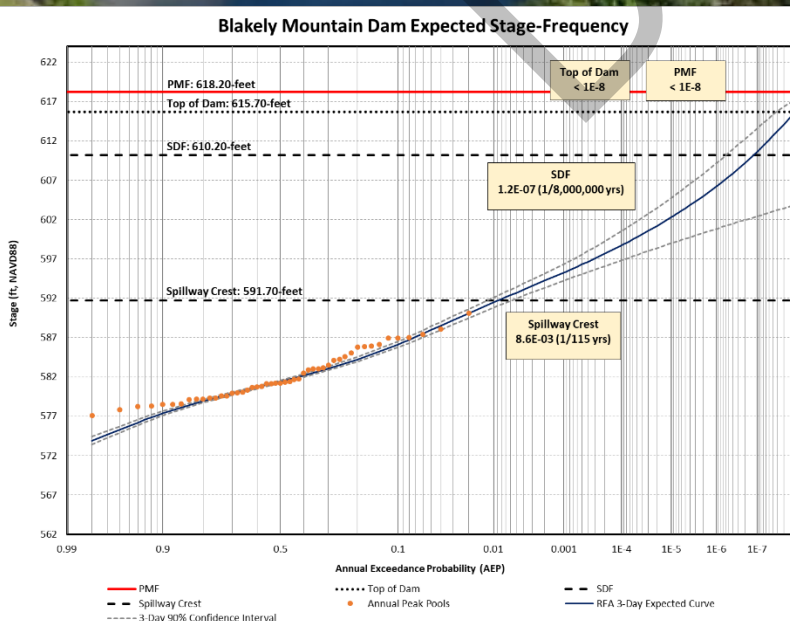


Blakely Mountain Dam Flood Hazard Assessment

Bayesian Approach to Estimating Stage Frequency Curves for Blakely Mountain Dam



REPORT DOCUMENTATION PAGE					Form Approved OMB No. 0704-0188	
The public reporting burden for this collection of information is estimated to average 1 hour per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing the burden, to Department of Defense, Washington Headquarters Services, Directorate for Information Operations and Reports (0704-0188), 1215 Jefferson Davis Highway, Suite 1204, Arlington, VA 22202-4302. Respondents should be aware that notwithstanding any other provision of law, no person shall be subject to any penalty for failing to comply with a collection of information if it does not display a currently valid OMB control number. PLEASE DO NOT RETURN YOUR FORM TO THE ABOVE ADDRESS.						
1. REPORT DATE (DD-MM-YYYY) 24-01-2020		2. REPORT TYPE Technical Report			3. DATES COVERED (From - To)	
4. TITLE AND SUBTITLE Blakely Mountain Dam Flood Hazard Assessment: Bayesian Approach to Estimating Stage-Frequency Curves for Blakely Mountain Dam				5a. CONTRACT NUMBER		
				5b. GRANT NUMBER		
				5c. PROGRAM ELEMENT NUMBER		
6. AUTHOR(S) Melissa Mika, SPK Haden Smith, RMC Carolyn Pearson, RMC Dana Moree, MVK				5d. PROJECT NUMBER		
				5e. TASK NUMBER		
				5f. WORK UNIT NUMBER		
7. PERFORMING ORGANIZATION NAME(S) AND ADDRESS(ES) USACE Risk Management Center 12596 West Bayaud Ave. Suite 400 Lakewood, CO 80228				8. PERFORMING ORGANIZATION REPORT NUMBER RMC-TR-2020-XX		
9. SPONSORING/MONITORING AGENCY NAME(S) AND ADDRESS(ES) USACE Risk Management Center 12596 West Bayaud Ave. Suite 400 Lakewood, CO 80228				10. SPONSOR/MONITOR'S ACRONYM(S)		
				11. SPONSOR/MONITOR'S REPORT NUMBER(S) RMC-TR-2020-XX		
12. DISTRIBUTION/AVAILABILITY STATEMENT Approved for public release; distribution is unlimited.						
13. SUPPLEMENTARY NOTES						
14. ABSTRACT This report documents the development of the stage-frequency curves for Blakely Mountain Dam in Arkansas, using a Bayesian approach incorporating both inflow volume and precipitation frequency analyses. The critical storm and flood characteristics of the Blakely Mountain Dam basin were identified, including flood duration, seasonality, and antecedent conditions. Volume-duration-frequency and precipitation-frequency relationships were developed. The reservoir stage-frequency curves, with uncertainty bounds, were generated using the Risk Management Center's Reservoir Frequency Analysis software (RMC-RFA) with the volume-duration-frequency information from RMC-BestFit as the primary input.						
15. SUBJECT TERMS Hydrologic Hazard Assessment, Risk Assessment, Flood Frequency, Precipitation Frequency, Inflow Volume-Frequency Analysis, Reservoir Stage-Frequency Analysis, Uncertainty Analysis						
16. SECURITY CLASSIFICATION OF:			17. LIMITATION OF ABSTRACT UU	18. NUMBER OF PAGES 73	19a. NAME OF RESPONSIBLE PERSON Haden Smith	
a. REPORT U	b. ABSTRACT U	c. THIS PAGE U			19b. TELEPHONE NUMBER (Include area code)	

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January 2020

*U.S. Army Corps of Engineers
Institute for Water Resources
Risk Management Center
12596 West Bayaud Ave. Suite 400
Lakewood, CO 80228*

Author(s)

Melissa Mika, Sacramento District, U.S. Army Corps of Engineers

Haden Smith, Risk Management Center, U.S. Army Corps of Engineers

Carolyn Pearson, Risk Management Center, U.S. Army Corps of Engineers

Dana Moree, Vicksburg District, U.S. Army Corps of Engineers

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Executive Summary

The purpose of this document is to develop the hydrologic loading curve (also referred to as a hazard curve or a stage-frequency curve) for Blakely Mountain Dam in western Arkansas. Blakely Mountain Dam is an earthen embankment dam used for flood control, hydropower, water supply, and recreation. This analysis is in support of the USACE Dam Safety Program, and recommends an expected stage-frequency curve for use in the Issue Evaluation Study (IES) hazard analysis of Blakely Mountain Dam. These curves provide dam risk assessments with a quantitative approach to estimating the annual exceedance probability (AEP) of a probable failure mode's loading condition.

The stage-frequency curve, with a 90% confidence interval (a 95% upper uncertainty bound and a 5% lower uncertainty bound), was developed using the current guidance, *RMC-TR-2018-03: An Inflow Volume-Based Approach to Estimating Stage-Frequency for Dams* (U.S. Army Corps of Engineers, 2018). The critical storm and flood characteristics of the basin were identified, including flood duration, seasonality, antecedent conditions (i.e., starting reservoir stage), and typical hydrograph shapes. Both an inflow volume-frequency and a precipitation-frequency approach were combined in a Bayesian frequency analysis using RMC-BestFit (alpha) (Smith & Skahill, 2019). The reservoir stage-frequency curves with uncertainty bounds were generated using the Risk Management Center's Reservoir Frequency Analysis software (RMC-RFA) (Smith C. H., 2017).

The best estimate hydrologic loading curve results from the analysis are shown in Figure 0-1. Table 0-1 summarizes the pertinent elevations for Blakely Mountain Dam and the AEP determined from the stage-frequency curve. The probability of overtopping the dam is estimated as an AEP less than 1E-08, as is the probability of exceeding the PMF, which was evaluated at an elevation of 618.2 feet-NAVD88 (U.S. Army Corps of Engineers, 2019). The spillway design flood (SDF) was established in 1950 at an elevation of 610.2, and has an estimated AEP of 1.2E-07. The spillway crest from the most recent survey is at an elevation of 591.7 feet-NAVD88 and has an estimated AEP of 8.6E-03.

Table 0-1. Annual exceedance probabilities (AEP) for critical reservoir elevations of Blakely Mountain Dam.

Critical Reservoir Elevation	Elevation (ft-NAVD88)	AEP	Return Period (years)
PMF (2019)	618.2	<1E-08	>100,000,000
Top of Dam (2018 survey)	615.7	<1E-08	>100,000,000
SDF (1950)	610.2	1.2E-07	8,000,000
Spillway Crest (2018 survey)	591.7	8.6E-03	115

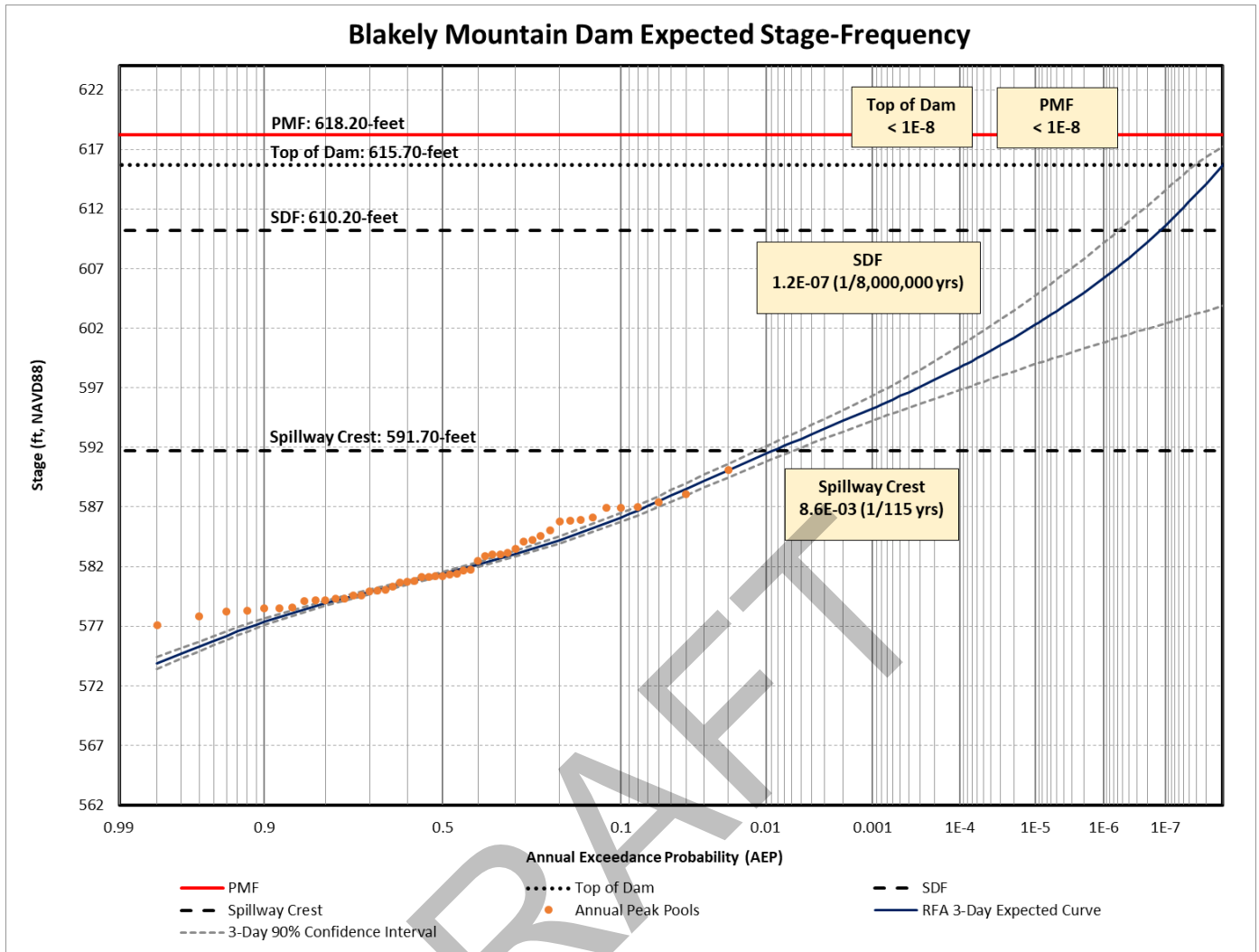


Figure 0-1. Final stage-frequency curve with uncertainty bounds for Blakely Mountain Dam.

Acknowledgements

This document was developed in coordination with the USACE Risk Management Center in support of the USACE Dam Safety Program. The team would like to acknowledge the technical contributions and guidance from John England (RMC), Haden Smith (RMC), Carolyn Pearson (RMC), and Mike Bartles (HEC). The team would also like to acknowledge the technical review performed by Simeon Benson (SWF).

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1 Introduction

1.1 Purpose

The U.S. Army Corps of Engineers (USACE) vision is to achieve safe, secure, and more resilient civil works infrastructure by enhancing its protection in order to prevent, deter, or mitigate the potential for dam failure. A major initiative within this effort consists of updating hydrologic data for more than 700 USACE impoundment structures and dams. It is expected that this effort, and the hydrologic results, can be utilized to reduce the uncertainty in the assessment of the hydrologic hazard for a given dam, as well as to provide a basis for design, if needed (U.S. Army Corps of Engineers, 2017).

Blakely Mountain Dam is a high hazard dam, located approximately 10 miles northwest of Hot Springs, Arkansas. The multi-purpose reservoir is used for flood control, hydropower, water supply, and recreation. The dam impounds Lake Ouachita and is operated in conjunction with two other USACE-owned dams for flood reduction at downstream control points. The energy company, Entergy, owns and operates the two reservoirs immediately downstream of Blakely Mountain Dam.

The purpose of this study is to better define the hydrologic risks related to Blakely Mountain Dam and the people and property downstream of Lake Ouachita. Hydrologic loading curves were developed to provide data for the estimation of the likelihood of failure for potential risk driver failure modes. The hydrologic loading curves (or hazard curves) represent the best estimate based on current knowledge and were developed with a risk-informed approach that incorporated all available hydrologic information.

1.1.1 Overview of Previous Studies

1.1.1.1 1946 Original Spillway Design

The Spillway Design Flood (SDF) for Blakely Mountain Dam was recommended during a conference held in the office of the Chief of Engineers on February 18-21, 1946. This design storm was approved by letter to the Division Engineer, Lower Mississippi Valley Division, dated February 1946.

This storm was developed from the 1927 flood, which was a flood of record for much of the lower Blakely Mountain Dam watershed. The original SDF precipitation was a 60-hour event with a total of 21.5 inches, of which 19.7 inches contributed to runoff. This design runoff produced a peak inflow of 418,000 cfs, a peak reservoir stage of 610.2 feet (NAVD88), and a peak outflow of 45,000 cfs. A Snyder Synthetic Unit Hydrograph was derived and peaked by 25 and 50 percent, with minimal change in total volume. Computations of the various arrangements of rainfall increments of the design storm were routed using the unit hydrograph to reveal the most critical precipitation pattern. An assumed infiltration rate of 0.20 inches per six-hour period was subtracted from the rainfall of the design storm. No infiltration was used for the reservoir pool area. The runoff used was 92 percent of the rainfall. The original SDF was routed through the reservoir assuming the reservoir level at an elevation of 592.0 feet with the outlet works closed during the passage of the flood.

1.1.1.2 2012 Issue Evaluation Study

A prior Issue Evaluation Study (IES) was completed in 2012 (U.S. Army Corps of Engineers, 2012). This study developed volume-duration-frequency and reservoir pool-frequency relationships. The volume-duration-frequency analysis was performed on the annual maximums for the log transform of the daily average flow data from 1961-2009 and fitted using a Log Pearson Type III (LPIII) distribution. The adopted loading curve was developed by fitting a curve to empirical annual maximum pool elevations (1961-2011) and assigning an AEP to (unofficial) PMF routing results.

1.1.1.3 2015 Semi-Quantitative Risk Assessment

A Semi-Quantitative Risk Assessment (SQRA) was completed in 2015 (U.S. Army Corps of Engineers, 2015). The 2015 analysis developed hydrologic loading curves using period-of-record inflow calculated from storage-outflow data. Skew was determined from Bulletin 17B regional skew maps and adjusted based on NOAA Atlas 14 frequency rainfall. The screening-level analysis developed estimates for the PMF ranging between 1/100,000 and 1/1,000,000 AEP, and estimates for overtopping frequency between 1/8,500 and 1/850,000 AEP.

1.2 Scope

This study builds on information generated in previous efforts, including the historical studies described above and is based on guidance from the Risk Management Center (RMC) Technical Report (TR) RMC-TR-2018-03, *An Inflow Volume-Based Approach to Estimating Stage-Frequency for Dams* (U.S. Army Corps of Engineers, 2018). The scope of the current investigation is as follows:

1. Determine critical storm and flood characteristics for the Blakely Mountain Dam basin, including flood duration, seasonality, and antecedent conditions.
2. Develop inflow volume-duration-frequency relationships for the watershed.
3. Develop an accurate representation of the reservoir.
4. Develop a reservoir stage-frequency curve with uncertainty bounds using the RMC Reservoir Frequency Analysis software (RMC-RFA) with the volume-duration-frequency information as the primary input.

These procedures are outlined below. A detailed description of the following procedures and results are provided in the respective sections of the report.

Flood Characteristics

- Determine the storm and flood characteristics for the watershed that will most affect the reservoir stage-frequency results, such as:
 - The critical inflow flood duration
 - The seasonality of floods
 - The antecedent conditions (reservoir starting stage)

Inflow Volume-Duration-Frequency Analysis

- Develop inflow volume-duration-frequency relationships for the critical flood duration by incorporating systematic, historical, and paleoflood data and a regional skew analysis using Bulletin 17C procedures.
- Develop inflow volume-duration-frequency relationships for the critical duration by incorporating all available data including information from a regional precipitation-frequency analysis and rainfall-runoff modeling using a Bayesian analysis framework.

Reservoir Representation

- Develop an accurate and reliable representation of the Blakely Mountain Dam, including stage-storage and stage-discharge relationships.

Reservoir Stage-Frequency Analysis

- Develop a volume-based stage-frequency curve using Monte Carlo sampling procedures.

Datum

The vertical datum of the elevations used throughout this report is based on the North American Vertical Datum of 1988 (NAVD88). Blakely Mountain Dam was designed and constructed using the National Geodetic Vertical Datum of 1929 (NAVD29). The conversion from NAVD29 to NAVD88 for Blakely Mountain Dam is +0.046 feet and is considered to be negligible.

2 Blakely Mountain Dam Drainage Basin

2.1 Watershed Description

Blakely Mountain Dam is an earthen embankment located on the headwaters of the Ouachita River Basin in west-central Arkansas approximately 10 miles northwest of Hot Springs, Arkansas (Figure 2-1). The dam collects runoff from an unregulated upstream basin of 1,105 square miles. The area lies entirely within the Ouachita Mountains and is characterized by roughly parallel ridge lines separated by deep valleys with steep, rocky inclines. Elevations in the basin range from 385 to 2,500 feet. Lake Ouachita drains an area roughly elliptical in shape, with an east-west width of approximately 60 miles and a north-south width of 20 miles. The area is wooded and is primarily located within the Ouachita National Forest, with portions of farmed land in the narrow valleys.

Downstream of the dam, the Ouachita River is approximately 430.4 miles long and is a tributary to the Black River, Red River, and ultimately the Mississippi River/Atchafalaya River distributary system. The entire Ouachita Basin drains over 24,200 square miles.

There are two hydroelectric dams owned and operated by Entergy of Arkansas immediately downstream of Blakely Mountain Dam. Carpenter Dam is approximately 19 miles downstream and forms the pool of Lake Hamilton. Lake Hamilton extends to the foot of Blakely Mountain Dam, with a constant pool of 400 feet in summer, and 395 during the winter and early spring. Rammel Dam is located approximately 12 miles downstream from Carpenter Dam and impounds Lake Catherine. Lake Catherine extends back to Carpenter Dam. Storage in Lake Catherine is small, and releases are simply passed through Rammel Dam.

The Ouachita River flows through the highly developed areas surrounding Hot Springs, Arkansas, which encompasses Lake Hamilton. Below Lake Catherine, land use becomes more rural. There are multiple federal levees downstream, between river miles 290 and 0, as well as a series of locks and dams.

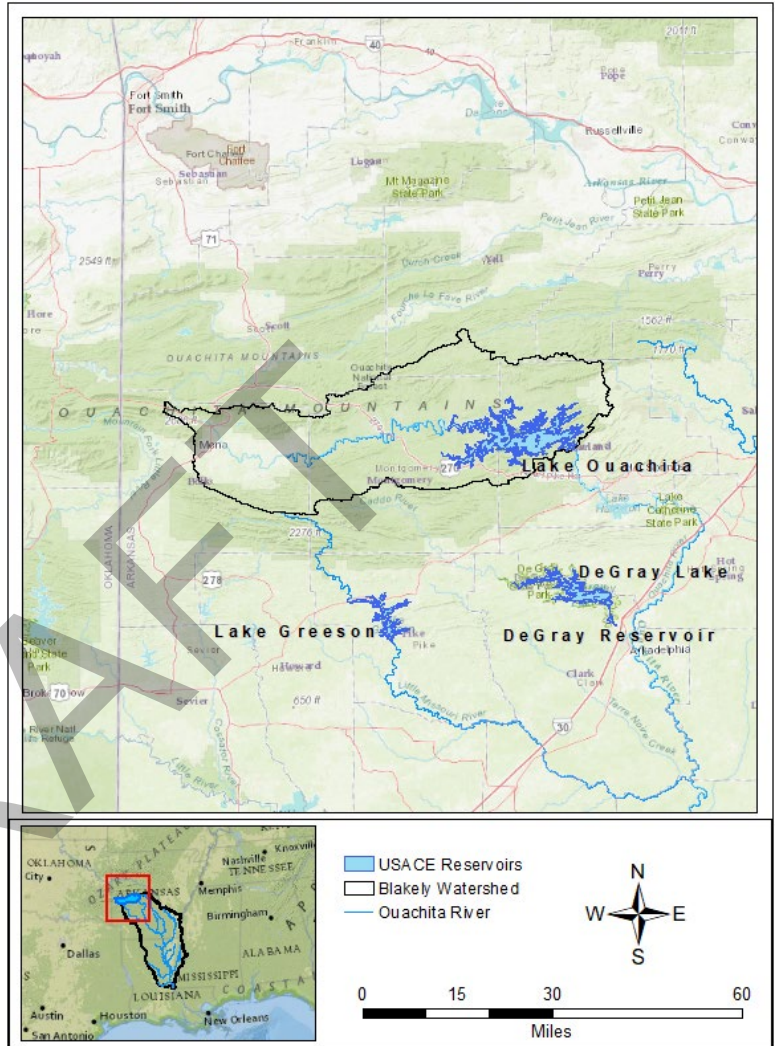


Figure 2-1. Lake Ouachita (Blakely Mountain Dam) watershed area map.

2.2 Storm and Flood Characteristics

The Ouachita River Basin in west-central Arkansas is typically subjected to damaging precipitation in the form of rain. Mean Annual Precipitation (MAP) over the Ouachita River basin is approximately 53 inches. Average annual rainfall recorded at gages throughout the Lake Ouachita Basin and Hot Springs Community vary from 55.44 to 59.19 inches. Snowfall may occur on average one to four times per year, with an average annual snowfall of 4 to 5 inches. Rainfall is present throughout the calendar year, with the highest amounts of precipitation occurring in the late spring and early winter months. Damaging storm events can occur at any time of the year, but are more common from winter to early summer. The average annual temperature is 62 degrees Fahrenheit, and the average monthly temperatures range from 43 degrees in December/January to 81 degrees in July/August.

Many major floods have occurred in the Ouachita River Basin throughout the period of record, with information dating to the early 1800's in the lower half of the basin. The earliest recorded storm event in the Blakely Mountain Dam watershed occurred in December 1848, with quantifiable information beginning in 1871. Two of the largest floods in gaged record were the April 1927 flood, which was used as the spillway design flood, and the December 1982 flood, which produced the highest observed pool. A table summarizing historical peak flow and peak elevation storm events is shown in Table 2-1.

The basin is fairly uniform with minimal impervious area, and stream flow is typically proportional to rainfall. However, it is common to have higher intensity rainfall occur over only a portion of the basin. When the more intense rain is in the eastern half of the basin, the inflow into the dam arrives more rapidly. It is also common for large events to be paired with an antecedent or subsequent inflow event.

Table 2-1. Historical peak flow and peak elevation events.

Date	Peak Flow (CFS)	Date	Elevation (FT-NAVD88)
May 1923*	>100,000	26 Apr 1927*	591.8
5 May 1982	300,000**	6 May 1982	590.1
30 Mar 1945*	123,000	20 May 1968	588.6
21 Apr 1927*	122,000	17 May 2009	588.1
5 Sep 2008	100,500	24 May 1990	587.4
29 Dec 2015	95,800	4 Nov 1984	587.0
15 May 1968	94,600	4 May 2013	586.95

* Pre-dam construction

** No direct or indirect estimate of peak flow available. Therefore, peak flow was estimated using a calibrated HEC-HMS model developed for this study.

2.3 Significant Historical Floods

Significant flood events in gaged record occurred during the years of 1923, 1927, 1945, 1982, and 1990. The April 1927 flood was used in-part to develop the spillway design flood for Blakely Mountain Dam. The December 1982 flood is the largest peak and volume within the systematic record, surpassing that of April 1927. More information on these events is detailed below.

Flood of May 1923

The 1923 flood occurred prior to the construction of Carpenter Dam downstream of Blakely Mountain Dam, and produced the maximum peak of record at Hot Springs, Arkansas (U.S. Army Corps of Engineers, 1950). The event was caused by heavy precipitation on May 13 and 14, with intense rainfall during a 6-hour period and an overall storm duration of approximately 24 hours. The average rainfall of the basin above Hot Springs was 5.5 inches. The storm was concentrated in the vicinity of Hot Springs, including 6.5 inches of rainfall within 6 hours during the night of May 14, and a total rainfall of 9.0 inches. The storm produced estimated peak discharges based on rating extensions of 109,000 cfs (stage of 36.8 feet) at Mountain Pine and 143,000 cfs (stage of 43.9 feet) at Hot Springs (U.S. Army Corps of Engineers, 1950).

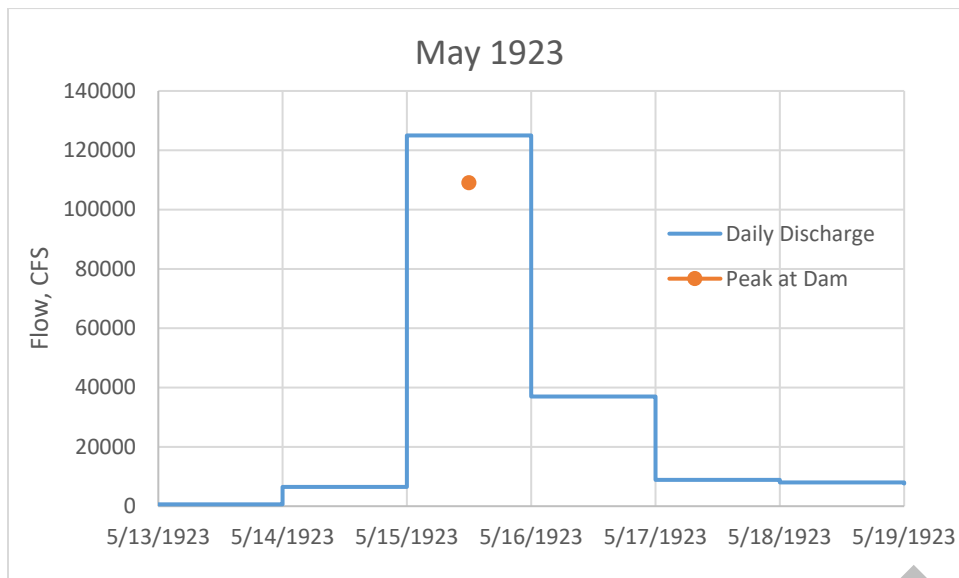


Figure 2-2. Estimated daily flow of the Ouachita River near Hot Springs (downstream of dam), and estimated peak flow at Blakely Mountain Dam site for the May 1923 historical flood event.

Flood of April 1927

The 1927 flood resulted from dual events consisting of an antecedent storm followed by a second, larger event, and was the most severe on record for the Ouachita River basin prior to the construction of Blakely Mountain Dam. This event was used as the basis for the design event for Blakely Mountain Dam. The 12-day event is documented in "Storm Rainfall in the United States" (U.S. Army Corps of Engineers, 1973) as two separate storms: LMV 4-8 which occurred from April 12-16, 1927 (centered in south-eastern Louisiana) and SW 2-4 which occurred from April 17-21, 1927 (centered at Jessieville and Story, Arkansas). The first of the two flood peaks produced an average rainfall of 5.7 inches over the basin. The second storm resulted in an additional average rainfall of 9.9 inches over the reservoir basin. The total rainfall at Story, Arkansas for the period April 11-21 was 18.9 inches. The peak 24 hour rainfall was 11.8 inches for the same location. Total average basin rainfall for Blakely Mountain Dam was 16.6 inches (April 11-21).

Heavy rainfall occurred over the entire watershed, as a result of a quasi-stationary storm front that followed three relatively rainy months and coincided with pre-existing high streamflow stages and antecedent soil moisture. The dam site was estimated to have a discharge of 122,000 cfs, which was calculated from rainfall/runoff, and calibrated to Arkadelphia and Camden gage data (Representatives, 1934, p. 86). The peak at Hot Springs was 42.4 feet determined from flood marks and 134,000 cfs determined from an extended rating curve. This flood produced the maximum stage of record at Arkadelphia at an estimated peak discharge of 133,000 cfs.

The 1927 flood event is shown in detail in Figure 2-3. Estimated isohyetal maps from the Definite Project Report (Plate 5) (U.S. Army Corps of Engineers, 1950) and the Water Control Manual (Plate 3) (U.S. Army Corps of Engineers, 1970) are shown in Figure 2-4 and Figure 2-5, respectively. The maximum depth-area-duration (DAD) table developed for storm SW 2-4 is reproduced in Table 2-2 (U.S. Army Corps of Engineers, 1973). The storm precipitation as basin-averaged values were determined from digitized isohyetal maps converted to 6-hour increments (Figure 2-6). The storm precipitation, as mass rainfall curves, are broken into the averaged rainfall above Mt. Ida (414 square miles), below Mt. Ida (694 square miles), and an average rainfall over the total basin (1,105 square miles).

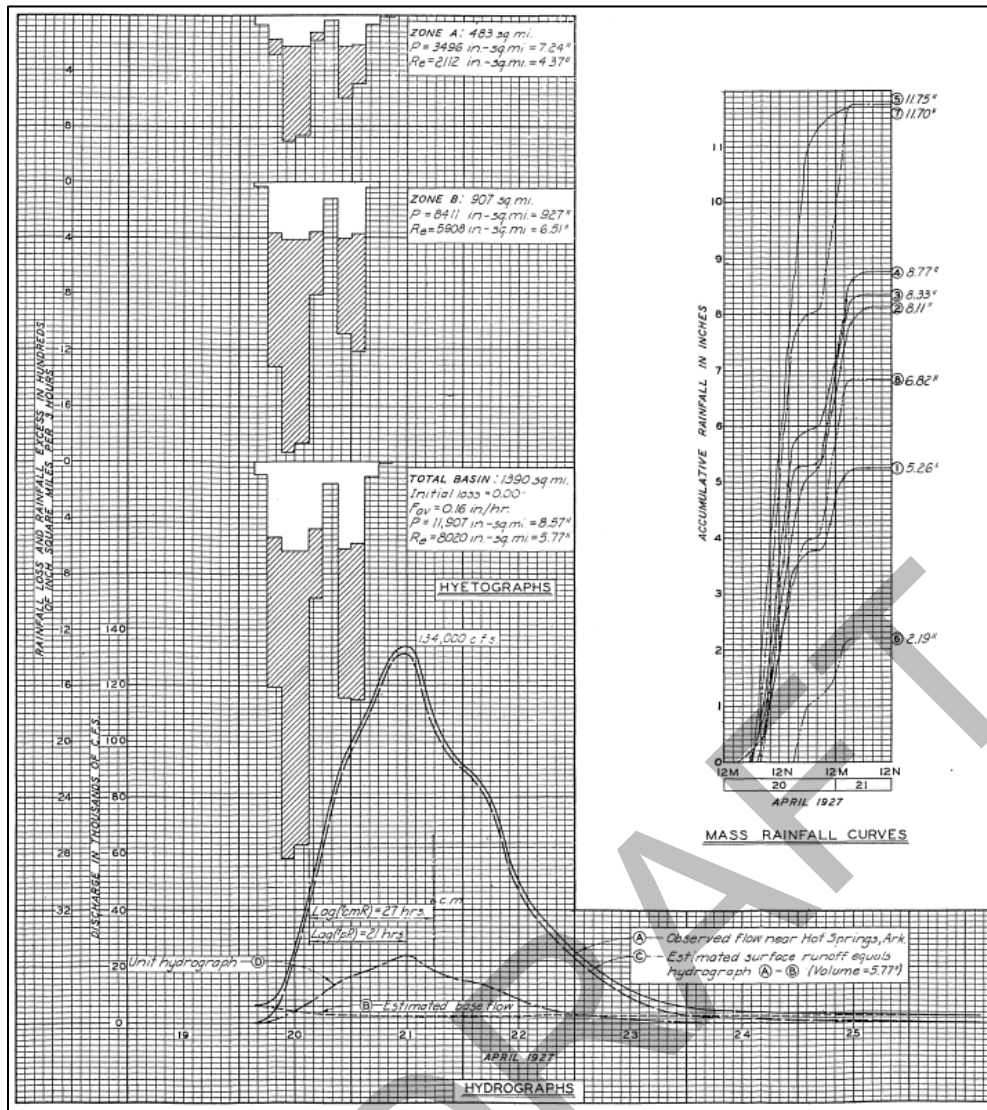


Figure 2-3. Mass rainfall curves, hyetographs, and hydrographs of the April 20-21, 1927 flood (U.S. Army Corps of Engineers, 1950).

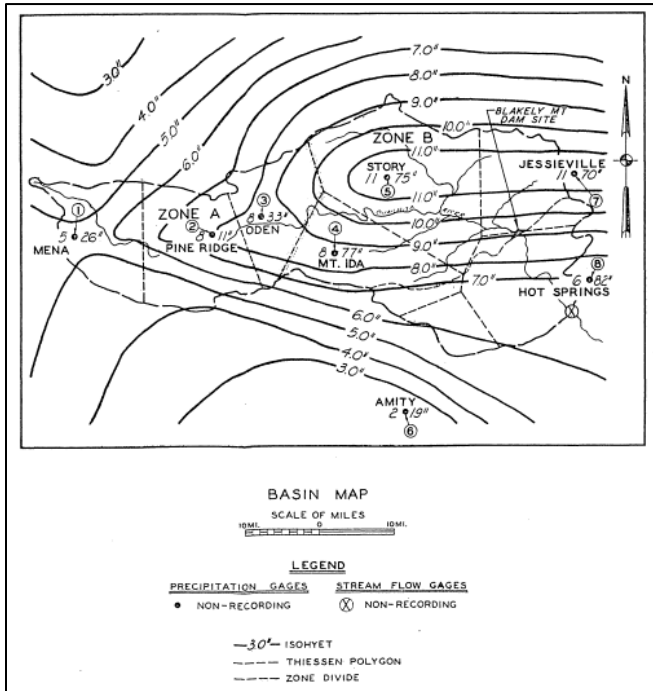


Figure 2-4. Isohyets for April 20-21, 1927 from Definite Project Report (U.S. Army Corps of Engineers, 1950).

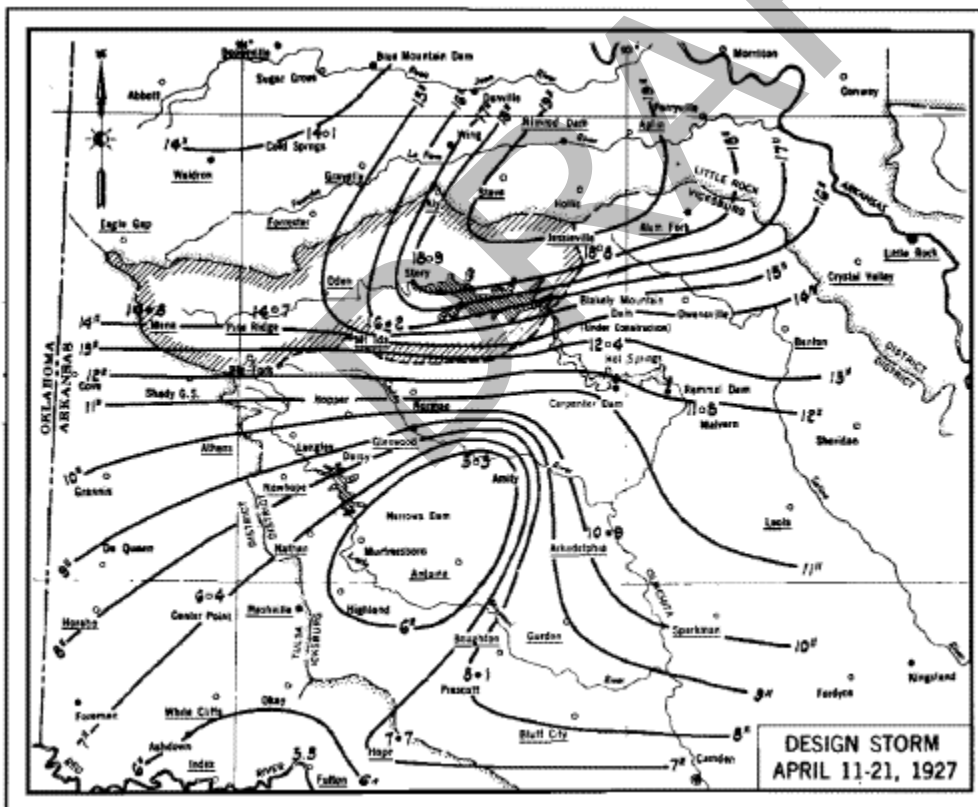


Figure 2-5. Isohyets for April 11-21, 1927 from the Water Control Manual (U.S. Army Corps of Engineers, 1970).

Bayesian Approach to Estimating Stage Frequency Curves for Blakely Mountain Dam

Table 2-2. Maximum depth-area-duration (DAD) data for the SW 2-4 storm centered near Jessieville, AR (U.S. Army Corps of Engineers, 1973).

Area (mi ²)	Depth of rainfall (in) for specified durations (hr)									
	6	12	18	24	30	36	48	60	72	96
10	7.0	10.9	11.3	11.8	11.8	11.8	12.0	12.5	12.5	12.5
100	6.3	9.6	10.7	11.5	11.7	11.7	11.8	12.3	12.3	12.4
200	6.0	9.2	10.4	11.2	11.6	11.6	11.6	12.1	12.1	12.2
500	5.4	8.4	9.7	10.6	11.0	11.0	11.0	11.5	11.5	11.6
1,000	4.8	7.5	8.9	10.0	10.3	10.3	10.3	10.8	10.8	10.9
2,000	4.0	6.2	7.9	9.0	9.4	9.4	9.4	9.8	9.8	10.0
5,000	2.9	4.7	6.3	7.2	7.7	7.7	7.7	8.1	8.1	8.7
10,000	2.2	3.7	5.2	5.7	6.3	6.4	6.4	6.6	6.6	7.6
20,000	1.6	2.8	4.1	4.6	5.0	5.1	5.1	5.2	5.4	6.5
50,000	1.1	1.7	2.6	3.1	3.3	3.4	3.4	3.6	3.9	5.0
100,000	0.7	1.2	1.7	2.0	2.1	2.2	2.2	2.6	2.8	3.7

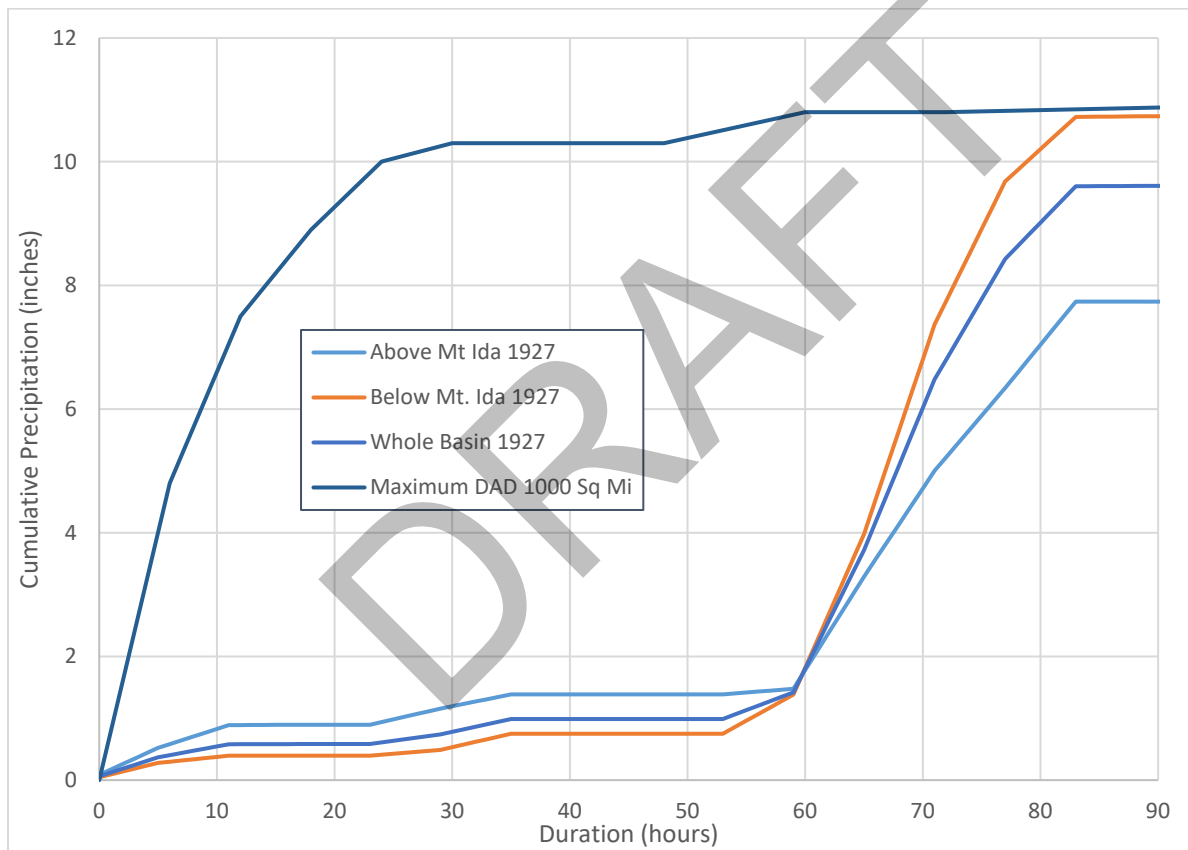


Figure 2-6. Peak 72-hour basin-specific mass rainfall curves for USACE storm SW 2-4.

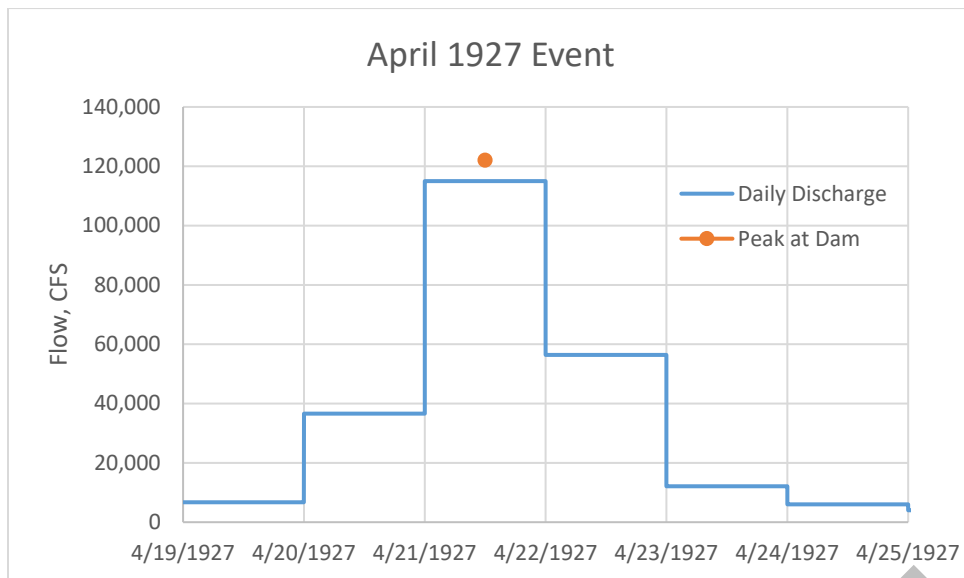


Figure 2-7. April 19-25, 1927 estimated peak and daily gaged inflow to Blakely Mountain Dam.

Flood of May 1930

This flood resulted from general rainfall over the entire Ouachita Basin, with the greatest concentration of rainfall in the upper portion of the basin. A series of storms on May 2-19, 1930 produced a total rainfall of 17.8 inches at Hot Springs, and a discharge of 66,000 cfs.

Floods of November 1931–February 1932

A series of storms in the Ouachita River basin upstream of the Blakely Mountain dam site occurred from November 1931 through February 1932, producing a total rainfall of 35.2 inches at Hot Springs over the 4-month period. The flood produced stages of 26.2 feet at Arkadelphia and 38.4 feet at Camden, with corresponding discharges of 89,100 and 102,000 cfs.

Flood of January 1937

This flood resulted from general rains produced by a storm front between tropical maritime and polar continental air masses over the basin from December 27, 1936 through January 25, 1937. The greatest concentration of rainfall occurred in upper parts of the Ouachita River Basin, with rainfall of 16.5 inches measured at Hot Springs. Peak stages were 22.4 feet at Mountain Pine, 26.0 feet at Arkadelphia, and 41.7 feet at Camden, with corresponding discharges of 39,100; 81,400; and 151,000 cfs, respectively.

Flood of February 1938

This flood was produced by heavy rainfall during February 14-19, 1938 when the river was above flood stage. Heavy rains in January produced a flood peak, which was then augmented by the February rainfalls. The flood produced a stage of 32.2 feet at Mountain Pine, with a discharge of 83,200 cfs.

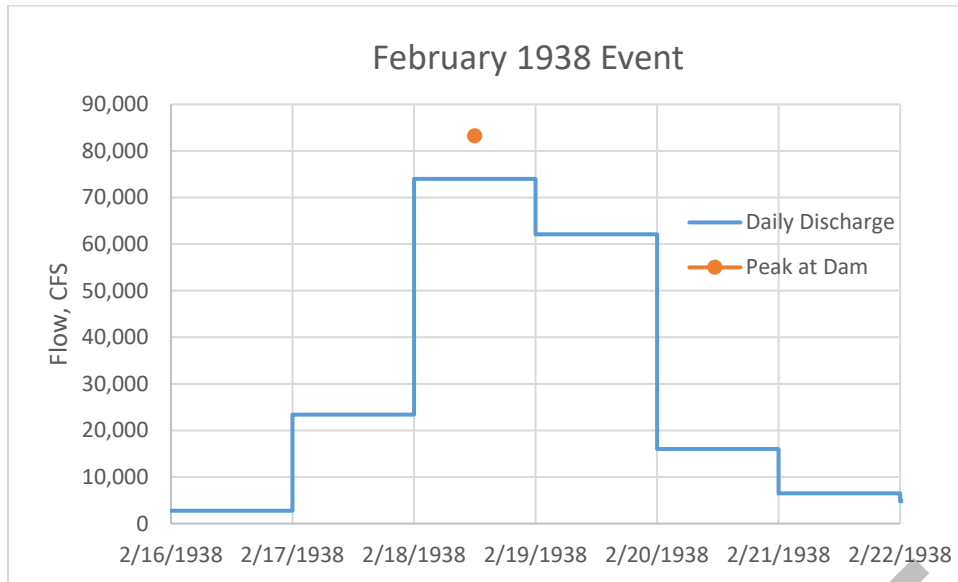


Figure 2-8. February 16-22, 1938 estimated peak and daily gaged inflow to Blakely Mountain Dam.

Flood of April 1939

This flood was the result of a series of rainfall over the basin above the dam site during the period March 25-April 21, 1939 culminating in a heavy two-day rainfall on April 16-17. The flood produced a maximum discharge of 94,900 cfs at Mountain Pine, 131,000 cfs at Arkadelphia, and 102,000 cfs at Camden.

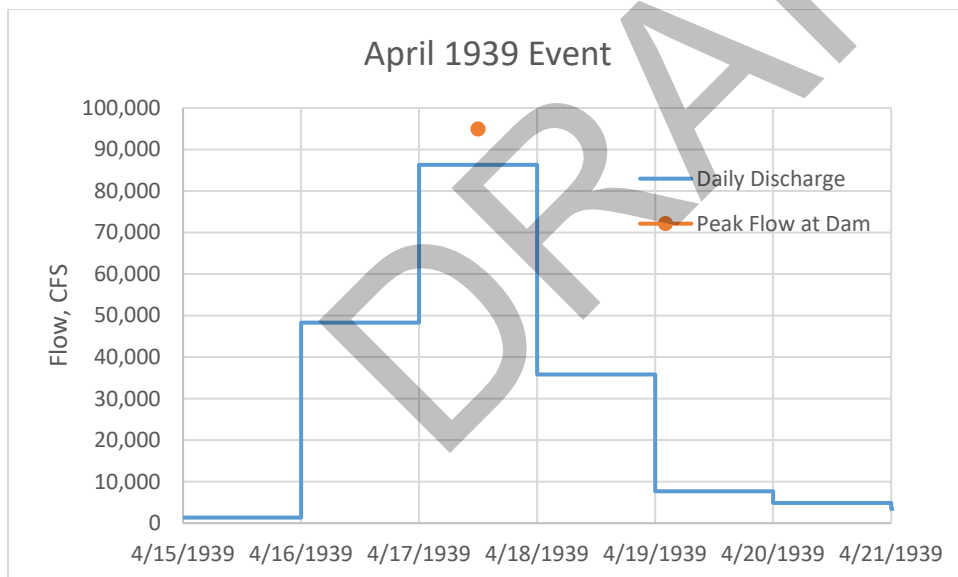


Figure 2-9. April 15-21, 1939 estimated peak and daily gaged inflow to Blakely Mountain Dam.

Flood of March 1945

The flood of March 1945, which had the largest peak discharge of record (highest gaged flow near the dam site), was of short duration and was caused by a series of high-intensity rains. The storm of March 28-April 2 had an average of 8.0 inches over the reservoir drainage area. The peak discharge of 123,000 cfs, with a stage of 38.55 feet at the Mountain Pine gage, occurred on March 30, 1945. Peak stage at the dam site was 422.66 feet and is estimated to correspond to 123,000 cfs.

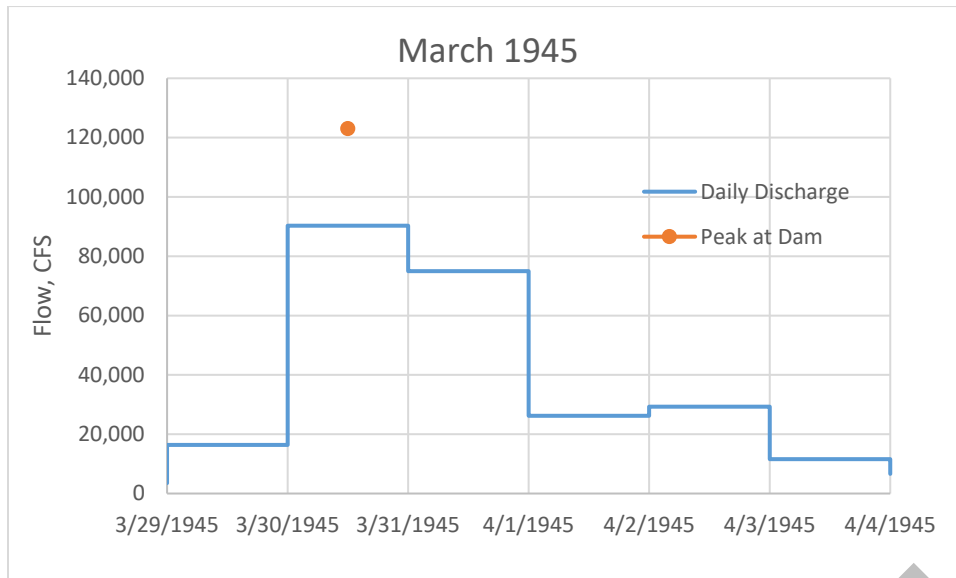


Figure 2-10. March 1945 estimated peak and daily gaged inflow to Blakely Mountain Dam.

Flood of December 1982

The December 1982 flood is the largest in recorded history at Blakely Mountain Dam. The storm began on December 1 when a deep, upper-level trough of low pressure over western Texas developed into a surface low and cold front system over southeast Texas (Amec Foster Wheeler Americas, Ltd., 2016). High moisture associated with southerly flow from the Gulf of Mexico caused multiple slow-moving thunderstorms to develop along the cold front. Heavy rainfall in Big Fork, Arkansas began on December 2. This was a frontal system with a Mesoscale Convective Complex (MCC). Cumulative rainfall and estimated inflow hydrographs are shown in Table 2-3 and Figure 2-11. The resulting inflows set a new record stage at Lake Ouachita.

Table 2-3. Depth-area-duration curves for the Big Fork, AR storm of 1982 (Amec Foster Wheeler Americas, Ltd., 2016).

Area (mi ²)	Depth of rainfall (in) for specified durations (hr)								
	6	12	18	24	30	36	48	60	72
10	6.6	9.7	13.1	14.2	14.5	14.7	15.1	15.3	15.5
100	6.0	9.0	12.1	13.7	14.1	14.4	14.7	14.9	15.0
200	5.8	8.9	12.0	13.2	13.7	14.1	14.1	14.3	14.5
1,000	5.1	7.7	10.9	12.2	12.6	12.9	13.3	13.5	13.6
5,000	4.0	6.6	9.5	10.8	11.3	11.7	11.9	12.0	12.1
10,000	3.5	5.8	8.4	9.6	10.0	10.8	10.9	11.1	11.2
20,000	2.8	4.7	6.9	8.1	8.7	9.2	9.7	9.9	10.0

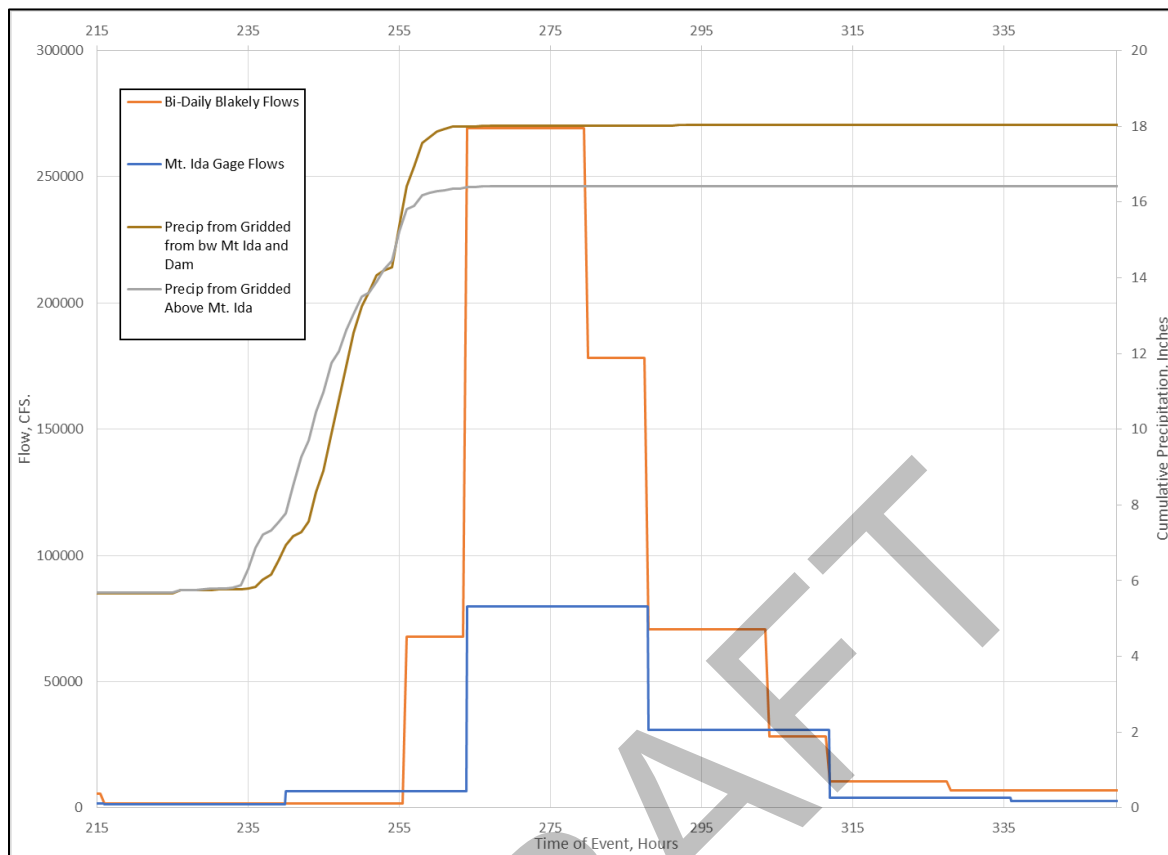


Figure 2-11. December 1982 inflow hydrograph (best available) at Mt. Ida and Blakely Mountain Dam.

Comparison of December 1982 and April 1927 Floods

The December 1982 event surpassed the April 1927 event in several portions of the rainfall mass curves and inflow hydrographs, both for total storm and basin-specific criteria, as shown in Figure 2-12 and Figure 2-13. Figure 2-12 is the basin averaged rainfall mass curves for each event. Figure 2-13 supplies the DAD curves developed for the Big Fork, Arkansas and SW 2-4 storms. Of note, the December 1982 maximum average rainfall depth exceeded the April 1927 event by about 25 percent for the critical duration period as shown in Table 2-4.

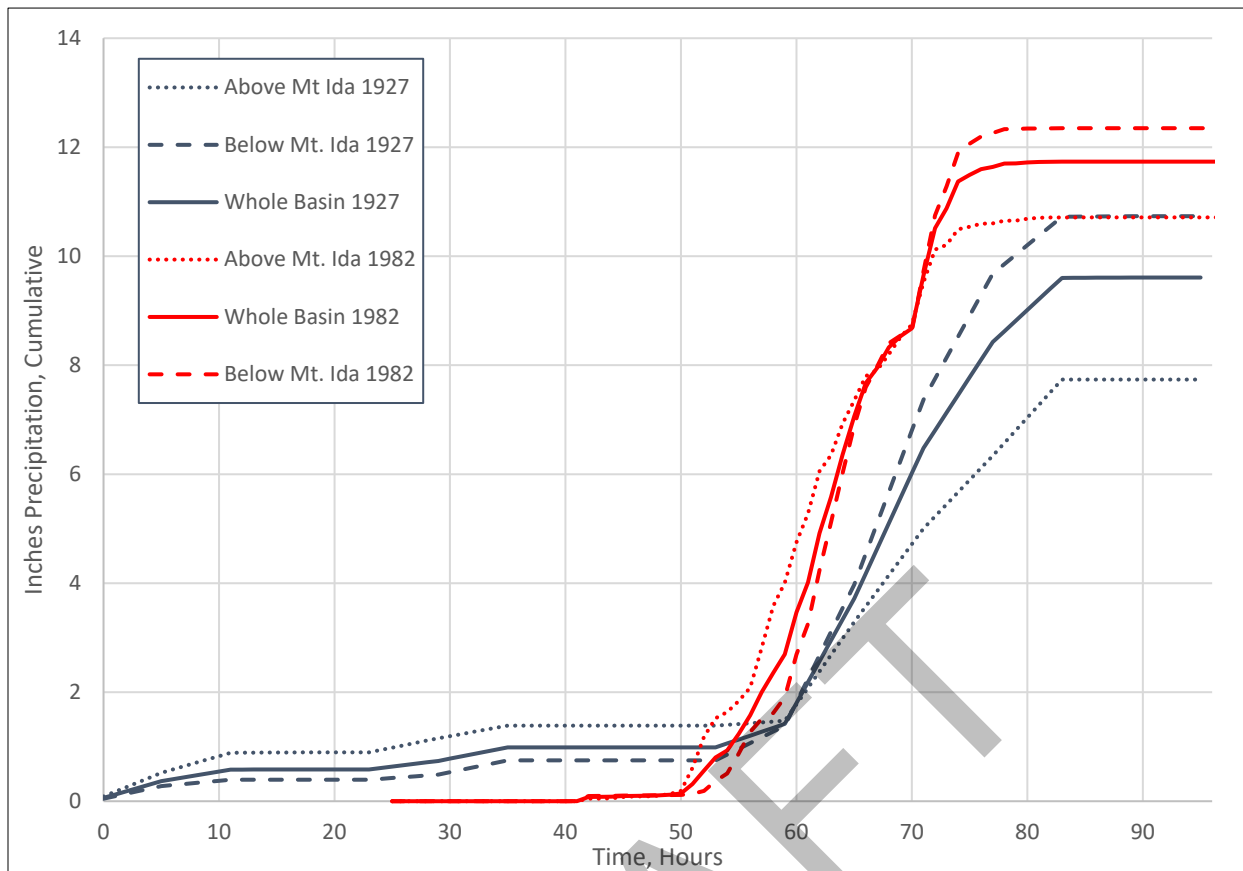


Figure 2-12. Peak 7-hour rainfall mass curves for floods of 1927 and 1982.

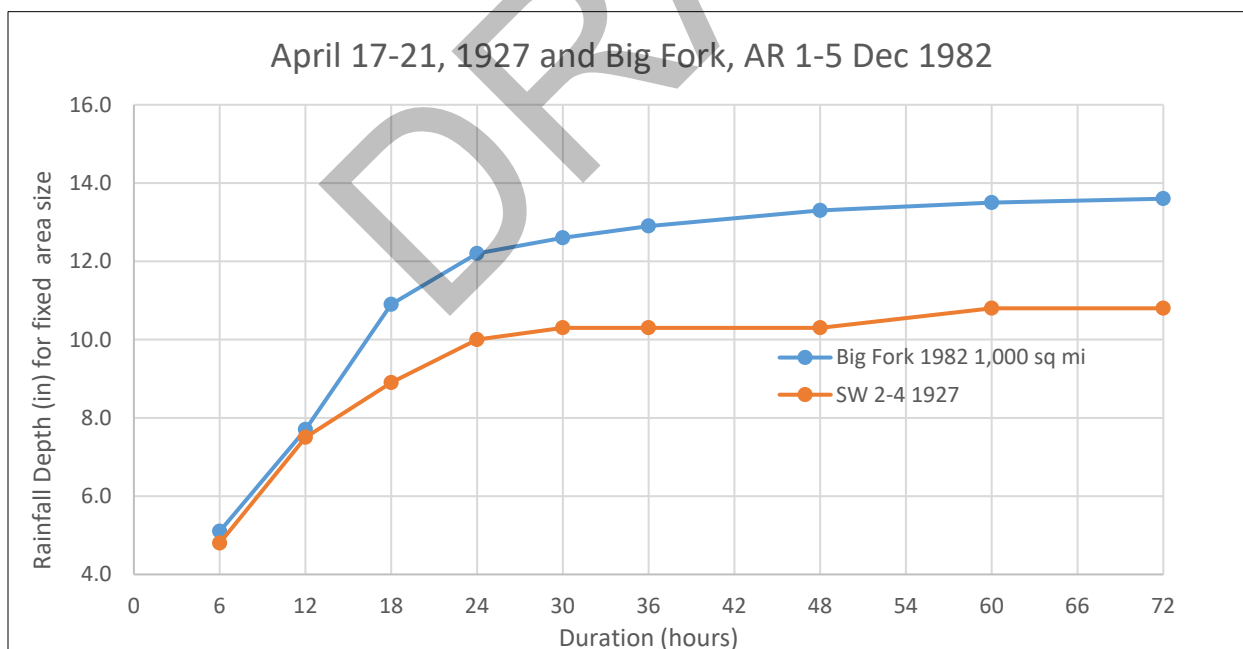


Figure 2-13. Maximum average 1,000 square-mile depth-duration curves for floods of 1927 and 1982.

Table 2-4. Ratio of maximum average 1,000 square-mile depth (December 1982 / April 1927 [SW 2-4] storms).

Rainfall Duration (hrs)	Maximum Average Depth Ratio (1982 storm / 1927 storm)
6	1.06
12	1.03
18	1.22
24	1.22
30	1.22
36	1.25
48	1.29
60	1.25
72	1.26

Flood of May 1990

A series of severe thunderstorms caused flash flooding near Hot Springs, Arkansas in May 1990. 12.97 inches of rainfall was recorded in Hot Springs within a 24-hour period. Maximum rainfall totals for the 6- to 24-hour periods at Remmel Dam and Carpenter Dam had recurrence intervals that exceeded 100 years. Floodwaters 2-4 feet deep flooded the historic downtown area of Hot Springs and caused extensive damage to public and private property (Southard, 1992). Blakely Mountain Dam gage recorded a stage of 6.6 inches corresponding to a calculated inflow of 50,000 cfs. A maximum stage of 4.7 inches was recorded at Mt. Ida. Estimated inflow at Remmel Dam at Jones Mill was 84,000 cfs. Blakely Mountain Dam released daily average flows between 20 and 6,000 cfs throughout the duration of the storm.

2.4 Reservoir Operations

Blakely Mountain Dam is an embankment dam that impounds Lake Ouachita. Lake Ouachita is a multi-purpose reservoir used for flood control, hydropower, water supply, and recreation. Spillway construction began in August 1947 and was completed in August 1948. Construction of the diversion tunnels began in July 1948. The dam was completed in 1952 and the powerhouse was completed in October 1955. Both units were placed online during October 1955. Blakely Mountain Dam has two 37,500 kW turbines, three slide gates (8 feet x 15 feet), and one uncontrolled spillway with a width of 200 feet at elevation 592.0 feet NGVD88. The dam is 1,100 feet in length.

Lake Ouachita is operated by the USACE Vicksburg District, with the energy company, Entergy, operating within the power pool for hydropower generation. General flood control operations are directed in the Blakely Mountain Dam Water Control Manual (WCM) (U.S. Army Corps of Engineers, 1970).

The reservoir was designed to control the 1927 historic flood of record and to regulate outflows to meet downstream channel capacity at Malvern, AR. The objective of the flood control pool is to store runoff from large storm events and to regulate releases such that discharges at the downstream control point, Malvern, AR, does not exceed the channel capacity of 20,000 cfs with dam releases limited to 15,000 cfs. In practice, discharges from Remmel Dam are used in lieu of Malvern flows. The project is operated using normal flood pool operations during emergency conditions, with allowance given to the Operations Manager to deviate from the guidelines in the event that it is necessary for emergency reasons for the safety of the structure and the protection of human life, or to avoid any other critical situation that may arise.

Table 2-5. Blakely Mountain Dam pertinent data.

General Project Information	
Dam Type	Earthfill
Completion Date	1955
Maximum Height	236 feet
Spillway Type	Uncontrolled
Outlet Controls	Single 19-foot conduit with three 8 ft x 15 ft slide gates at an intake elevation 480.0 ft
Upstream Projects, River Mile, Drainage Area	None (Unregulated)
Downstream Projects, River Mile, Drainage Area	Carpenter Dam (Entergy), RM 411, 411 sq. mi. Rommel Dam (Entergy), RM 399, 46 sq. mi. Thatcher Lock and Dam, RM 281 Felsenthal Lock and Dam, RM 226 Columbia Lock and Dam, RM 117 Jonesville Lock and Dam, RM 25
Hydrology	
Watershed	Ouachita
River	Ouachita River, RM 430.4
Drainage Area	1,105 sq. mi.
Spillway Design Flood (SDF) Peak Inflow	418,200 cfs
Spillway Design Flood (SDF) Maximum Pool Elevation	610.2 feet (NAVD88)
Probable Maximum Flood (PMF) Peak Inflow ¹	991,500 cfs
Probable Maximum Flood (PMF) Maximum Pool Elevation ¹	618.2 feet (NAVD88)
Pool of Record Elevation	590.1 feet (NAVD88)
Operating Levels	
Dam Crest Elevation	616.0 feet (NAVD88) (Design) 615.7 feet (NAVD88) (2018 Survey)
Spillway Crest Elevation	592.0 feet (NAVD88) (Design) 591.7 feet (NAVD88) (2018 Survey)
Flood Control Pool	592.0 feet (NAVD88)
Power Pool	578.0 feet (NAVD88)
Conservation Pool	535.0 feet (NAVD88)
Power Conduit Intake Elevation	480.0 feet (NAVD88)
Minimum Pool	435.0 feet (NAVD88)

¹Probable Maximum Flood (PMF) best estimate taken from the PMF report (U.S. Army Corps of Engineers, 2019).

3 Data Sources

Reservoir stage and inflow records need to reflect current conditions and operations within the basin for the period of record. However, systematic record lengths are often insufficient to determine empirical stage-frequency relationships for the infrequent reservoir stages of interest. Additional data, including historical floods and paleoflood information, can be used to extend the observed stage record. This chapter describes the systematic period of record (POR) data used in the stage-frequency curves, as well as supplemental data that were collected to extend the systematic record, including historical data, paleoflood data, and regional skew.

3.1 Systematic Data

3.1.1 Streamflow Gaging Records

Consistent gaging efforts began in the upper Ouachita Basin in 1923, with fragmentary records maintained prior to 1923 (Frame, 1930). Due to changing river conditions and the construction of hydro-electric dams, river gages were added and discontinued. Daily discharge estimates for the Ouachita River near Blakely Mountain Dam have been compiled from five gage stations, in order to form a more complete record of annual maximum daily flows, summarized in Table 3-1. The stations were chosen in order to gain a complete record of all large floods that exceeded approximately 80,000 cfs for the gaging period of record.

Table 3-1. Annual maximum daily discharge records on the Ouachita River near Blakely Mountain Dam used for frequency analysis.

USGS Gaging Station No.	Station Name	Drainage Area (mi ²)	Period of Record (Water Year)	Records used for Frequency Analysis (Water Year)	Maximum 1-Day Discharge (ft ³ /s) (year)
USACE Records	Blakely Mountain Dam	1,105	1961-Present	1961-Present	--
USACE Records	Upper Pool Gage at Blakely Mountain Dam near Hot Springs, AR	1,105	1943-1961	1951-1960	76,200 (1960)
07357501	Ouachita River at Blakely Mountain Dam near Hot Springs, AR	1,105	1942-1953 1953-1977	--	--
07357500	Ouachita River near Mountain Pine, AR (Later renamed Lake Ouachita near Hot Springs Arkansas)	1,100	1936-1951	1936-1950	90,300 (1945)
07358000	Ouachita River near Hot Springs, AR	1,405	1923 - 1930	1923 - 1930	125,000 (1923)
07359001	Ouachita River below Rammel dam at Jones Mill, AR	1,550	1928-2003	1931-1935	104,000 (1945)

The Hot Springs gage (07358000) was in place from September 1922 until 1930, when the location was inundated by the filling of the downstream dam. The gage was located in a stable channel with a stream bed of gravel and rock, with the channel being obstructed by outcropping rock dikes. Flow measurements were made from bridge or by wading. Stage measurements were made from a chain gage on the downstream side of the bridge. Records below 60,000 cfs are considered good. An extension rating curve determined flows above 60,000 cfs.

There were multiple gages near Rammel Dam (07359001), downstream of Blakely Mountain Dam for 1925-1927 and 1928-present. The gage was located in a stream bed of gravel. Flow measurements were made from cable or by wading. Stage measurements were made using a water recorder, with daily discharges determined with a discharge integrator. Peak stages are determined by high flood marks. Records below 70,000 cfs are considered good. An extension rating curve determined flows above 70,000 cfs using an unknown method.

The Ouachita River near Mt. Pine (07357500) was an automatic gage in service from 1936-1951 and was located on the Ouachita River in Goring Bend, 4 miles northwest of Mt. Pine, AR. The site is now inundated by Lake Ouachita after the construction of Blakely Mountain Dam. The gage had a drainage area of 1,100 square miles.

Several USACE gages were in place starting with the design of Blakely Mountain Dam. The Blakely Mountain dam site gage (1943-1951) had a drainage area of 1,105 square miles. This was replaced by the Blakely Mountain Upper Pool Gage, which was read by the construction division at the project site. This gage consists of a staff gage in two sections. This gage was a temporary gage, which lasted from 1950 until completion of Blakely Mountain Dam. The gage was replaced with a permanent stage gage. Before 1952 the gage reported flow, but in 1953 began reporting stage only. An additional downstream gage (Ouachita River below Blakely Mountain Dam Site) reported discharge. More information on these gages is located in the report, *Paleoflood Analysis for Blakely Mountain Dam, Garland County, Arkansas* (U.S. Army Corps of Engineers, 2019).

3.1.2 Reservoir Stage and Inflow Records

Inflow to the reservoir was determined for the period 1953 to present by calculating inflow as the sum of the outflow and the change in storage. Outflow was calculated from gate settings, and the change in storage was determined from an existing stage-storage relationship. Calculated inflows for the period 1961 through 2019 was provided by the USACE Vicksburg District. Hardcopy records of stage and outflow (used to calculate inflow) from 1953-1961 were also provided. Daily stage data were available from 1961 to present, and hourly stage data from 1999 to present.

3.2 Historical Data

Historical data gathered from government and local historical records is valuable for extending the systematic period of record. For estimating probabilities of extreme floods, especially beyond the systematic record, Bulletin 17C recommends collecting historical flood data and using it in flood frequency (England, et al., 2019). Information on historical flood events is detailed in the below paragraphs. More information on these events and historical records is located in the paleoflood report, *Paleoflood Analysis for Blakely Mountain Dam, Garland County, Arkansas* (U.S. Army Corps of Engineers, 2019).

Historical information, settlements, and flood history were initially searched and gathered from the USACE Mississippi Valley Division; the National Archives at Atlanta; the Smithsonian Archives; the Arkansas State Archives; Hot Springs National Park; various newspapers; interviews with the National Weather Service in Little Rock, AR; and, internet searches. The major sources of information used were settlement dates, the U.S. Army Annual Reports of the Chief of Engineers, and reports submitted to the U.S. House of Representatives (House Documents or '308' Reports).

According to a report to the U.S. House of Representatives, "The earliest discharge records on the Ouachita were made in connection with the surveys of 1872. However, nothing further was done until the surveys of 1895-97, when measurements were taken at a large number of places along the river" (U.S. House of Representatives, 1934, p. 35). The earliest documents focused on navigation and do not recommend extensive work along the river, until the "permanent improvement of the upper river is practical at a reasonable cost" (U.S. Army Corps of Engineers, 1891). However, the 1891 Annual Report of the Chief of Engineers had the first crude high water profile, a combination of the 1871 and 1874 flood events (U.S. Army Corps of Engineers, 1891). Then, by 1909, there are reports to Congress with preliminary design work for a dam near the current Blakely Mountain Dam site, and designs are mentioned again in 1938 (U.S. House of Representatives, 1912; U.S. House of Representatives, 1938).

These reports state, "Destructive floods have occurred in 1882, 1894, 1905, 1907, 1915, 1920, 1923, 1927, and 1930. Records and data are incomplete for floods prior to 1915. Floods of less magnitude and duration occur nearly every year. [...] The most destructive flood for which records are available was that produced by the 1927 storm. [...] The 1882 flood was greater than the 1927 flood, but records are not available" (U.S. House of Representatives, 1934, p. 27). It is known that there were also flood events along the upper Ouachita River in 1848 and 1871. Because of the mountainous terrain, it is common to have limited flooding on the Ouachita River in the reach upstream of Malvern, as given in Exhibit O of U.S. House of Representatives (1934). Notable flood events before 1923 (beginning of gage data) include the following.

December 1848: The Rockport Bridge was built in 1846-7. This was a well-built, lattice-style wooden bridge and was, "thought to be 8 feet over the high water mark, but on day of destruction the water was 11 foot higher than it had known to be before" (Bennett & Smith, 1988). This event was also part of a larger storm system which formed the Great Flood of 1849 on the Mississippi River.

May 1871: This event caused the Ouachita River to rise 12 feet in Arkadelphia, Arkansas (an estimated additional 35,000–50,000 cfs) (River News, 1871). It was a flash flood on Hot Spring Creek as well: “throughout the day there had been an occasional slight mist and sprinkle of rain, but at about 3:40 o'clock in the evening rain began to fall in vast volume and continued for about twenty-five minutes, when it abated for about five minutes, and then commenced again as hard as before. It soon became apparent that the creek was rapidly rising, and that another hour's rain [illegible] that of the past half hour would send it high out of its usual bed. But the expectations of all in that respect were more than realized, for is a little over an hour after the heavy rain first began to fall, many of the homes in the valley were threatened with destruction.” (Taughner, 1871)

May 1882: This is an event that seems to parallel the 1927 event in many ways. This flood was part of a larger regional flood event lasting 5 months with 22,000 square miles being flooded (including the Mississippi River). The Army Signal Corps reported, “Through the 12th, heavy rains fell in much of the state, with the heaviest totals in the west and central portions. The rains sent the Ouachita, Saline, Arkansas, White, Saint Francis, and Mississippi Rivers into flood. Hot Springs suffered severe flash flooding on the 11th (National Weather Service, 2015). It is unknown what percentage of the damages was from flash flooding versus riverine flooding from the vicinity of Blakely Mountain Dam. At Mt. Ida on May 8, the maximum recorded daily rainfall was 3.7 inches, and on May 9 there was a reported 8.26 inches at Malvern (National Weather Service, 2015). The 1934 House Document high water profiles show the 1882 and 1927 events at a similar magnitude for the 2,311 square mile drainage area surrounding Arkadelphia, Arkansas (National Weather Service, 2019). In the 5,360 square mile drainage area surrounding Camden, Arkansas, the Ouachita River stage reached 46 feet, which is the flood of record (National Weather Service, 2019).

3.3 Paleoflood Data

To further extend the period of record, a limited, reconnaissance-level paleoflood analysis was completed on the reach immediately downstream of the Blakely Mountain Dam. This effort is further detailed in the paleoflood report, *Paleoflood Analysis for Blakely Mountain Dam, Garland County, Arkansas* (U.S. Army Corps of Engineers, 2019).

The terrain in the study area was analyzed for geomorphic features that would suggest the occurrence of an ancient flood event or features that appear unaffected by high floodwaters. Field Reconnaissance of these features resulted in the collection of charcoal samples from a location with a Paleo Stage Indicator (PSI) and in a location seemingly unaffected by floodwaters, which is a Non-Exceedance Bound (NEB). Charcoal sample age was estimated via numerical radio carbon analysis, which was validated by the relative geomorphic position and relative soil-profile development of the soil type.

The geomorphic and stratigraphic results from this paleoflood analysis provided information for calibration of a HEC-RAS model for the flood of record, evidence of a large, prehistoric flood between 1,000 and 5,000 years ago, with a best estimate age of 3,200 years, and evidence of the maximum limiting flood stage within the past approximately 5,000 to 9,000 years (best estimate age of 7,200 years). The presence of a low historic terrace along both the western and eastern banks of the Ouachita River provided a nearly continuous down-valley profile of historic high stages for about 3.5 miles downstream from Blakely Mountain Dam. The terrace approximates the high river stages during the 1945 flood of record (prior to the construction of Blakely Mountain Dam), and serves to aid in calibration of the HEC-RAS hydraulic model along this reach of river. This helps to constrain likely roughness and channel gradient scenarios used in modeling larger, prehistoric discharges.

Discharge estimates were estimated from 2D hydraulic simulations with a steady inflow hydrograph and model geometry calibrated to the March 1945 flood peak. The geometry was also altered to estimate natural pre-settlement conditions. Discharge ranges include variation in roughness coefficients, depths to inundate terraces, and variation in geomorphic surfaces. Table 3-2 provides the best estimate and associated ranges of ages and peak discharges for the “Upper Red Cup” NEB and the “Big Old Tree” PSI, as previously described. The recommended paleoflood (PSI and NEB) ages for use in frequency analyses were determined through sensitivity analyses and calibration. The young estimate for each, corresponding to an age of approximately 1,000 years for the PSI and 5,000 for the NEB, were recommended for use in the inflow volume-frequency analyses.

Table 3-2. Paleoflood data for Blakely Mountain Dam frequency analysis (U.S. Army Corps of Engineers, 2019).

Geomorphic Datum	Elevation of Datum (centerline distance)	Estimated Age of Datum	Recommended Age of Datum for Frequency Analysis	Low Estimate Peak Discharge (cfs)	Best Estimate Peak Discharge (cfs)	High Estimate Peak Discharge (cfs)
Upper Red Cup (Ht1) NEB	438.5 ft (@7400 ft)	7,200 years ago (5,000 to 9,000 years ago)	5,000 years ago	350,000	400,000	450,000
Big Old Tree (Ht2) PSI	419.5 ft (@7400 ft)	3,200 years ago (1,000 to 5,000 years ago)	1,000 years ago	150,000	165,000	180,000

3.4 Regional Skew

Regional skew information was used to weight the at-site skew for the inflow volume-frequency analysis. The USGS recently performed a regional skew analysis and peak flow-frequency study for sites in Arkansas using data through WY 2013 (Wagner, Krieger, & Veilleux, 2016), using methods consistent with Bulletin 17C (England, et al., 2019). Peak-flow data through 2013 for Arkansas, Oklahoma, and Louisiana was analyzed using 180 stations to develop a regional skew model. A constant model for peak regional skew was selected, with a regional skew of -0.17. This constant model had an Average Variance of Prediction (AVP) at a new site (AVP is equivalent to the Bulletin 17C Mean Square Error) equal to 0.12, which corresponds to an effective record length of 59 years.

Flood Region B, Subregion 1, which includes the Blakely Mountain Dam watershed, is the Ouachita Mountains physiographic province. In this region, 36 gages were analyzed with basin areas ranging from 0.04 to 2,680 square miles and elevations ranging from 1,527 to 494 feet. The Ouachita River at Mount Ida (410 square miles; USGS station 07356000) was used in the regional skew analysis. The peak station skew and skew mean-square error (MSE) estimates for this station were -0.15 and 0.08, respectively, for the 72-year systematic record that was analyzed (1942–2013) (Wagner, Krieger, & Veilleux, 2016). The skew and MSE computed for this station is consistent with the regional skew study results.

4 Inflow Volume-Frequency Analysis

The stage-frequency relationships for Blakely Mountain Dam was determined using an inflow volume-based approach as outlined in RMC-TR-2018-03 (U.S. Army Corps of Engineers, 2018). Estimating the annual exceedance probability (AEP) for extreme floods such as the Probable Maximum Flood (PMF) is an important element of Dam Safety studies. For the risk assessment of dams, the probability of failure is often conditional on the magnitude of the hydrologic loading. Therefore, it is imperative to ascertain credible estimates of exceedance probabilities of extreme floods that could lead to failure.

In the U.S., high hazard dams are designed to pass the PMF, which typically has an AEP of 1:10,000 (1E-4) or less frequent. Most projects in the USACE portfolio have limited flood information, with many sites having less than 100 years of gaged flow data. As a result, the greatest source of uncertainty in AEP and quantile estimates in a flood frequency analysis is often inadequate data. In order to reduce epistemic uncertainties in the estimated AEPs for extreme floods, such as the PMF, it is important to incorporate as much hydrologic information into the frequency analysis as reasonably possible.

The paleoflood frequency analysis for Blakely Mountain Dam was performed following the procedures outlined in Bulletin 17C (England, et al., 2019), which estimates flood frequency curves using the Expected Moments Algorithm (EMA). All EMA frequency analyses were performed using the PeakfqSA software as detailed in the paleoflood report (U.S. Army Corps of Engineers, 2019).

The final inflow volume-frequency analysis results for this study were based on a Bayesian analysis framework, originally profiled by Viglione et al. (2013), for combining limited at-site flood data with temporal information on historic and paleofloods, regional information on the skewness coefficient and precipitation-frequency, and causal information on the flood processes. This framework is implemented in the Bayesian estimation and fitting software RMC-BestFit¹. This framework was additionally used in recent dam risk assessments, including the Willamette Basin (U.S. Army Corps of Engineers, 2019) and Grapevine Dam (U.S. Army Corps of Engineers, 2018)².

The Bayesian flood-frequency method used for the volume-frequency analysis was necessary to integrate the regional precipitation-frequency results into the final stage-frequency relationship. Similar to the EMA methodology used in the paleoflood frequency analysis, the Bayesian method is capable of incorporating non-standard data, such as censored historical or paleoflood data, into the parameter estimation procedure. The Bayesian method is also capable of incorporating causal information, such as rainfall-runoff results and expert opinion into the frequency analysis; whereas EMA is not capable of incorporating the causal rainfall-runoff information in a formal, probabilistic manner.

The following sections details the inputs used by both PeakfqSA and RMC-BestFit. This will include the critical duration, systematic record, historical events, paleoflood study results, regional skew, and rainfall-runoff. The chapter begins by outlining the PeakfqSA results, followed by the RMC-BestFit results. A sensitivity analysis on the reservoir stage-frequency curve was performed using the PeakfqSA and RMC-BestFit results (section 5.7).

4.1 Critical Inflow Duration Analysis

The critical inflow duration is defined as the inflow duration that typically results in the highest water surface elevations for the reservoir. Accurate assessment of reservoir stage-frequency relies on estimates of annual maximum stages; therefore, the critical duration must be considered to properly identify the maximum peak stages for the reservoir. For this analysis, the critical duration represents a single, independent inflow event. Prior analyses found the critical inflow duration to be heavily dependent on the antecedent flood event; however, in this analysis, only the peak event was considered, consistent with RMC-RFA procedures. Events were selected such that the storms were independent and are representative of large inflow events to Blakely Mountain Dam. Both rainfall and inflow were graphically analyzed

¹ The U.S. Army Corps of Engineers (USACE) Risk Management Center (RMC), in collaboration with the Engineer Research and Development Center (ERDC) Coastal and Hydraulics Laboratory (CHL), developed the Bayesian estimation and fitting software (RMC-BestFit) to enhance and expedite flood hazard assessments within the Flood Risk Management, Planning, and Dam and Levee Safety communities of practice. The alpha version was used for this study. The final version of the software is planned for release and certification in 2020.

² The Bayesian methodology and implementation using RMC-BestFit have been reviewed and approved by RMC lead engineers and ERDC personnel, and the software has been numerically validated against three established software programs. The implementation of the software for the Willamette Basin and Grapevine Dam studies have undergone Agency Technical Review (ATR) approval.

to determine the critical inflow duration to Blakely Mountain Dam based on the 2 largest volume events and the 3 largest independent peak pool events with hourly data: 1927, 1982, 2011, 2013, and 2015. Inflow was plotted with reservoir elevation to determine at which duration the peak stage occurs.

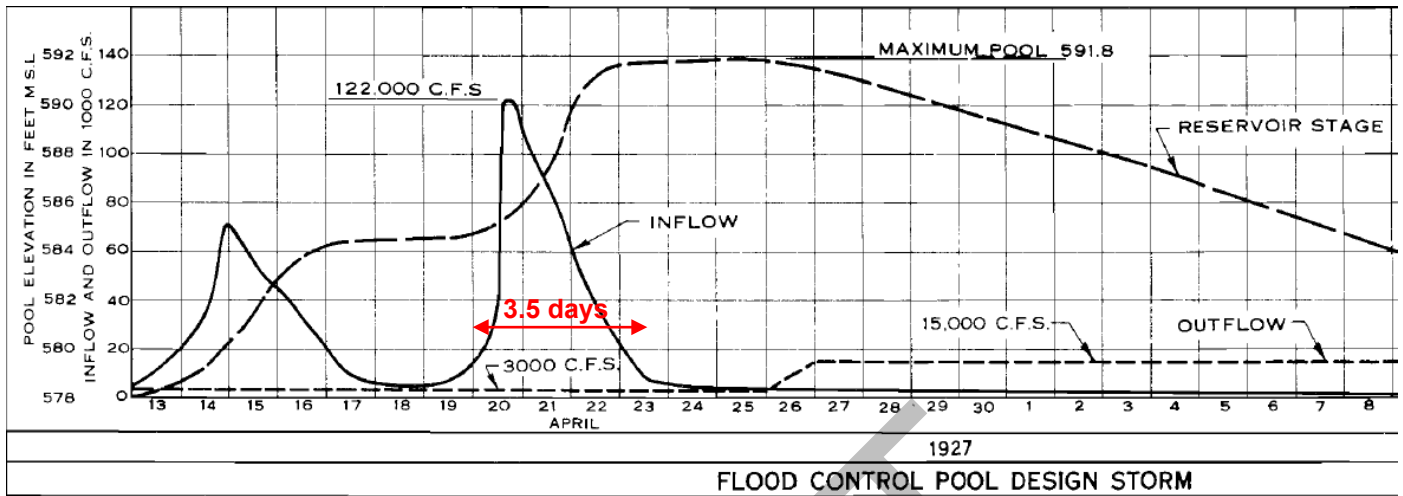


Figure 4-1. Comparative hydrographs for the 1927 flood at Blakely Mountain Dam (Plate 7-1) (U.S. Army Corps of Engineers, 1970). Critical inflow duration of 3.5 days.

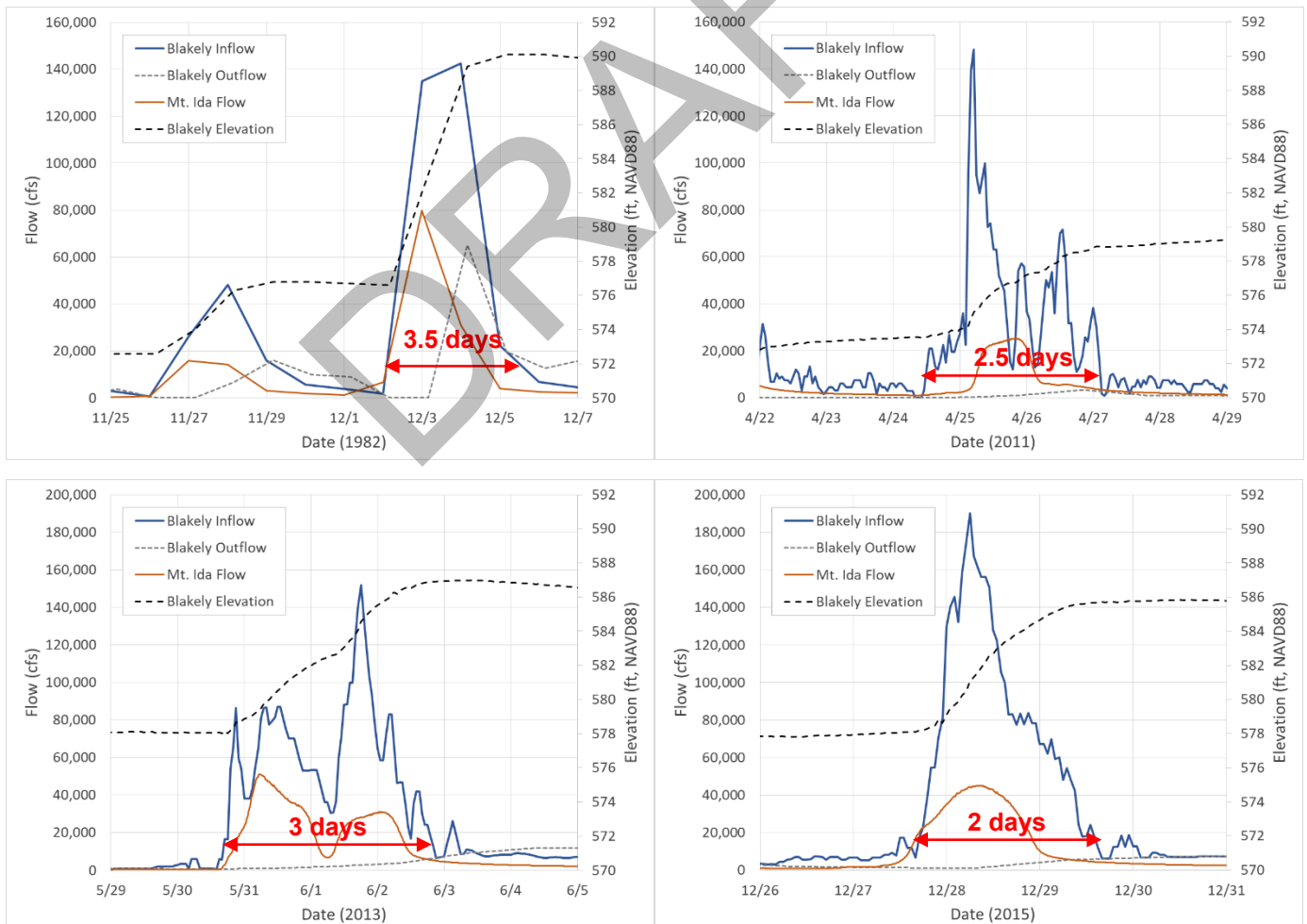


Figure 4-2. Critical inflow duration analysis for events of 1982, 2011, 2013, and 2015. Critical durations range from 2.0 to 3.5 days.

For the representative events analyzed, the critical duration ranged from 2.0 to 3.5 days. Plots of reservoir inflow, Mt. Ida gage flow, reservoir elevation, reservoir outflow, and critical duration are shown in Figure 4-1 and Figure 4-2 for the events analyzed. The critical inflow duration for Blakely Mountain Dam was determined to be 3-days based on evaluation of the historical winter and spring runoff events.

4.2 Frequency Analysis with EMA

The currently accepted flood-frequency methodology in the United States, described in Bulletin 17C (U.S. Geological Survey, 2018), recommends fitting the Log-Pearson Type III (LPIII) distribution using the Expected Moments Algorithm (EMA), which is capable of incorporating historical and paleoflood information into flood frequency studies. The following subsections describe the inflow frequency analysis using EMA and PeakfqsA.

4.2.1 Frequency Analysis with Systematic Data

Systematic data is collected at regular, prescribed intervals under a defined protocol (U.S. Geological Survey, 2018). The total gaged record length near Blakely Mountain Dam is 96 years (1923-2018), including a perception threshold to account for unobserved data (5 years). Assessment of the perception threshold for missing data and broken records is based on the April 1927 flood 3-day maximum (~66,000 cfs). A more complete record of systematic annual maximum daily flows was developed by combining information from five gaging stations. There are no significant tributaries between the five gages. Although there are slight differences in drainage areas between three of the gaging sites, no adjustments were made to transfer the 11 years of downstream records to upstream of Lake Ouachita. There were an additional six gages in this region of the Ouachita River: two gages to provide supplemental data during the design of the dam, and four gages near Carpenter and Remmel Dams. The annual maximum estimated daily flows for the systematic period, with combined gages and reservoir records from WY 1923 to WY 2018 (96 years of record) is shown in Figure 4-3.

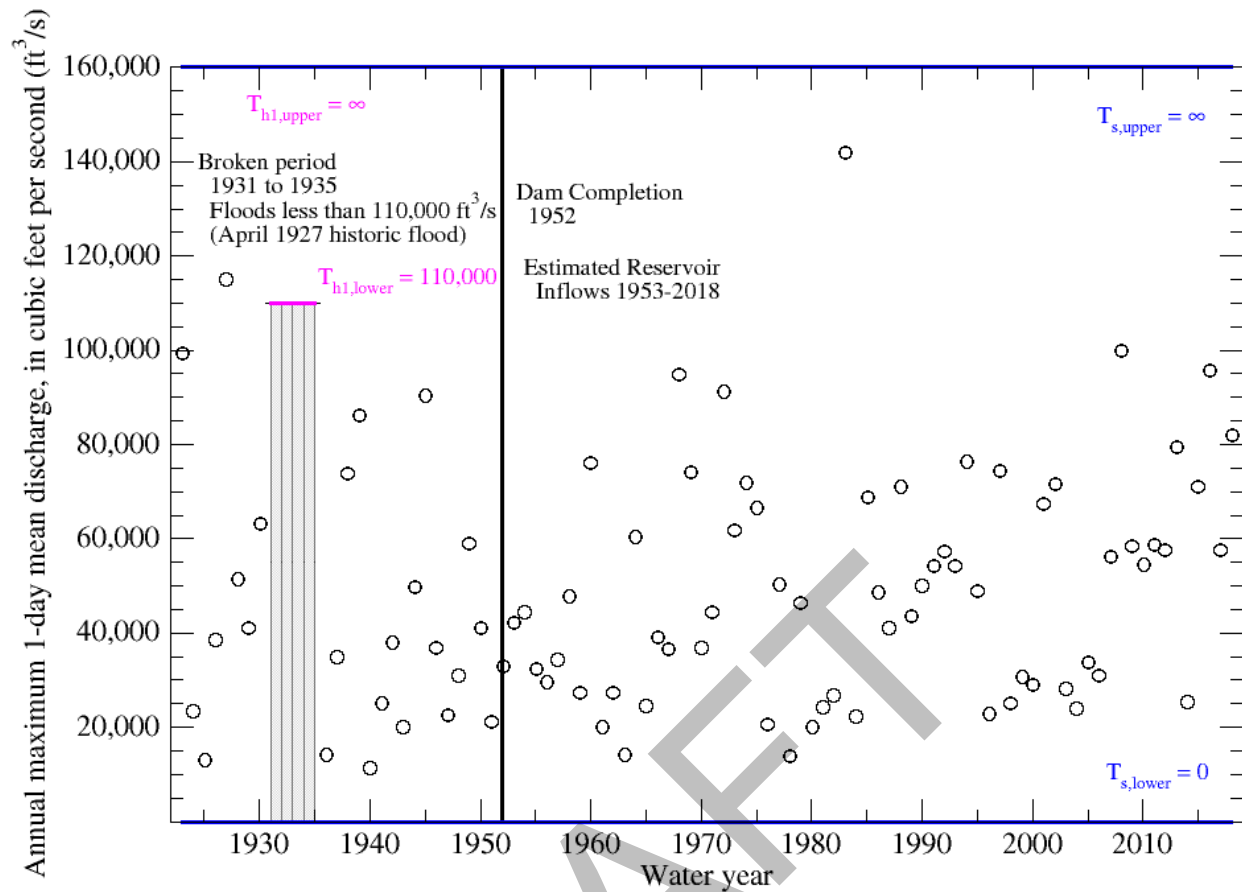


Figure 4-3. Unregulated annual maximum 1-day flows for the systematic record at Blakely Mountain Dam (U.S. Army Corps of Engineers, 2019).

An annual maximum 1-day flow frequency curve using systematic flow records was developed using Bulletin 17C with EMA methodology in PeakfqSA. The fitted moments using the EMA procedure is shown in Table 4-1, and the 1-day inflow-volume frequency relationship with confidence intervals is shown in Figure 4-4.

Table 4-1. Moments using EMA procedures based on systematic observations.

Inflow Duration	Mean of Logs (cfs)	Standard Deviation of Logs (cfs)	Coefficient of Skewness	ERL (years)
1-day	4.628	0.237	-0.247	92

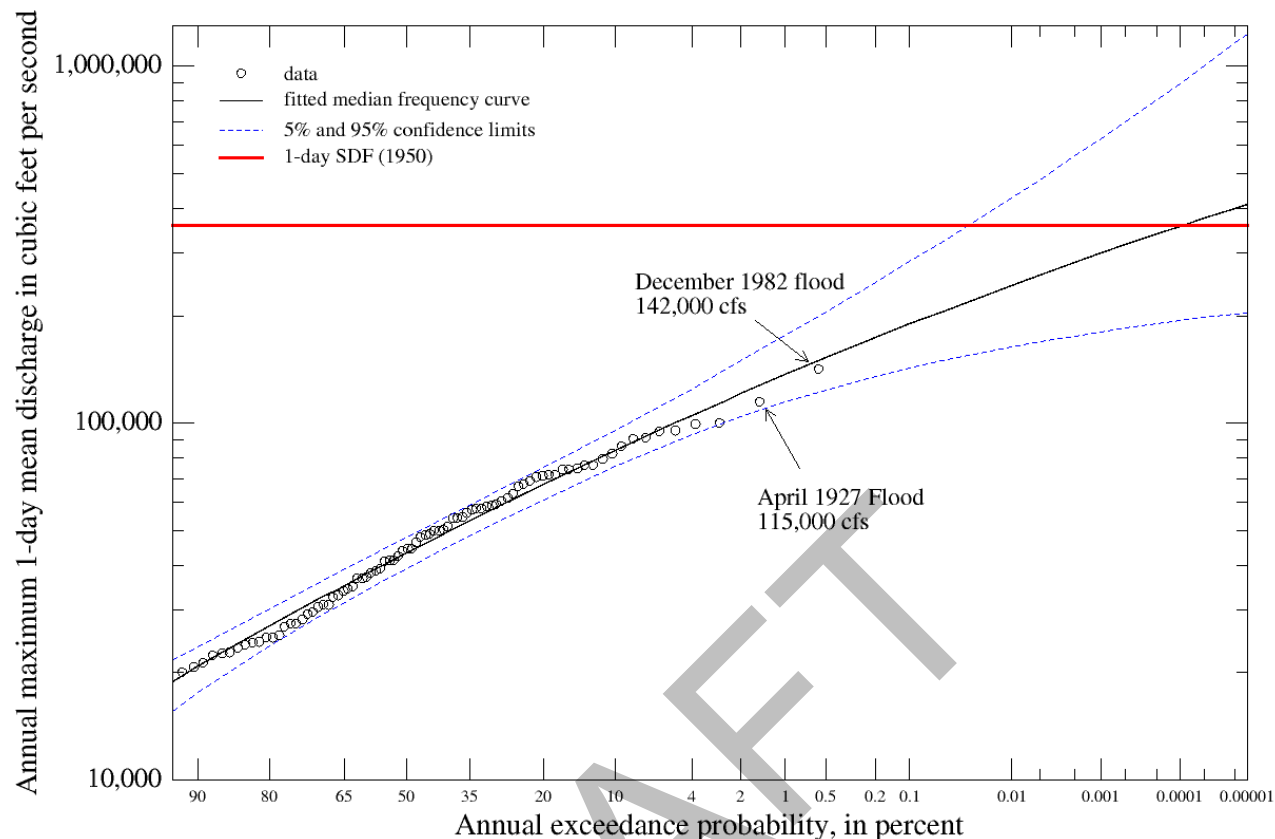


Figure 4-4. Unregulated annual maximum 1-day inflow volume-frequency curve and 90 percent confidence interval based on the systematic record at Blakely Mountain Dam (U.S. Army Corps of Engineers, 2019).

4.2.2 Frequency Analysis with Historical Data

Temporal information expansion is directed toward collecting information on the flood behavior before (or after) the systematic data period (Viglione, Merz, Salinas, & Bloschl, 2013). As discussed in section 3, the historical period useful for flood frequency begins in 1870. The April 1927 flood discharge was used as a perception threshold, as it is the earliest well-known, large-scale regional flood and recorded event. This is based on the assumption that had there been 3-day flows larger than about 66,000 cfs or peak flows higher than 122,000 cfs during the period of 1870 to 1922 (53 years), the events would have been recorded or noted in historical documents. The 1882 flood event is similar in magnitude and potentially larger than the late April 1927 flood (Representatives, 1934, p. 27), and was included as a historical flood. The 1882 historical flood lower estimate (Q_l) is set equal to 1927 (66,000 cfs); the upper estimate (Q_u) is 30 percent larger (86,000 cfs). A time series of the systematic and historical flood data is shown in Figure 4-5. The inclusion of the 1882 flood (possibly larger than December 1982) and historical period increases the extreme flood data and confidence in frequency estimates.

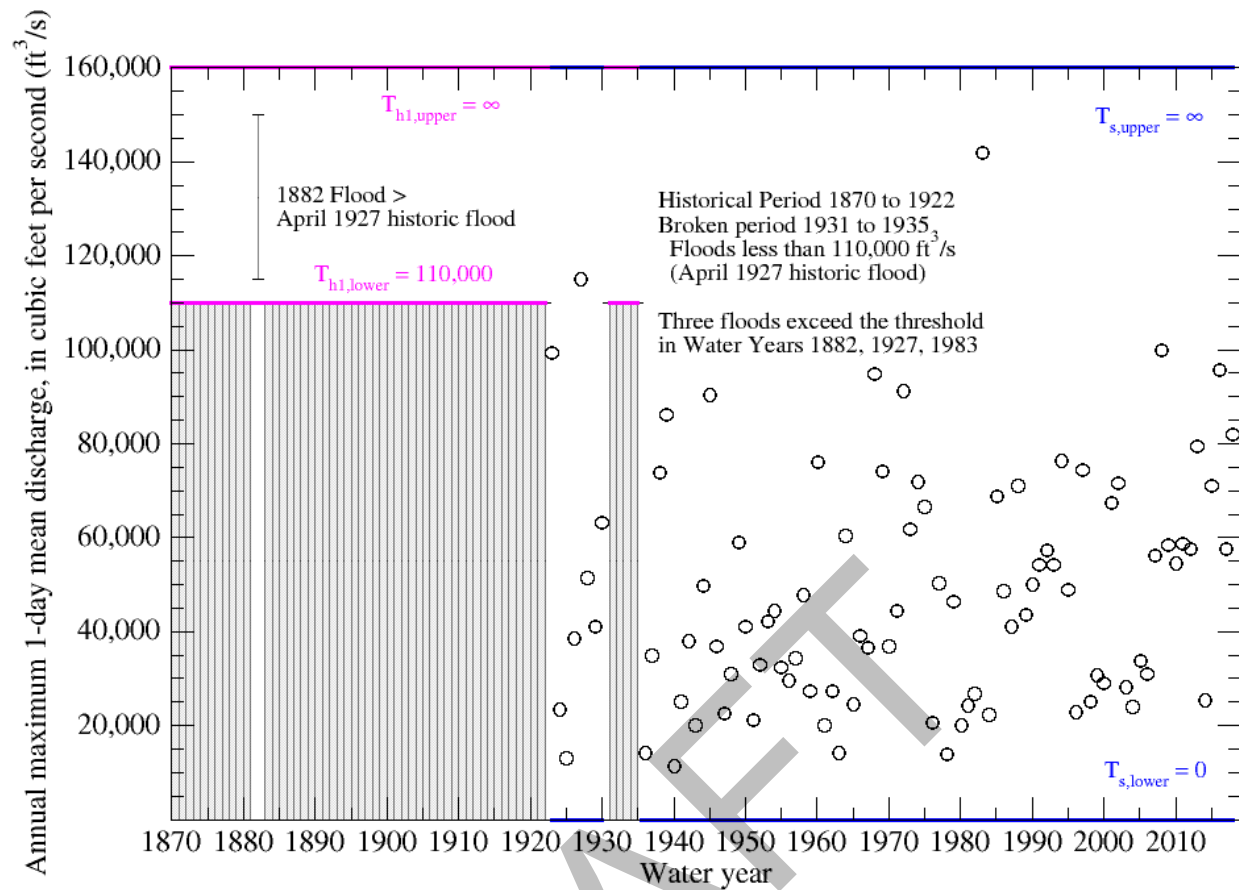


Figure 4-5. Unregulated annual maximum 1-day flows for the systematic record and historical period at Blakely Mountain Dam (U.S. Army Corps of Engineers, 2019).

An annual maximum 1-day flow frequency curve using systematic records and historical flood data is shown in Figure 4-6. Input data include 149 years (1870-2018), with 57 years censored below the perception threshold. The largest floods are December 1982 (WY 1983), the 1882 historical flood, and April 1927. The moments for the fitted frequency curve are shown in Table 4-2. The addition of the historical data slightly decreased the mean, standard deviation, and skew of the curve.

Table 4-2. Moments using EMA procedures for systematic and historical data.

Inflow Duration	Mean of Logs (cfs)	Standard Deviation of Logs (cfs)	Coefficient of Skewness	ERL (years)
1-day	4.625	0.234	-0.279	136

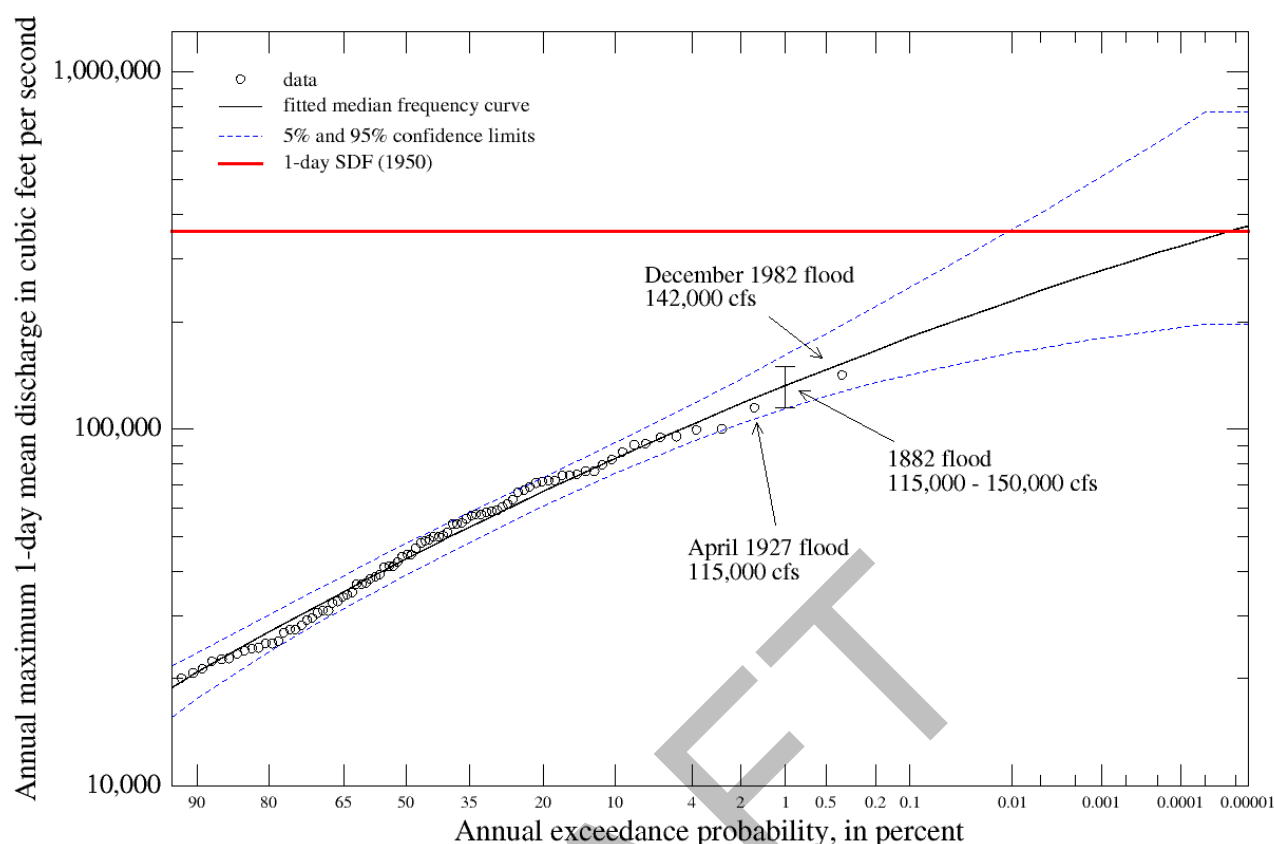


Figure 4-6. Unregulated annual maximum 1-day inflow volume-frequency curve and 90 percent confidence interval based on the systematic and historical flood record at Blakely Mountain Dam (U.S. Army Corps of Engineers, 2019).

4.2.3 Frequency Analysis with Paleoflood Data

Paleoflood data for the frequency analysis were included as best estimates for ages and discharges with perception thresholds. The PSI is represented as a flow interval in year 1020 (~1,000 years before present). The inclusion of the paleoflood, which is larger than the December 1982 flood, and longer paleoflood time frame, increases the extreme flood data and confidence in frequency estimates. A time series of the systematic, historical, and paleoflood data (best estimates) for frequency analysis is shown in Figure 4-7.

4.2.4 1-Day Flow Frequency with Regional Skew

An annual maximum 1-day flow frequency curve incorporating systematic records, historical data, and paleoflood data was developed. A sensitivity analysis was run on the paleoflood estimates to optimize interpretation of the paleoflood data and to quantify uncertainty associated with the estimated paleoflood discharges and ages (U.S. Army Corps of Engineers, 2019). The inclusion of paleoflood information reduces knowledge uncertainty in flood frequency estimates while bringing the largest observed events into longer-time context.

Regional skew was additionally incorporated into the frequency analysis. As discussed in section 3, regional skew was determined from a USGS regional study of Arkansas, Oklahoma, and Louisiana (Wagner, Krieger, & Veilleux, 2016). For this analysis, the USGS regional skew and MSE for peak flows was adopted for 1-day flow frequency at Blakely Mountain Dam. While the USGS study did not analyze skew and MSE for longer flood durations (such as 3-day) and did not include historical information, the skew and MSE estimates are approximate and are considered to be appropriate for the level of effort in this report.

A weighted skew was determined for the frequency analyses using at-site skew and the reported regional skew of $G = -0.17$ from the USGS study and a mean-square error (MSE) of 0.12. The moments of the fitted curve using Bulletin 17C with EMA procedures is summarized in Table 4-3 and plotted as a curve in Figure 4-8.

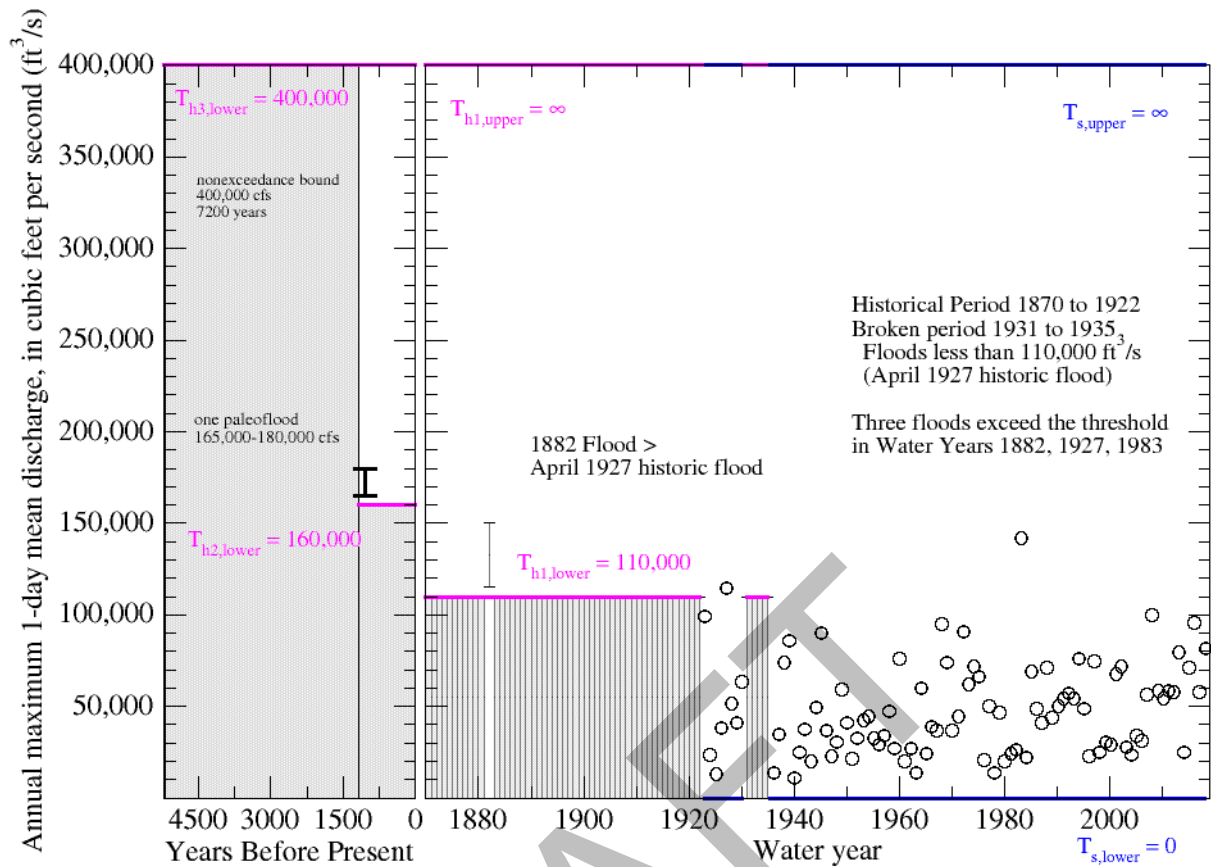


Figure 4-7. Unregulated annual maximum 1-day flows for the systematic, historical, and paleoflood periods at Blakely Mountain Dam. A scale break is used to differentiate the longer paleoflood period from the systematic and historical period (U.S. Army Corps of Engineers, 2019).

Table 4-3. Moments using EMA procedures for systematic, historical, paleoflood, and regional skew data.

Inflow Duration	Mean of Logs (cfs)	Standard Deviation of Logs (cfs)	Coefficient of Skewness	ERL (years)
1-day	4.624	0.235	-0.244	387

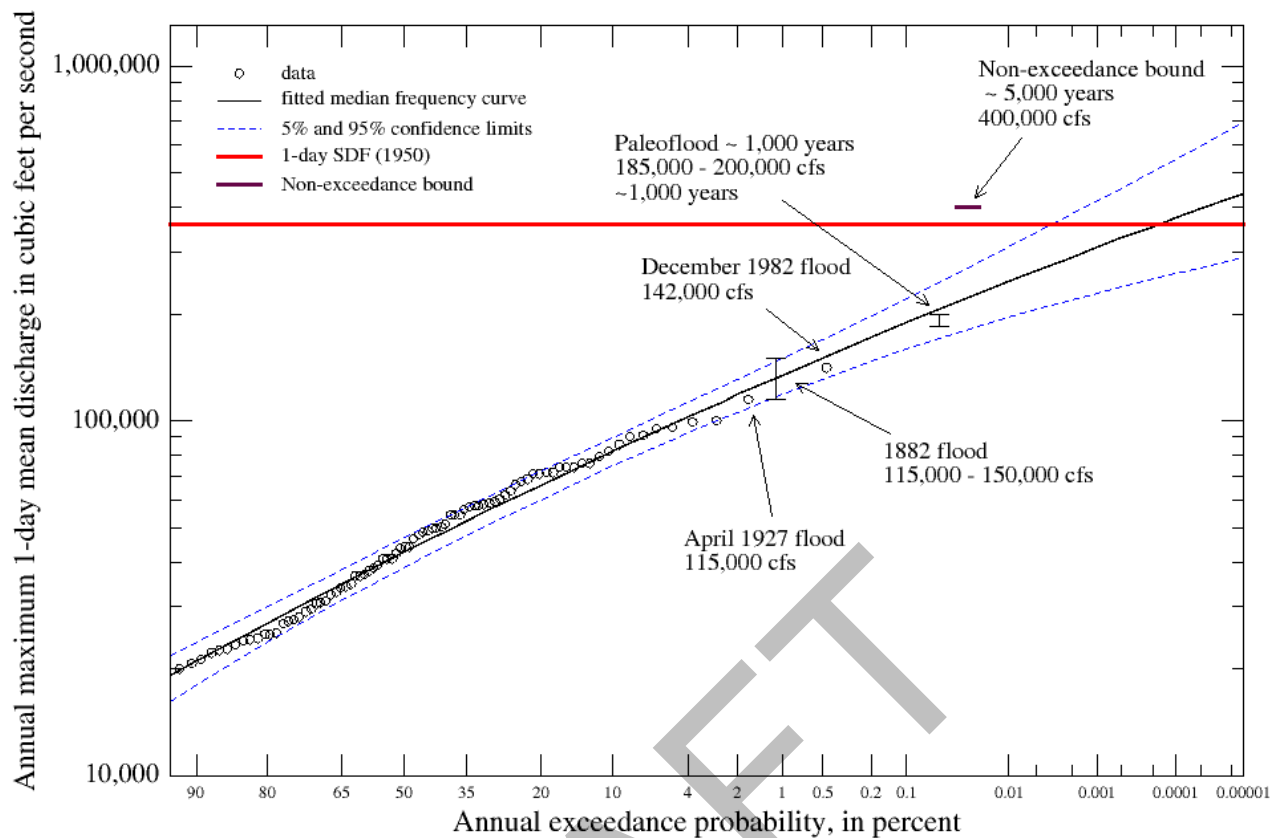


Figure 4-8. Unregulated annual maximum 1-day inflow volume-frequency curve and 90 percent confidence interval based on the systematic, historical, and paleoflood record with regional skew at Blakely Mountain Dam (U.S. Army Corps of Engineers, 2019).

4.2.5 3-Day Flow Frequency with Regional Skew

A 3-day inflow volume-frequency curve was developed to support the hydrologic hazard analysis, in which the critical inflow duration was determined to be 3 days. An index flood method³ was used to shift the mean of the data downward to match the 3-day mean, while preserving the standard deviation and skew of the 1-day data. In this case, the resulting 3-day flow frequency curve was based on the 1-day curve to directly translate the paleoflood information to a relevant volume frequency curve. A comparison of 1-day and 3-day inflow statistics using the systematic record (1927–2018) was used as a basis for estimating the 3-day flow frequency curve by applying a ratio to the 1-day fitted curve moments. A more detailed description of this process is in *Paleoflood Analysis for Blakely Mountain Dam, Garland County, Arkansas* (U.S. Army Corps of Engineers, 2019).

Additionally, regional skew was incorporated into the 3-day frequency analysis. A weighted skew was determined for the frequency analyses using at-site skew and the reported regional skew of $G = -0.17$ from the USGS study and a mean-square error (MSE) of 0.12. Table 4-4 provides a comparison of the 3-day fitted curve moments with variation in the determination of weighted skew: at-site skew (no regional skew), weighted skew using regional skew and Bulletin 17B methods, and weighted skew using regional skew and EMA methods. A sensitivity analysis of these curves was performed to assess the level of uncertainty associated with the weighted skew (section 5.7).

The 3-day flow frequency curve, based on translating the 1-day curve using an at-site mean ratio and preserving the weighted skew, is shown in Figure 4-9. The standard deviation decreased slightly to preserve mean and skew. The 1950 SDF 3-day mean flow of 185,000 cfs (U.S. Army Corps of Engineers, 1950) (Appendix I, Table 8) is shown as a horizontal line.

³ The index flood method was used in lieu of a peak-to-ratio for historic and paleoflood events and statistical smoothing across durations, both of which are approved methods.

Table 4-4. Moments using EMA procedures for systematic, historical, and paleoflood data with varying methods of computing skew.

Inflow Duration	Weighted Skew Method	Mean of Logs (cfs)	Standard Deviation of Logs (cfs)	Coefficient of Skewness	ERL (years)
3-day	EMA at-site	4.409	0.224	-0.283	387
3-day	EMA with regional (B17B MSE method)	4.408	0.223	-0.268	387
3-day	EMA with regional (EMA MSE method)	4.408	0.222	-0.244	387

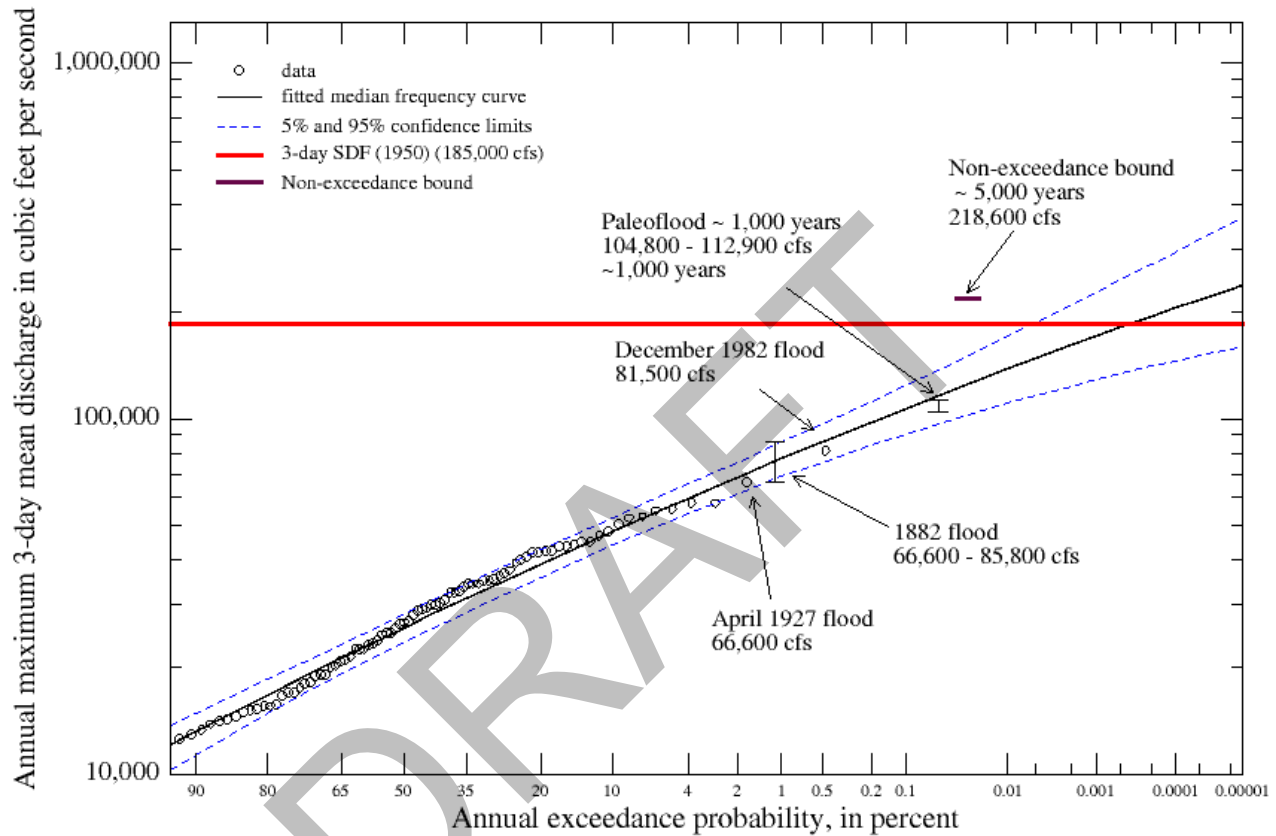


Figure 4-9. Unregulated annual maximum 3-day inflow volume-frequency curve and 90 percent confidence interval based on the systematic, historical, and paleoflood record with regional skew at Blakely Mountain Dam (U.S. Army Corps of Engineers, 2019).

4.3 Frequency Analysis with RMC-BestFit

The Bayesian approach offers a framework that is well suited to incorporate different sources of information, such as systematic flood records, historical floods, regional information, and other hydrologic information along with related uncertainties (Viglione, Merz, Salinas, & Blöschl, 2013). The Bayesian approach allows hydrologists to formally include causal information on the flood processes and their own expertise into the frequency analysis by choosing *a priori* distributions. The ability to combine this information with the observed data is even more important because samples are typically limited in size in hydrological studies. Consequently, the Bayesian approach is more flexible and versatile than classical approaches, such as EMA.

Inflow volume-frequency analyses were performed with RMC-BestFit for systematic data, historical and paleoflood data, regional data on skew, and causal information on the flood processes. Only the 3-day inflow volume-frequency was

evaluated with RMC-BestFit, and each analysis used the LPIII distribution to be consistent with current U.S. flood frequency practice and EMA⁴. The following sections describe each step of the Bayesian flood frequency analysis.

4.3.1 Systematic Data

All of the data used in the Bayesian flood frequency analyses were the same as described in chapter 3, *Data Sources*, and section 4.2, *Frequency Analysis with EMA*. Figure 4-10 shows the Bayesian results for the 3-day volume frequency curve using systematic data only. The distribution corresponding to the posterior mode of the parameters is shown as a solid black line in Figure 4-10 and is labelled as “Posterior Mode” in the legend. The posterior mode is provided for consistency with the maximum likelihood method. The expected probability distribution is shown as a blue dash-dot line and is labelled “Posterior Predictive” in the legend. The 90% credible intervals (which are equivalent to 90% confidence intervals) are shown as a shaded range. The observed annual maximum flood events are plotted using the Hirsch-Stedinger plotting positions (Hirsch & Stedinger, 1987) with a plotting parameter $\alpha = 0.4$, corresponding to a Cunnane formula⁵. Figure 4-10 also lists the first three moments (of log) for the LPIII distribution corresponding to the posterior mode, and the effective record length (ERL), which in this case is 91 years of systematic data.

As can be seen, there is tremendous uncertainty concerning the AEP of the SDF runoff volume. There are well more than four orders of magnitude in uncertainty in AEP between the 90% credible intervals for the SDF. There are over three orders of magnitude between the expected and most likely curves. This high level of uncertainty can lead to over- or under-design of dam and levee safety modifications. Hence, there is a need to collect historical and paleoflood data to extend the record and reduce uncertainty.

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⁴ Model selection uncertainty was not considered in this analysis. The authors recognize that the tail behavior of the distribution is heavily influenced by the choice of LPIII. Adoption of a different probability model would yield a different expected value and associated confidence intervals.

⁵ Please see Appendix 5 of the Guidelines for Determining Flood Flow Frequency Bulletin 17C (England, et al., 2019) for more details on how the plotting positions are determined when incorporating historical flood data and perception thresholds.

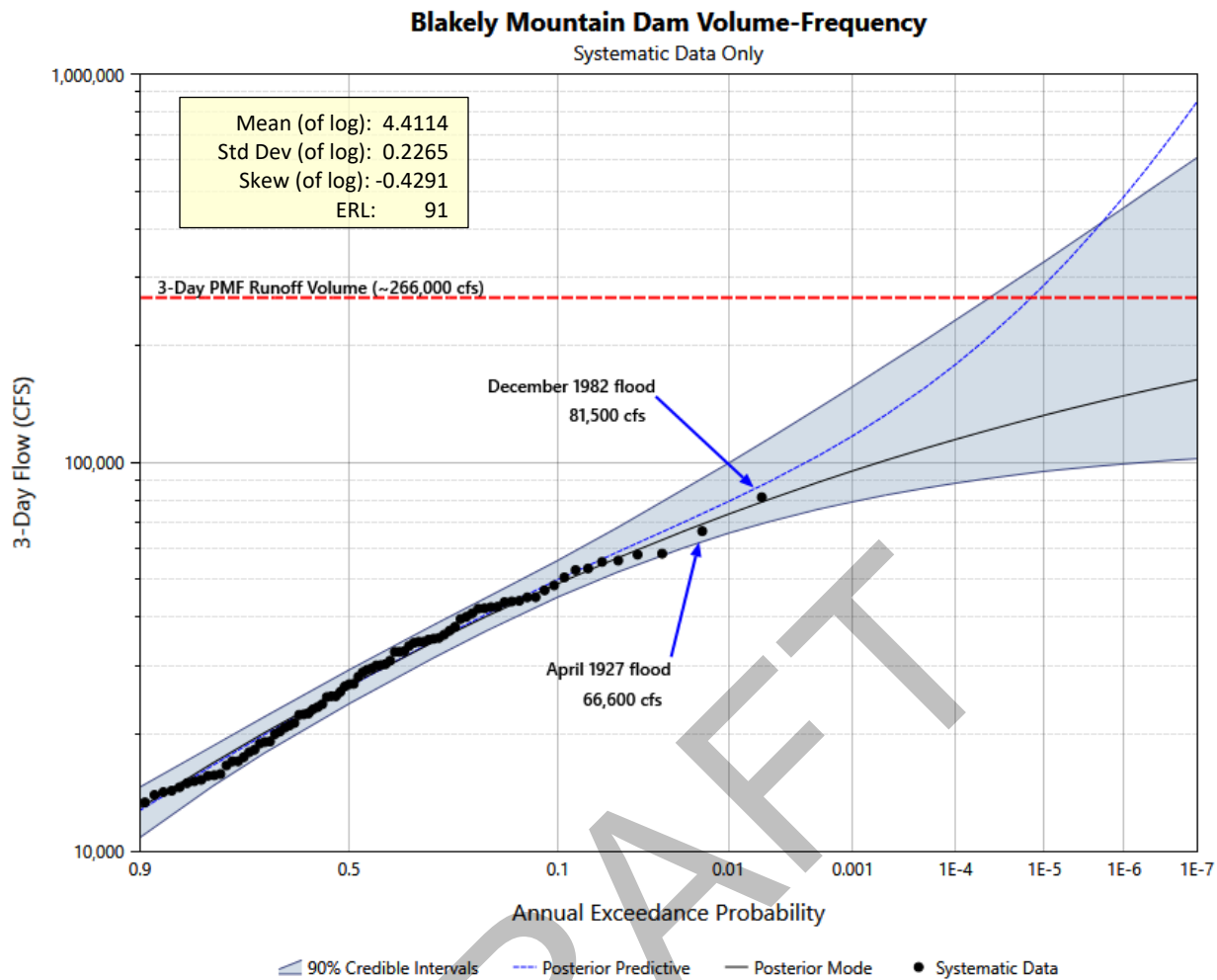


Figure 4-10. Bayesian results for the unregulated annual maximum 3-day inflow volume-frequency curve and 90 percent confidence interval based on the systematic data only.

4.3.2 Historical and Paleoflood Data

Figure 4-11 shows a chronology plot of the systematic, historical, and paleoflood data. Figure 4-12 shows the fit obtained by applying the Bayesian analysis method with the inclusion of historical and paleoflood data, assuming non-informative, flat priors for the LP III distribution parameters.

In order to assess the “value” of including additional hydrologic information, such as historical and paleoflood data, into the flood frequency analysis, *Cohn, Lane, & Baier* (1997) propose an equation to compute the *effective record length* (ERL). For the purposes of this study, the equation was augmented to achieve greater accuracy. ERL is computed at multiple quantiles and then averaged as shown in Equation 1.

$$ERL = \frac{1}{N_q} \sum_{i=1}^{N_q} N_s \frac{Var [\hat{X}_i | N_s, \theta_T]}{Var [\hat{X}_i | N_T, \theta_T]} \quad \text{Equation 1}$$

where ERL is the effective record length in years; N_q is the number of exceedance probabilities for which the variance is computed; $\hat{X}_i$ is the flood magnitude for a given exceedance probability; $Var [\cdot]$ is the variance in flow magnitude for a given exceedance probability and given flood record; N_s is the systematic record length; N_T is the total record length (systematic plus the historical and paleoflood events, etc.); and θ_T are the LP III parameters given the total record length.

Using this approach, the ERL was estimated to be 307 years. In other words, the value of the historical and paleoflood data was equivalent to having observed an additional 200 years of events. As can be seen in Figure 4-12, the uncertainty in the AEP of the SDF has been reduced considerably with the inclusion of temporal information. There is less than two orders of magnitude between the expected and most likely curves.

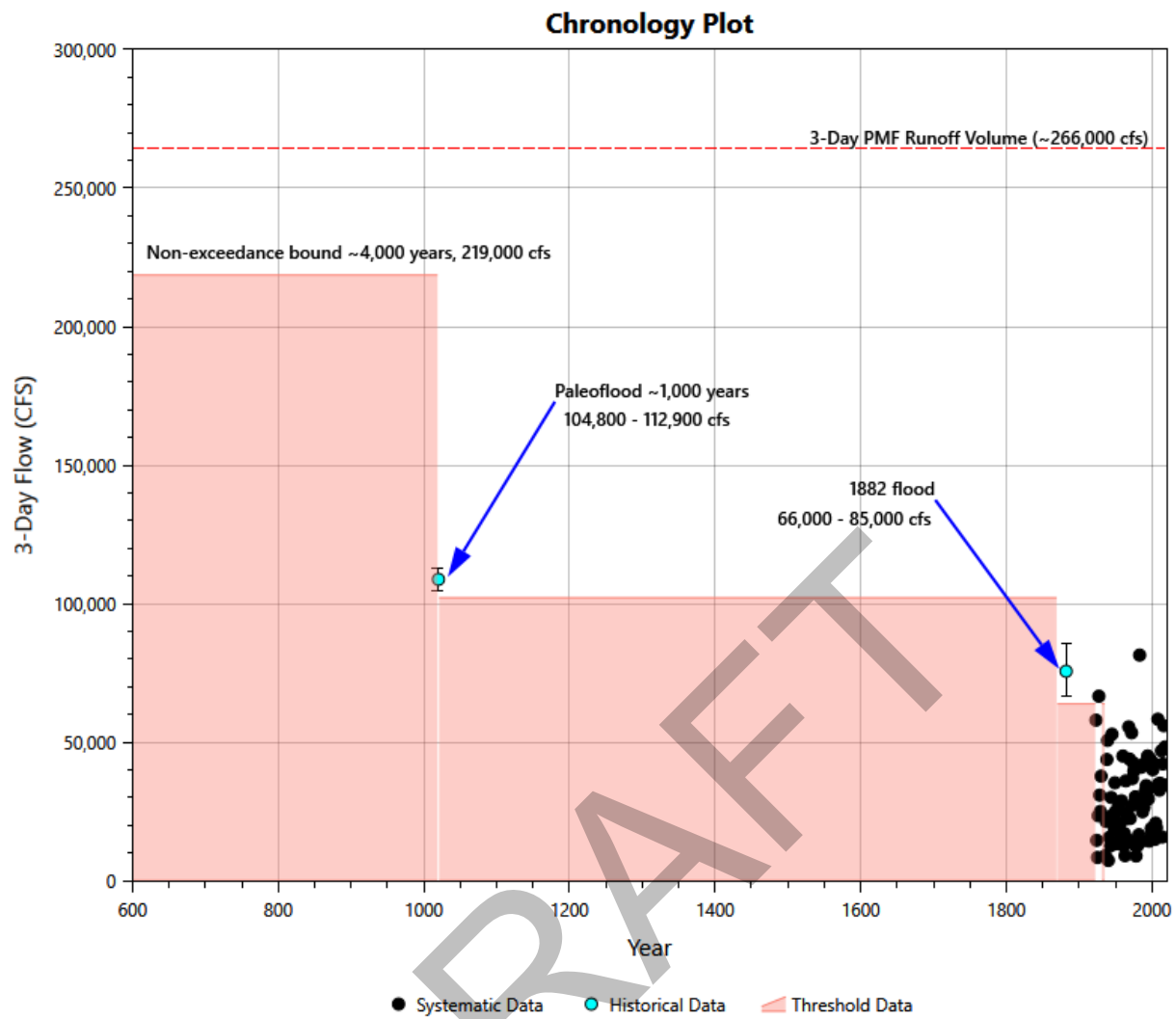


Figure 4-11. Chronology plot showing the systematic, historical, and paleoflood data. Perception thresholds are shown as shaded boxes.

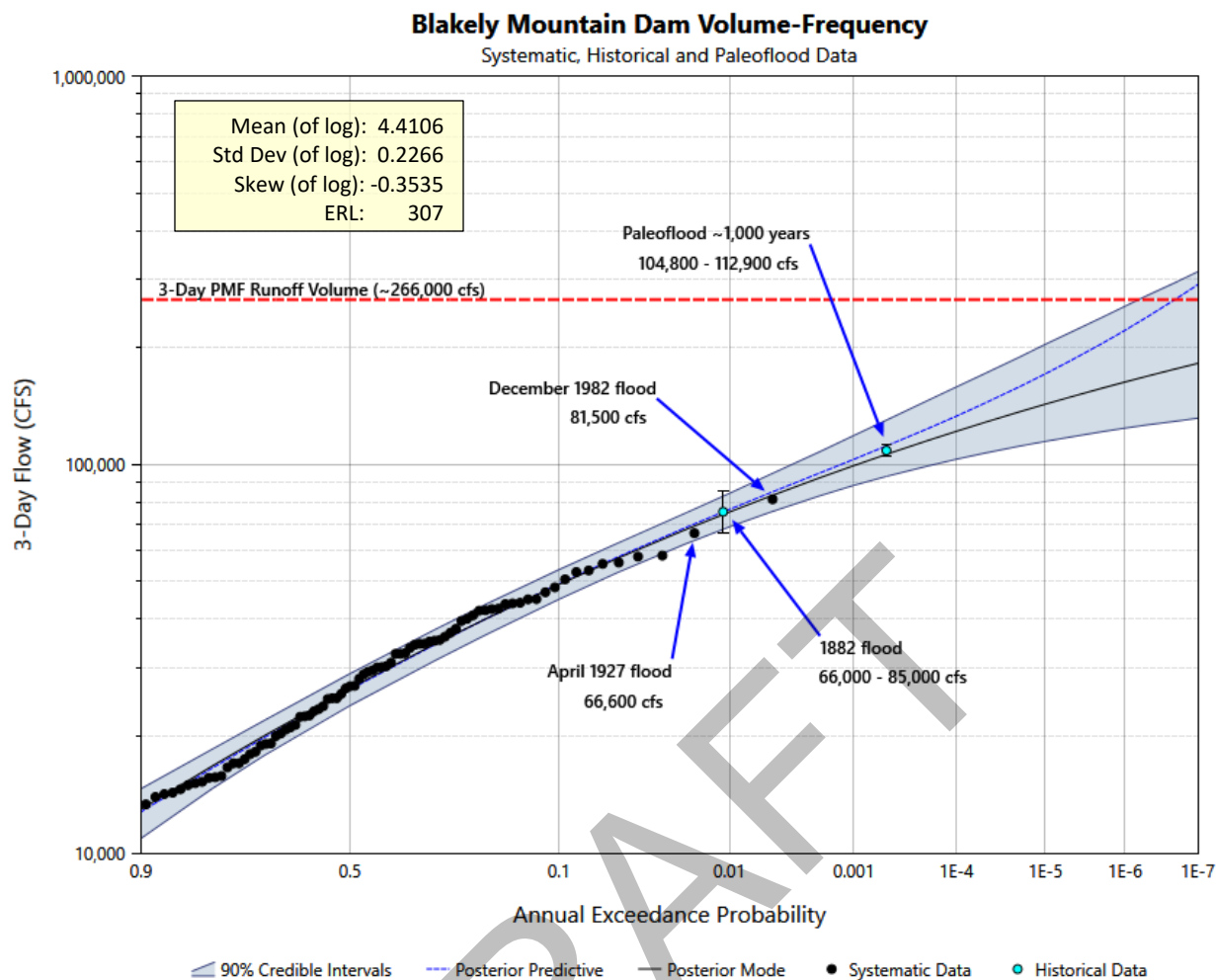


Figure 4-12. Bayesian results for the unregulated annual maximum 3-day inflow volume-frequency curve and 90 percent credible intervals based on systematic, historical and paleoflood data.

4.3.3 Regional Skew Data

As discussed in section 4.2.5, *3-Day Flow Frequency with Regional Skew*, the USGS study estimated regional skew with an expected value of -0.17, mean-square error (MSE) of 0.12, and ERL of 59 years (Wagner, Krieger, & Veilleux, 2016). This regional skew information was incorporated into the Bayesian analysis using a Normally distributed prior for skew with a mean of -0.17 and a variance of 0.12⁶. Figure 4-13 shows the fit obtained by applying the Bayesian analysis method with the inclusion of systematic, historical, paleoflood data, and regional skew using an informative, Normally distributed prior for skew. The ERL is 322 years, which is slightly longer than the ERL using only systematic, historical, and paleoflood data. The most likely curve is slightly shifted to left, closer to the expected curve, indicating a reduction in knowledge uncertainty. However, overall the inclusion of regional skew has a minimal effect.

This is expected behavior based on large sample Bayesian inference. As discussed in *Bayesian Data Analysis* (Gelman, et al., 2014), the data likelihood dominates the prior distribution with large at-site sample sizes. The asymptotic results of large sample theory formalize the notion that the importance of the prior distribution diminishes as the sample size increases. However, when sample sizes are small, the prior distribution is a critical part of the model specification (Gelman, et al., 2014). This concept is illustrated in Figure 4-14 and Figure 4-15.

Figure 4-14 shows a comparison of the marginal distributions for skew when evaluating the systematic data only. The black curve is the distribution when using a non-informative, flat prior. The blue curve is the distribution when using a Normal distribution prior for skew with a mean of -0.17 and variance of 0.12. As shown, with a relatively limited sample,

⁶ This approach for incorporating regional skew in a Bayesian analysis consistent with practice (Kuczera G. , 1983), (Kuczera G. , 1999), and (Reis & Stedinger, 2005).

the regional prior for skew has a substantial effect, pulling the posterior distribution of skew towards the regional estimate of -0.17. However, as can be seen in Figure 4-15, when evaluating systematic, historical, and paleoflood data, the regional skew has minimal effect on the posterior, marginal distribution of skew. This behavior confirms the predictions of large sample theory.

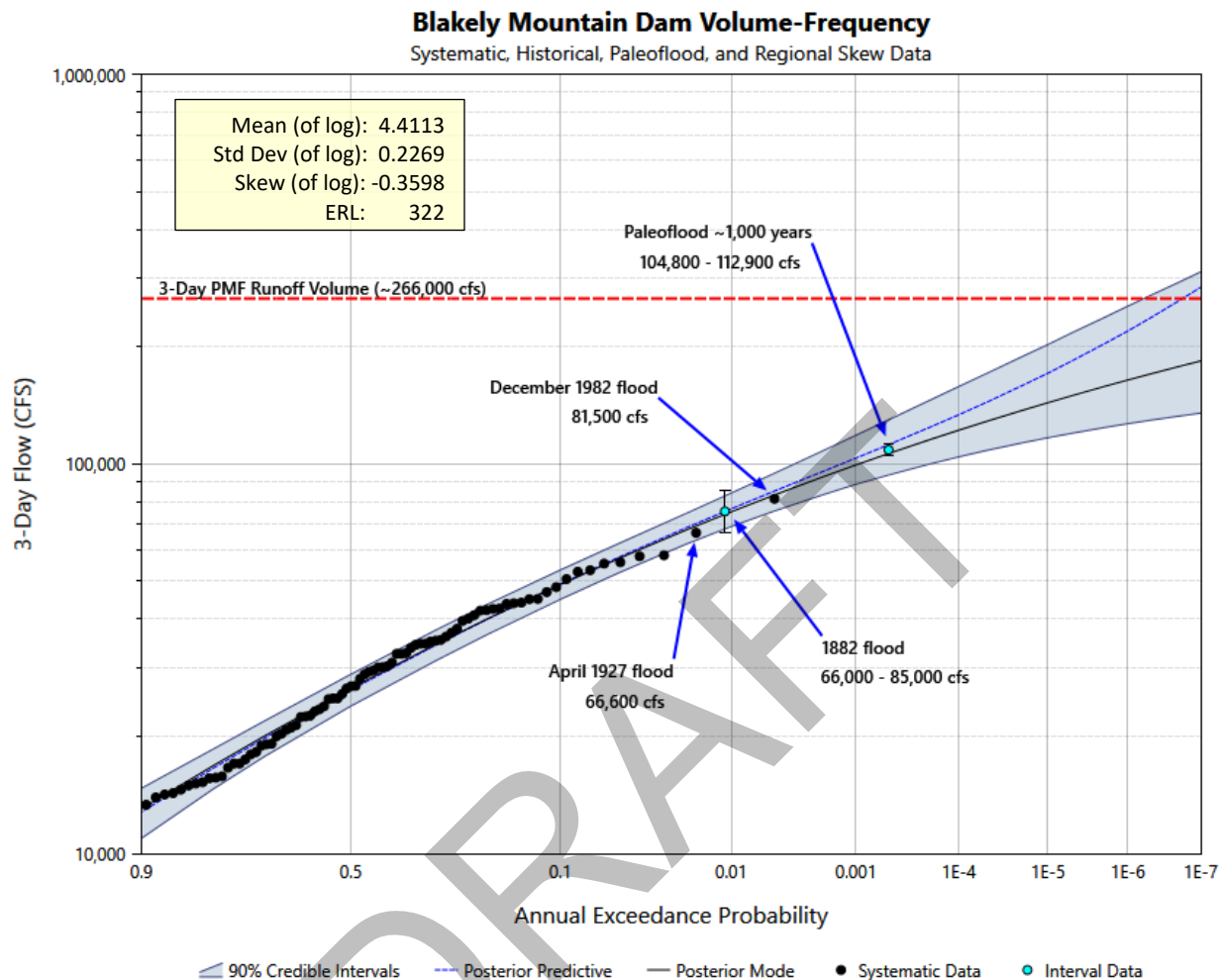


Figure 4-13. Bayesian results for the unregulated annual maximum 3-day inflow volume-frequency curve and 90 percent credible intervals based on systematic, historical, paleoflood, and regional skew data.

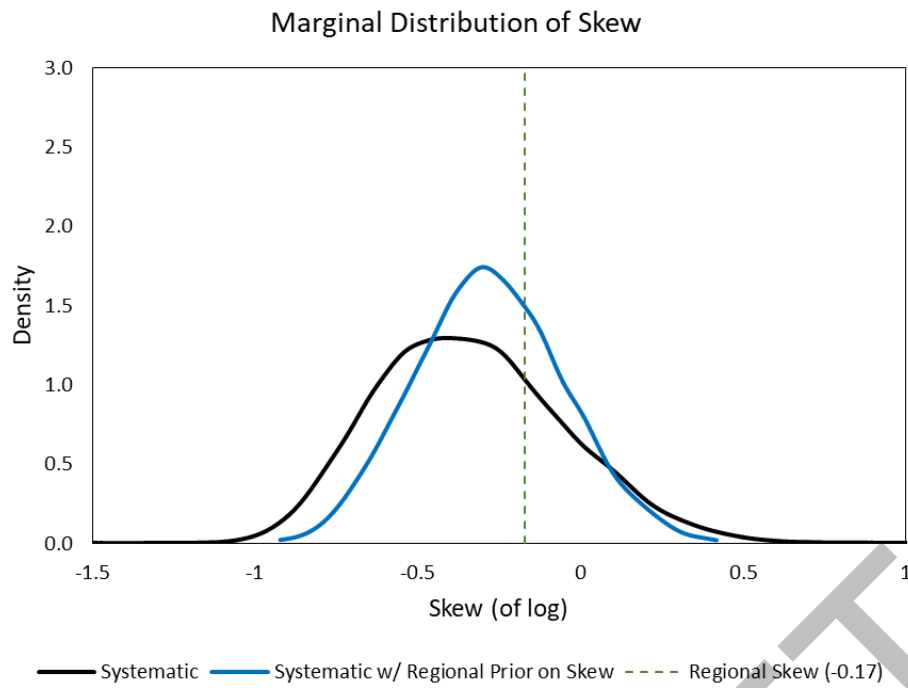


Figure 4-14. A comparison of the marginal distributions of skew using systematic data only. The black curve is the distribution when using a non-informative, flat prior. The blue curve is the distribution when using a Normally distributed prior for skew with a mean of -0.17 and variance of 0.12.

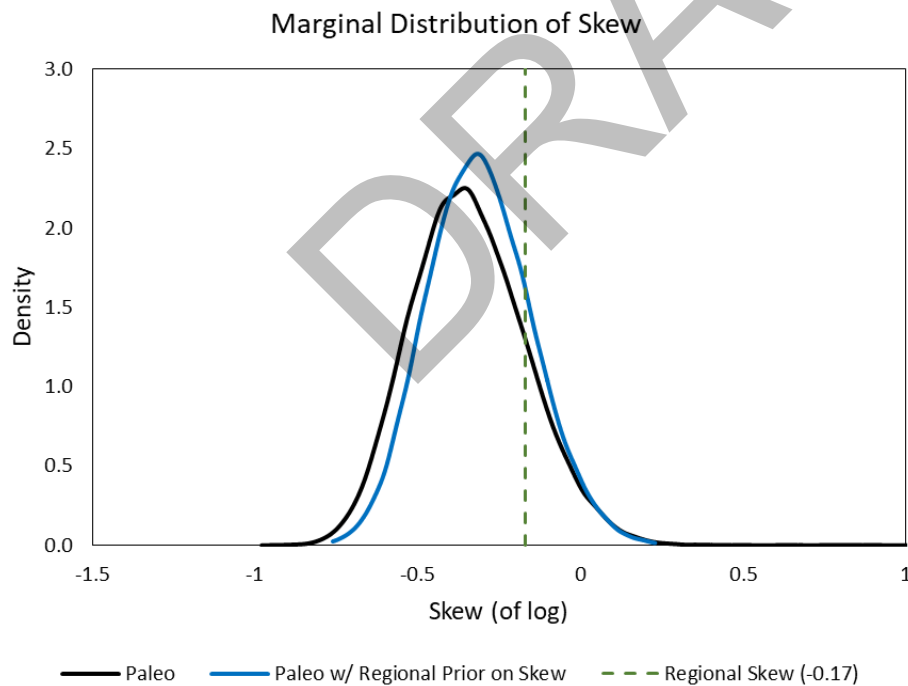


Figure 4-15. A comparison of the marginal distributions of skew using systematic, historical and paleoflood data. The black curve is the distribution when using a non-informative, flat prior. The blue curve is the distribution when using a Normally distributed prior for skew with a mean of -0.17 and variance of 0.12.

4.3.4 Regional Precipitation-Frequency and Rainfall-Runoff Data

Spatial information expansion is based on using flood information from neighboring catchments to improve flood frequency estimates at the site of interest. As discussed in the previous section, spatial information in the form of a regional skew estimate was incorporated for the Blakely Mountain Dam frequency analysis. However, a much greater amount of spatial data was obtained by utilizing available regional precipitation-frequency information.

Causal information expansion analyses the generating mechanisms of floods in the catchment of interest (Merz & Blöschl, 2008). Causal information was developed by modeling rainfall-frequency events using a calibrated hydrologic model. The main benefit of modeling regional rainfall-frequency information is that available rainfall records are often much longer than the flow records.

There were two sources of regional precipitation-frequency information available for the Blakely Mountain Dam watershed: 1) NOAA Atlas 14, and 2) data developed for the Trinity River Basin in Texas, which is described in “Regional Extreme Precipitation-Frequency Analysis for the Trinity River Basin” (MetStat, Inc., 2018). The regional frequency analysis performed for the Trinity River Basin included the geographical region where the Blakely Mountain watershed is located (see location map in Figure 4-16). The Trinity River Basin study incorporated advanced techniques, such as storm typing; therefore, the results are considered to provide a better extrapolation to rare exceedance probabilities than NOAA Atlas 14.

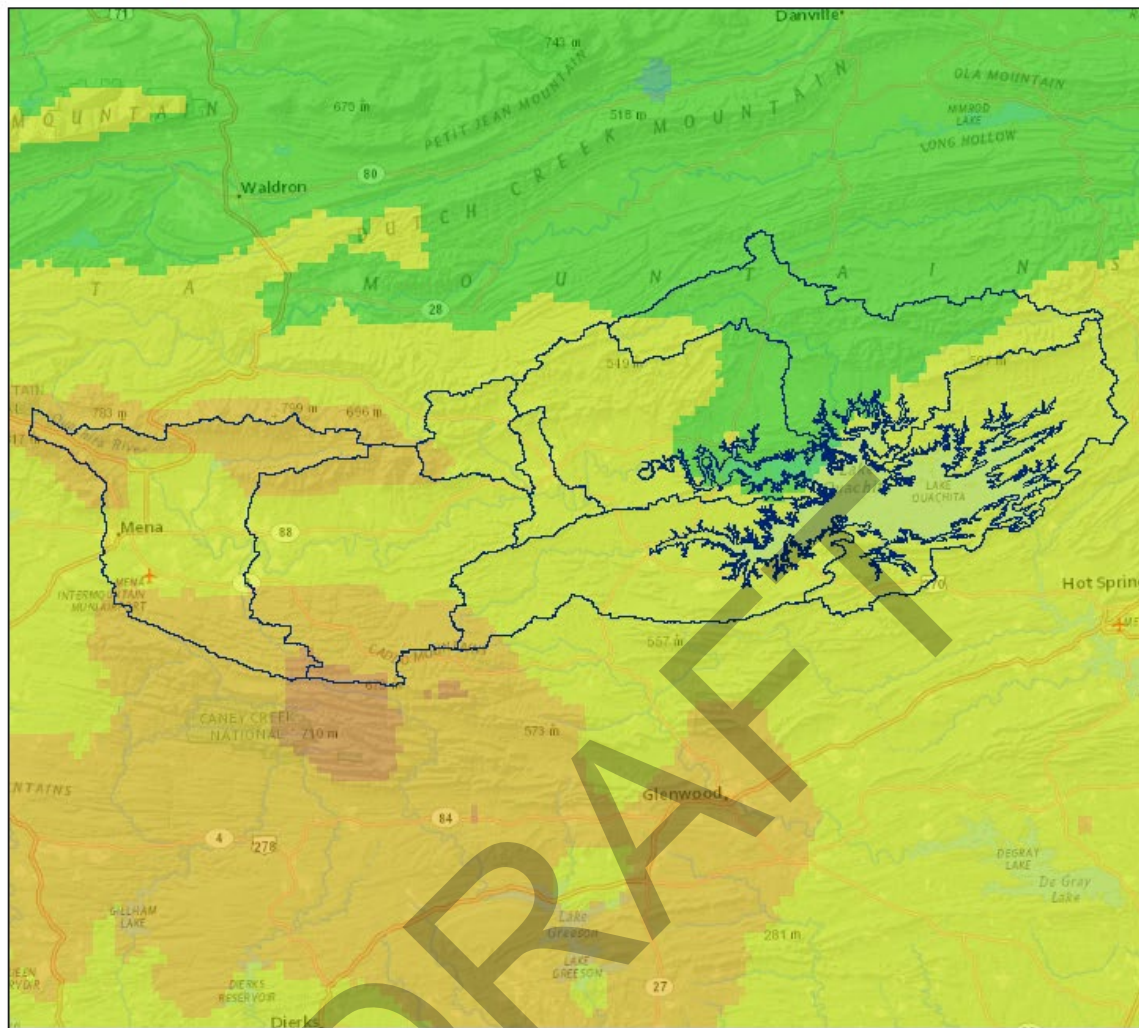
The regional precipitation-frequency data was collected for the watershed. 3-day point rainfall-frequency estimates were converted to basin-average rainfall using an areal reduction factor. Historical hyetographs were scaled to the 3-day basin-average rainfall-frequency events and routed using a calibrated HEC-HMS model. The modeled 3-day rainfall-runoff for each frequency event was recorded. Finally, a distribution of rainfall-runoff at the 1E-4 AEP was estimated and incorporated into the Bayesian analysis as described in (Coles & Tawn, 1996) and (Viglione, Merz, Salinas, & Blöschl, 2013). The following sub-sections detail each step in the process.

4.3.4.1 Data Collection and Analysis

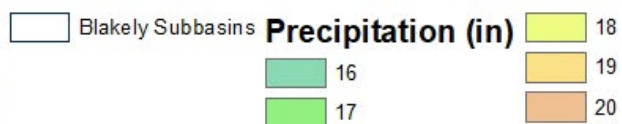
For each of the desired frequency events, 72-hour gridded precipitation data from NOAA Atlas 14 and the Trinity River Basin study were imported into HEC-MetVue software, with the Blakely Mountain HEC-HMS model watershed shapefile, to enable computation of average precipitation for each subbasin. For example, Figure 4-16 depicts the Blakely Mountain Dam watershed subbasins and the Trinity Basin study 1,000-year, 72-hour gridded precipitation-frequency data. The spatial extents of the regional Trinity Basin precipitation-frequency study included the Blakely Mountain Dam watershed. For comparison, Figure 4-17 displays the same spatial extents with the 1,000-year, 72-hour NOAA Atlas 14 gridded precipitation-frequency.

While the Trinity Basin regional precipitation-frequency study extents include the Blakely Mountain Dam watershed and thus the data were directly applied for this analysis, a visual comparison was conducted with the NOAA Atlas 14 gridded precipitation-frequency data. The plotted comparison of precipitation-frequency values for the NOAA Atlas 14 and Trinity Basin study are displayed in Figure 4-18. Precipitation-frequency values for the compared probabilities (0.1 through 10⁻³ AEP) were within 1.0%. Expected precipitation-frequency values from the Trinity basin study were applied to the Blakely Mountain Dam frequency analysis.

Blakely Mountain Dam Precipitation-Frequency Map



Trinity Basin Study 1000-Year, 72-Hour

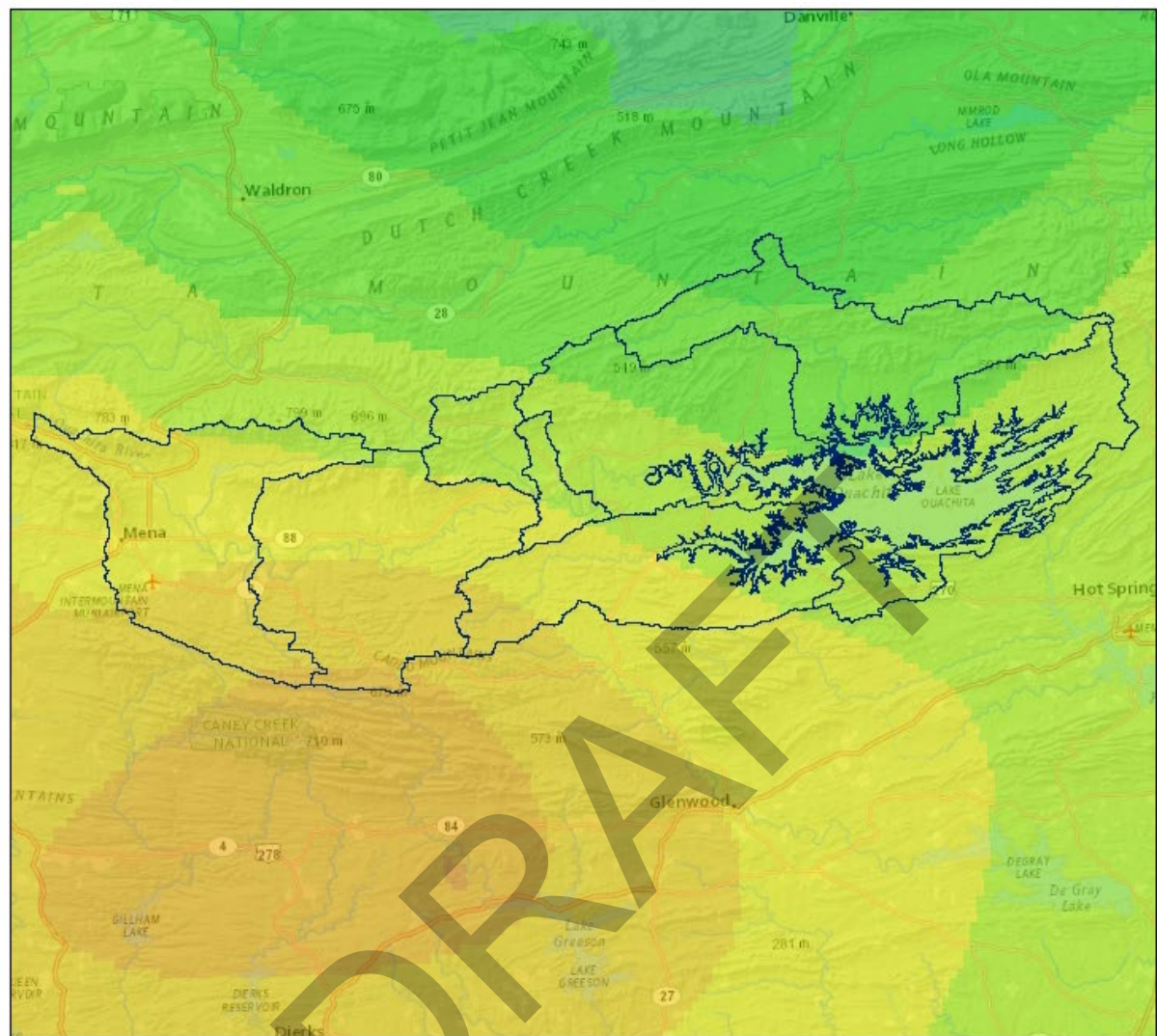


0 35 70 140 210 Miles

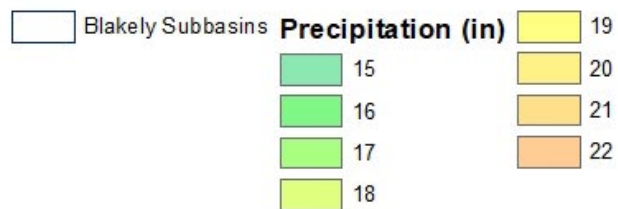


Figure 4-16. Trinity Basin study 1,000-year, 72-hour gridded precipitation-frequency map.

Blakely Mountain Dam Precipitation-Frequency Map



NOAA Atlas 14 1000-Year, 72-Hour



0 35 70 140 210 Miles

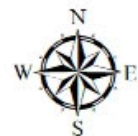


Figure 4-17. NOAA Atlas 14 1,000-year, 72-hour gridded precipitation-frequency map.

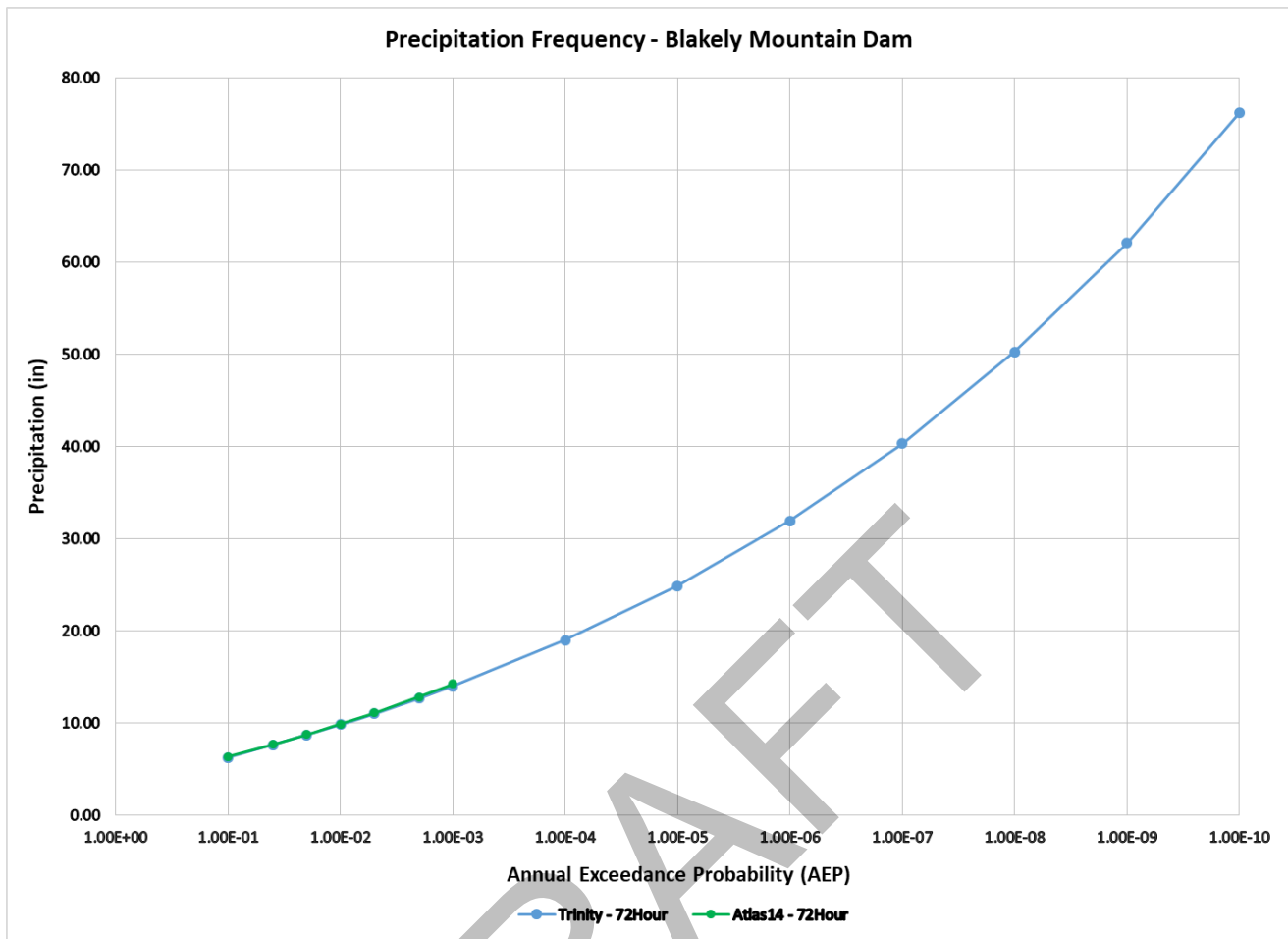


Figure 4-18. Blakely Mountain Dam precipitation-frequency.

As the Trinity River Basin study gridded precipitation data was based on point precipitation, an areal reduction factor (ARF) was applied from HMR52 (Figure 4-19). A critical storm duration of 72-hours was used with the watershed area of 1,102 square miles to compute an ARF of 0.786. An ARF of 0.8 was used to reduce basin average precipitation for the Trinity Basin study data. Reduced basin average precipitation depths for the Trinity Basin study are displayed in Table 4-5. The 10,000-year ($1E-4$ AEP), 72-hour precipitation-frequency from the Trinity Basin study was selected for use in the frequency analysis for Blakely Mountain Dam (Figure 4-20). The rationale for selecting this frequency event is discussed in section 4.3.4.5.

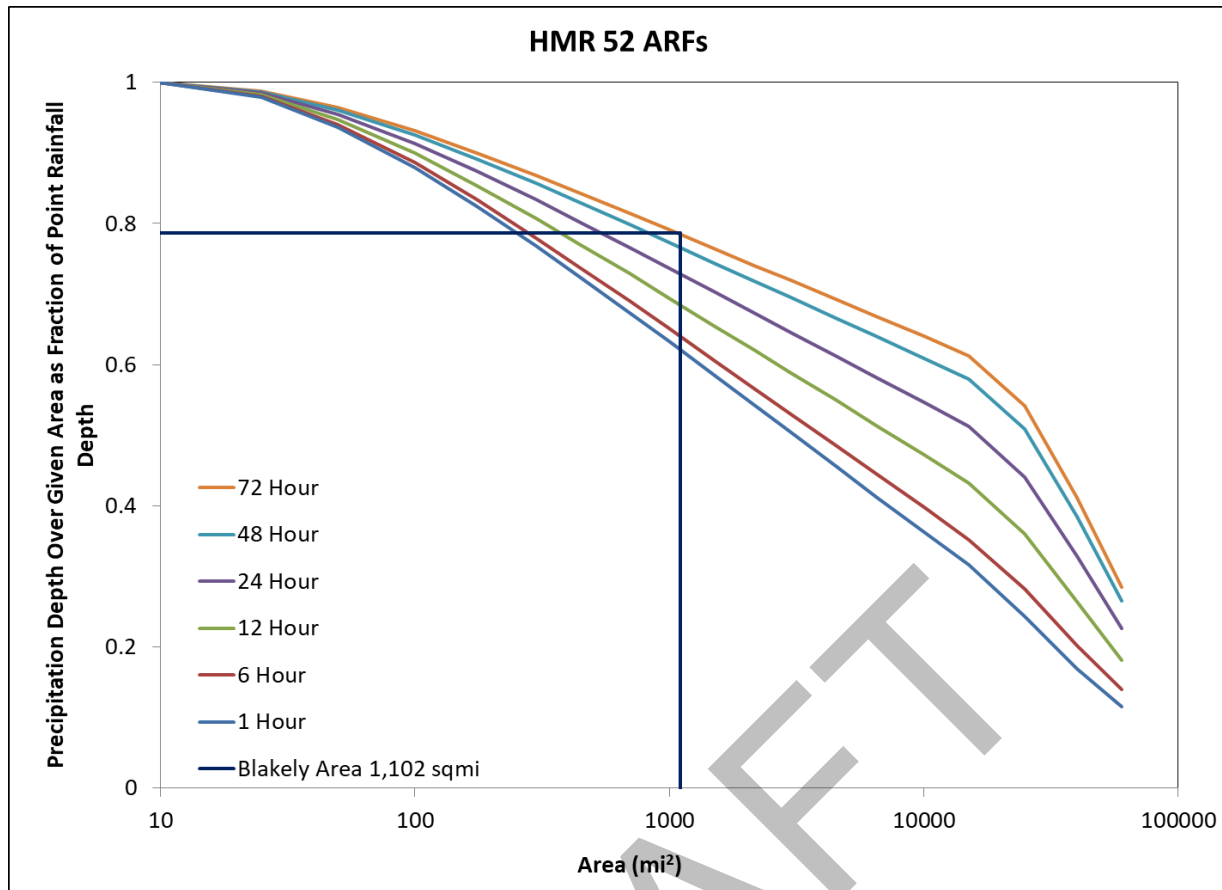
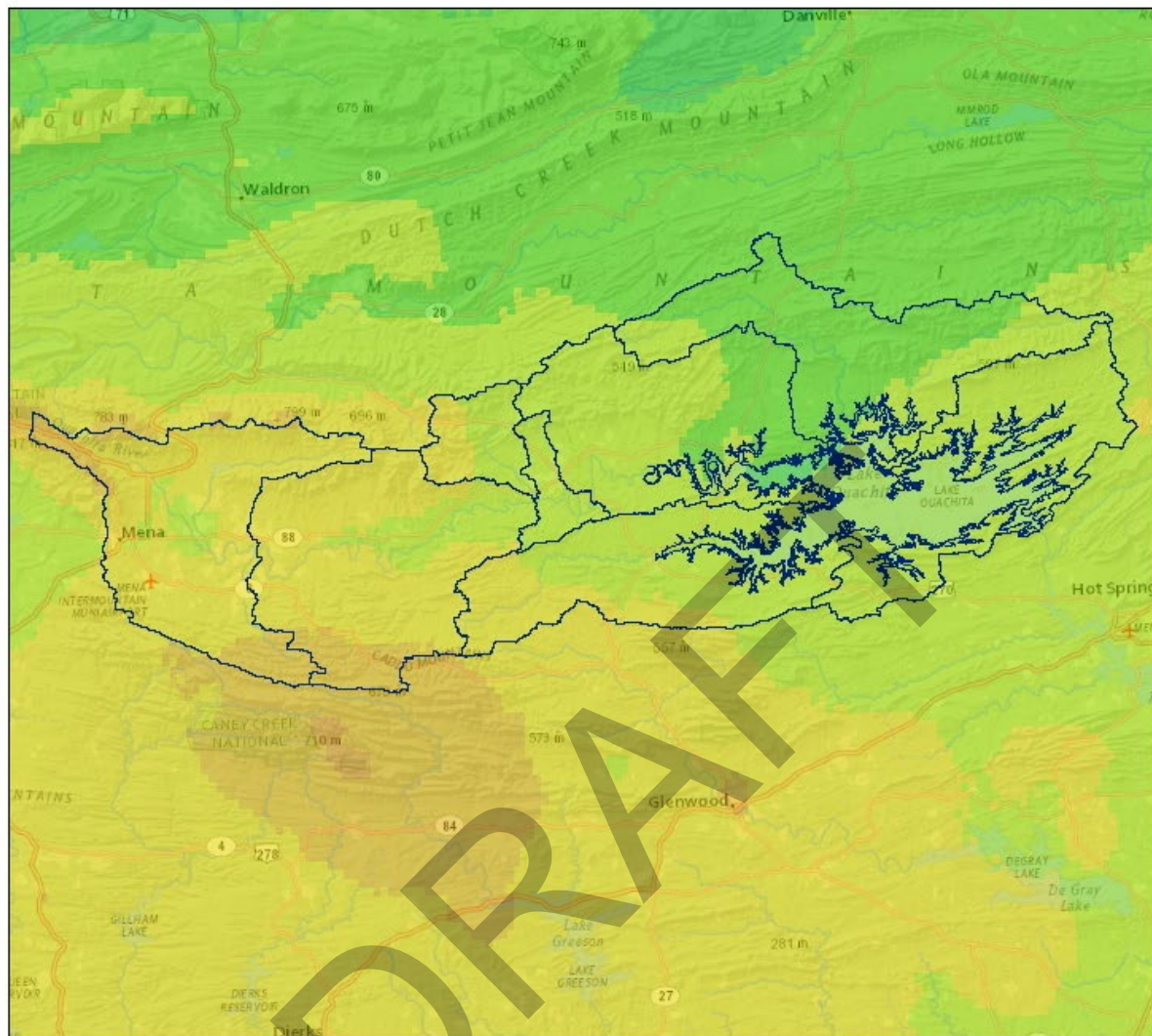


Figure 4-19. HMR52 areal reduction factors.

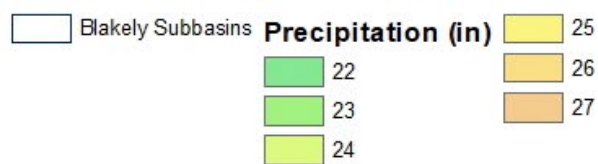
Table 4-5. Basin Average Precipitation Depths for the Trinity Basin

	Trinity Basin Maximum 72-Hour Precipitation-Frequency												
	GIS Basin Name	Blakely Creek	Crystal Mountain	Gaston Mountain	Iron Fork Mountain	Middle Fork	MT.IDA (GAGE)	North Fork	Oden	Ouachita Lake	South Fork	Basin Average Precip	Reduced Basin Averages
	Precipitation Unit	(in)	(in)	(in)	(in)	(in)	(in)	(in)	(in)	(in)	(in)	(in)	(in)
AEP	Return Year	PER-CUM	PER-CUM	PER-CUM	PER-CUM	PER-CUM	PER-CUM	PER-CUM	PER-CUM	PER-CUM	PER-CUM		
1E-10	10-10 Yr	94.48	95.96	97.65	97.20	93.30	95.33	91.71	95.22	94.82	96.21	95.20	76.16
1E-09	10-9 Yr	77.01	78.10	79.61	79.38	76.01	77.63	74.81	77.73	77.15	78.33	77.60	62.08
1E-08	10-8 Yr	62.32	63.12	64.45	64.36	61.48	62.77	60.58	63.01	62.33	63.32	62.80	50.24
1E-07	10-7 Yr	49.98	50.55	51.70	51.71	49.28	50.30	48.62	50.61	49.90	50.73	50.37	40.30
1E-06	10-6 Yr	39.61	40.01	40.98	41.05	39.03	39.83	38.55	40.16	39.48	40.16	39.92	31.93
1E-05	10-5 Yr	30.89	31.16	31.97	32.07	30.42	31.04	30.08	31.37	30.74	31.29	31.13	24.90
1E-04	10-4 Yr	23.56	23.74	24.39	24.51	23.19	23.65	22.95	23.96	23.41	23.84	23.75	19.00
1E-03	10-3 Yr	17.40	17.51	18.02	18.13	17.12	17.46	16.95	17.72	17.26	17.59	17.54	14.03
0.002	500 Yr	15.74	15.84	16.31	16.41	15.49	15.79	15.34	16.04	15.61	15.92	15.87	12.69
0.005	200 Yr	13.68	13.76	14.18	14.27	13.45	13.72	13.33	13.95	13.56	13.83	13.79	11.03
0.01	100 Yr	12.21	12.28	12.66	12.75	12.01	12.25	11.90	12.45	12.10	12.34	12.31	9.85
0.02	50 Yr	10.81	10.87	11.21	11.29	10.63	10.84	10.53	11.03	10.70	10.92	10.90	8.72
0.04	25 Yr	9.48	9.52	9.82	9.90	9.31	9.50	9.23	9.67	9.38	9.57	9.55	7.64
0.1	10 Yr	7.79	7.82	8.07	8.14	7.64	7.80	7.58	7.95	7.70	7.86	7.85	6.28

Blakely Mountain Dam Precipitation-Frequency Map



Trinity Basin Study 10,000-Year, 72-Hour



0 35 70 140 210 Miles



Figure 4-20. Trinity Basin Study 10,000-year, 72-hour gridded precipitation-frequency selected for use in the Blakely Mountain Dam frequency analysis.

4.3.4.2 Scaled Frequency Hyetograph Development

Three of the largest historic storms (December 1982, May/June 2013, and December 2015) observed in the Blakely Mountain Dam watershed were selected to provide representative temporal and spatial precipitation patterns for synthetic frequency hyetographs. For each observed storm event, hourly rainfall hyetographs were provided by the USACE Vicksburg District for each subbasin within the watershed. Basin average precipitation was computed by taking the sum of the average rainfall weighted by subbasin area (sum product) for each hour of the storm. The observed maximum 72-hour basin average was then computed. The 72-hour maximum cumulative rainfall in inches for 1982, 2013, and 2015 were 11.74, 9.45, and 7.05 inches, respectively. Figure 4-21 to Figure 4-23 display the basin average historical storm hyetographs and cumulative rainfall for the three storms.

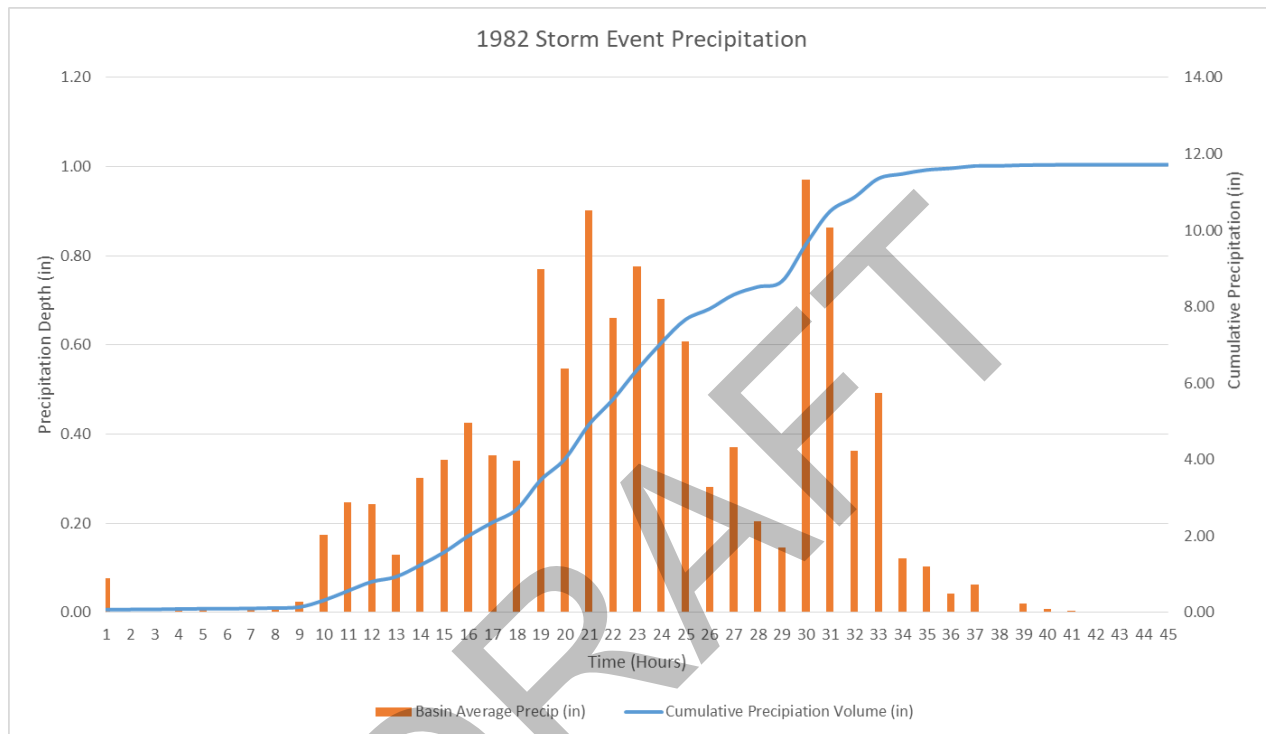


Figure 4-21. 1982 historic storm hyetograph and cumulative precipitation (record flood).

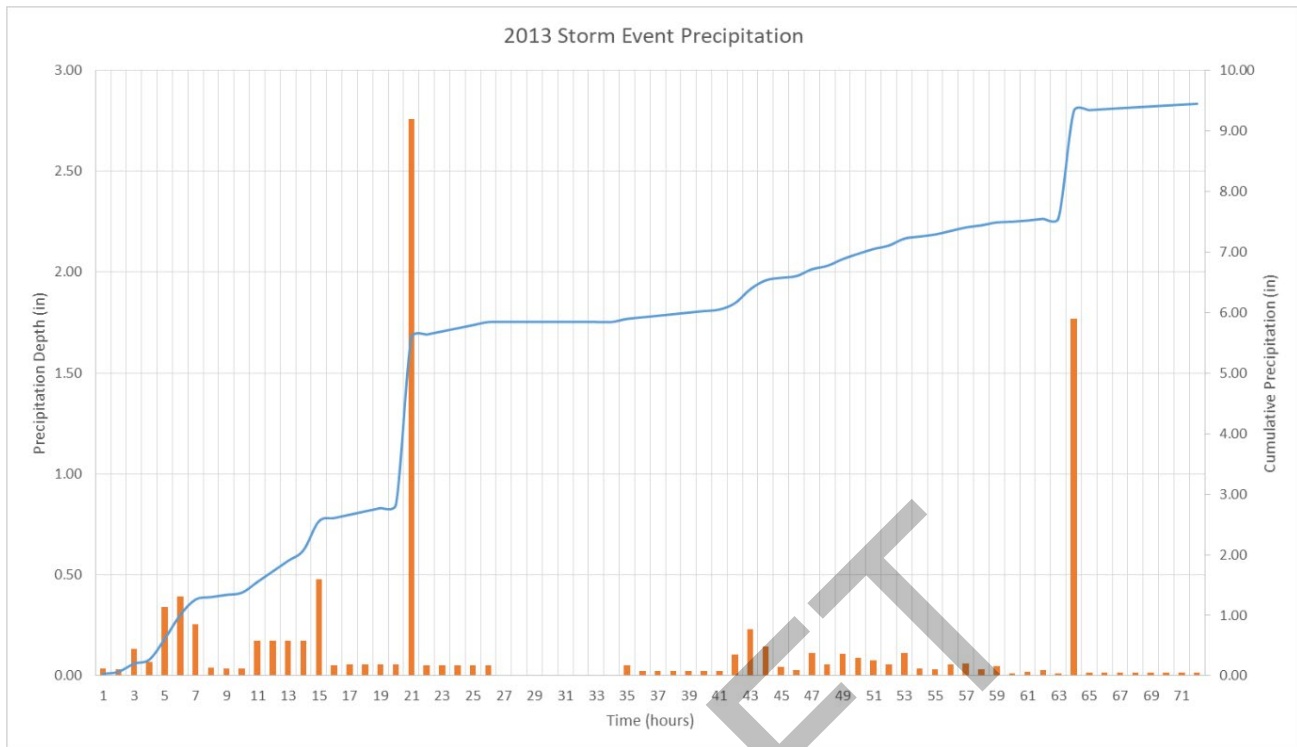


Figure 4-22. 2013 historic storm hyetograph and cumulative precipitation.

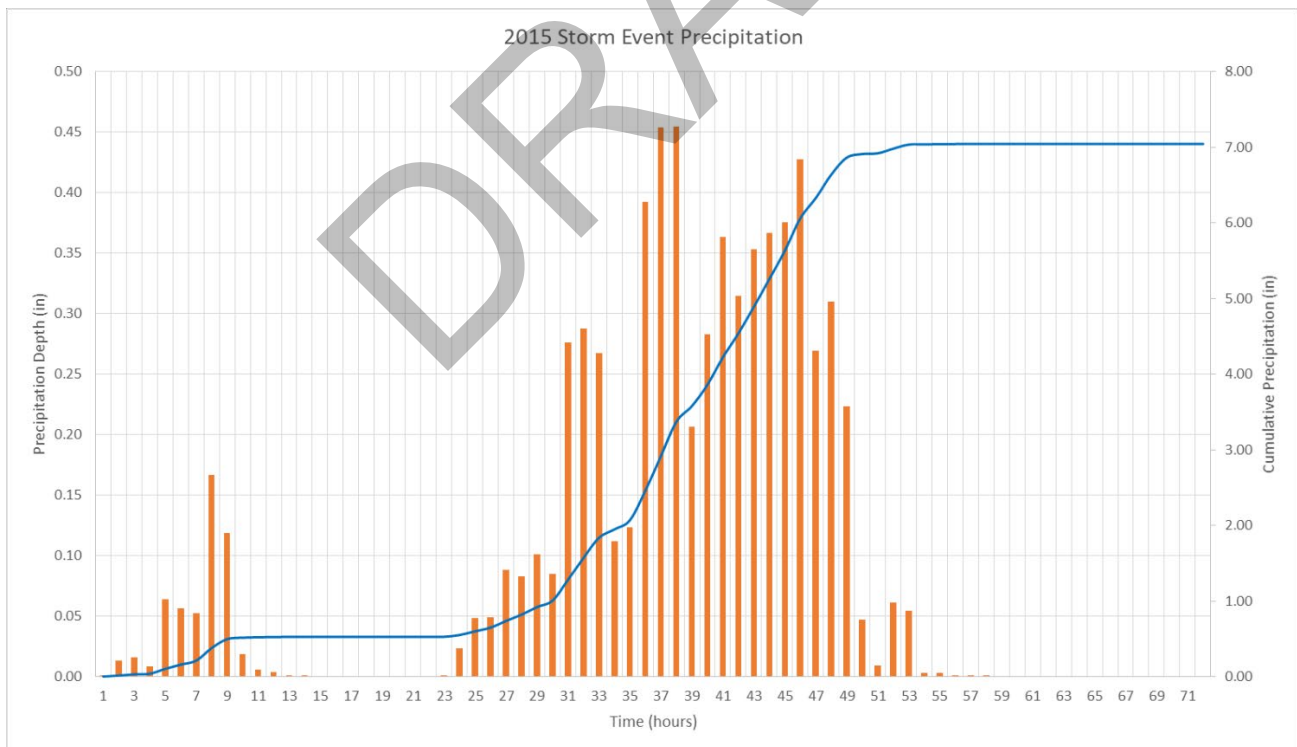


Figure 4-23. 2015 historic storm hyetograph and cumulative precipitation.

For each observed storm, a set of normalized subbasin hyetographs (historic storm templates) were then developed by taking each hourly subbasin rainfall ordinate and dividing it by the total basin average for that hour. These historic storm templates were used to create synthetic storm hyetographs for the basin.

A range of precipitation frequency events from the Trinity Basin study were selected to inform the development of synthetic hyetographs, including 0.1, through 1E-8 AEP. These events were selected to provide a wide range of precipitation-frequency input to the Bayesian volume-frequency analysis. Table 4-6 displays the Trinity Basin study 72-hour basin average precipitation depths in inches for each frequency event.

A scale factor was developed for each frequency event by dividing the Trinity Basin study maximum 72-hour basin average precipitation depths shown in Table 4-6 by the observed storm 72-hour maximum basin depths (1982: 11.74-inches, 2013: 9.45-inches, and 2015: 7.05-inches).

Table 4-6. Trinity Basin study precipitation frequency depths.

AEP	Frequency Event (Return Years)	Trinity 72-hour Basin Avg. Depth (in)	1982 Scale Factor (11.74 in)	2013 Scale Factor (9.45 in)	2015 Scale Factor (7.05 in)
0.1	10	6.28	0.53	0.66	0.89
0.05	20	7.33	0.62	0.78	1.04
0.02	50	8.72	0.74	0.92	1.24
0.01	100	9.85	0.84	1.04	1.40
5E-03	200	11.03	0.94	1.17	1.57
2E-03	500	12.69	1.08	1.34	1.80
1E-03	1,000	14.03	1.20	1.49	1.99
5E-04	2,000	15.61	1.33	1.65	2.22
2E-04	5,000	17.58	1.50	1.86	2.50
1E-04	10,000	19.00	1.62	2.01	2.70
5E-05	20,000	20.85	1.78	2.21	2.96
2E-05	50,000	23.20	1.98	2.46	3.29
1E-05	100,000	24.90	2.12	2.64	3.53
5E-06	200,000	27.10	2.31	2.87	3.85
2E-06	500,000	29.89	2.55	3.16	4.24
1E-06	1,000,000	31.93	2.72	3.38	4.53
5E-07	2,000,000	34.53	2.94	3.66	4.90
2E-07	5,000,000	37.85	3.22	4.01	5.37
1E-07	10,000,000	40.30	3.43	4.27	5.72
5E-08	20,000,000	43.37	3.69	4.59	6.16
2E-08	50,000,000	47.32	4.03	5.01	6.72
1E-08	100,000,000	50.24	4.28	5.32	7.13

To develop scaled frequency hyetographs for each subbasin, first the observed Blakely Mountain basin average rainfall was multiplied by a scale factor to develop hourly synthetic basin average precipitation. Then, individual subbasin scaled frequency hyetographs were created by multiplying each hourly ordinate of the normalized storm templates by the scaled basin average precipitation. This procedure was conducted for each AEP listed in Table 4-7. These scaled frequency hyetographs served as the forcing input to a hydrologic routing model to create frequency inflow volumes to Blakely Mountain Dam.

4.3.4.3 Hydrologic Routing Model Formulation

The rainfall-runoff software HEC-HMS v. 4.3 was used to develop the hydrologic routing model for the Blakely Mountain Dam PMF study. The calibrated and validated HEC-HMS model from the Blakely Mountain IES PMF study was used to determine excess rainfall-volume and baseflow for each subbasin within the Blakely Mountain watershed for discrete runoff events (U.S. Army Corps of Engineers, 2018).

Modeling methods included the initial and constant loss method, Clark Unit Hydrograph method, exponential recession baseflow method, and the Muskingum-Cunge reach routing method. Parameters and model geometry include the most current data available, including the delineation of subbasins, reservoir storage curves, and dam discharge rating curves. Figure 4-24 depicts the HEC-HMS modeling process flow chart. The model geometry for the Blakely Mountain Dam HEC-HMS model is displayed in Figure 4-25.

The validated model parameters are summarized in Table 4-7 and Table 4-8, and include the two constant loss assumptions used to perform the precipitation-frequency analysis, as well as the Muskingum-Cunge routing parameters. The constant loss values for the HEC-HMS validation model (low constant loss value of 0.03 in/hr and high constant loss value of 0.07 in/hr) were selected from the low and high constant loss calibration parameters respectively. For a comprehensive description of the model development such as watershed delineation and initial model parameterization, refer to the report RMC-TR-2018-XX, "Probable Maximum Flood Analysis for Blakely Mountain Dam" (U.S. Army Corps of Engineers, 2019).

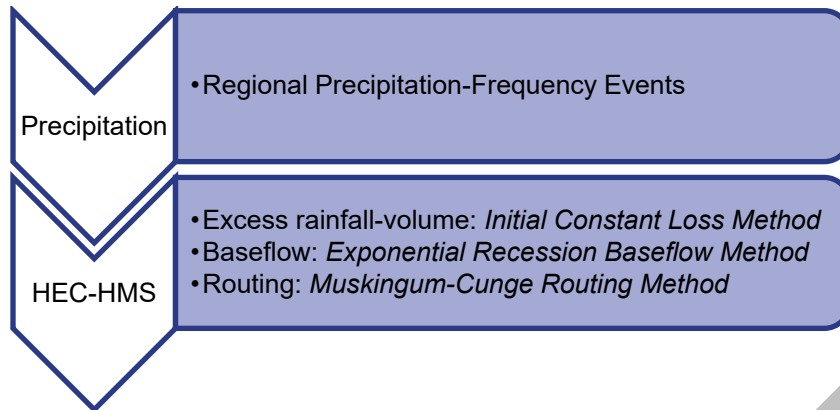


Figure 4-24. HEC-HMS modeling flow chart.

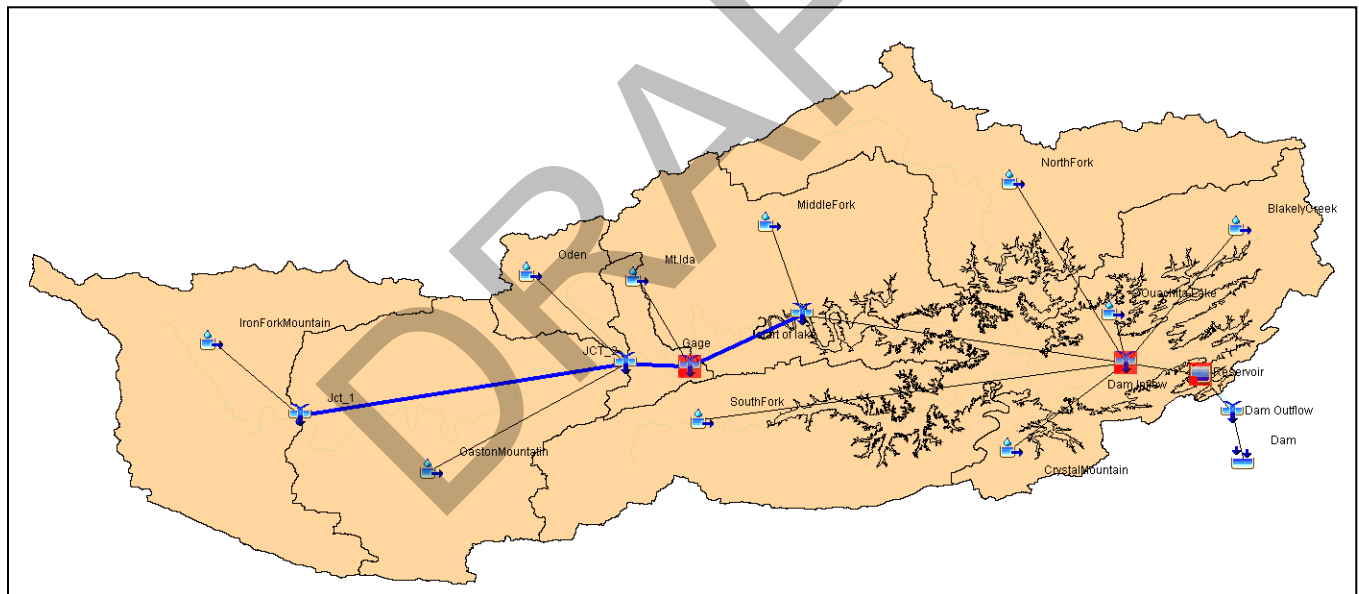


Figure 4-25. Blakely Mountain Dam HEC-HMS watershed model.

Table 4-7. HEC-HMS basin model parameters.

Validation Parameters							
Subbasin ID	Initial and Constant-Rate Loss			Clark's Unit Hydrograph		Exponential Recession Baseflow	
	High Inflow Low Constant Loss Rate (in/hr)	Low Inflow High Constant Loss Rate (in/hr)	Impervious (%)	Tc (hr)	R (hr)	Recession Constant	Ratio to Peak
IronForkMountain	0.03	0.07	1.5	10.0	7.0	0.8	0.05
GastonMountain	0.03	0.07	0.0	18.0	12.0	0.8	0.05
Oden	0.03	0.07	0.0	3.0	2.0	0.8	0.05
Mt.Ida	0.03	0.15	0.0	7.5	5.0	0.8	0.05
MiddleFork	0.02	0.07	0.0	10.0	7.0	0.8	0.05
NorthFork	0.02	0.15	0.0	9.0	6.0	0.8	0.05
SouthFork	0.02	0.15	1.0	8.0	6.0	0.8	0.05
Ouachita Lake	-	-	-	118.0	79.0	-	-
BlakelyCreek	0.02	0.15	0.0	2.0	1.4	0.8	0.05
CrystalMountain	0.02	0.07	0.0	2.0	1.4	0.8	0.05

Table 4-8. HEC-HMS Muskingum-Cunge routing parameters.

Reach ID	Length (ft)	Slope (ft/ft)	Manning's n	Index Flow (cfs)	L.B. Manning's n	R.B. Manning's n	Invert (ft)
Ouachita_US_Oden	136,000	0.0009	0.03	6500	0.09	0.09	825
Ouachita_US_MtIda	44,000	0.0005	0.03	6500	0.09	0.09	677
MiddleFork_US_Lake	31,881	0.0001	0.03	6500	0.09	0.09	629

4.3.4.4 Hydrologic Routing Results

Hydrologic modeling conducted for the Blakely Mountain Dam precipitation-frequency analysis made use of an *AEP-neutral* approach (Nathan & Bowles, 1997) in flood simulations where typical values of hydrometeorological conditions were used for antecedent soil moisture, initial reservoir level, etc. Typical values of hydrometeorological conditions generally yield reasonable flood-frequency estimates in humid climates where “typical” may be interpreted as near-mean values.

Frequency inflow hydrographs were developed from discrete runoff events using three historic storm shape templates and two constant loss rate assumptions. This set of synthetic flood simulations produced an estimate of a given flood output for each AEP, such as inflow flood peak discharge and runoff volume, for each of the three historical storm templates. For example, Figure 4-26 depicts routed inflow hydrographs for all three observed storm templates for the 0.1 AEP frequency event. Hydrographs in the plot are offset for ease of visualization.

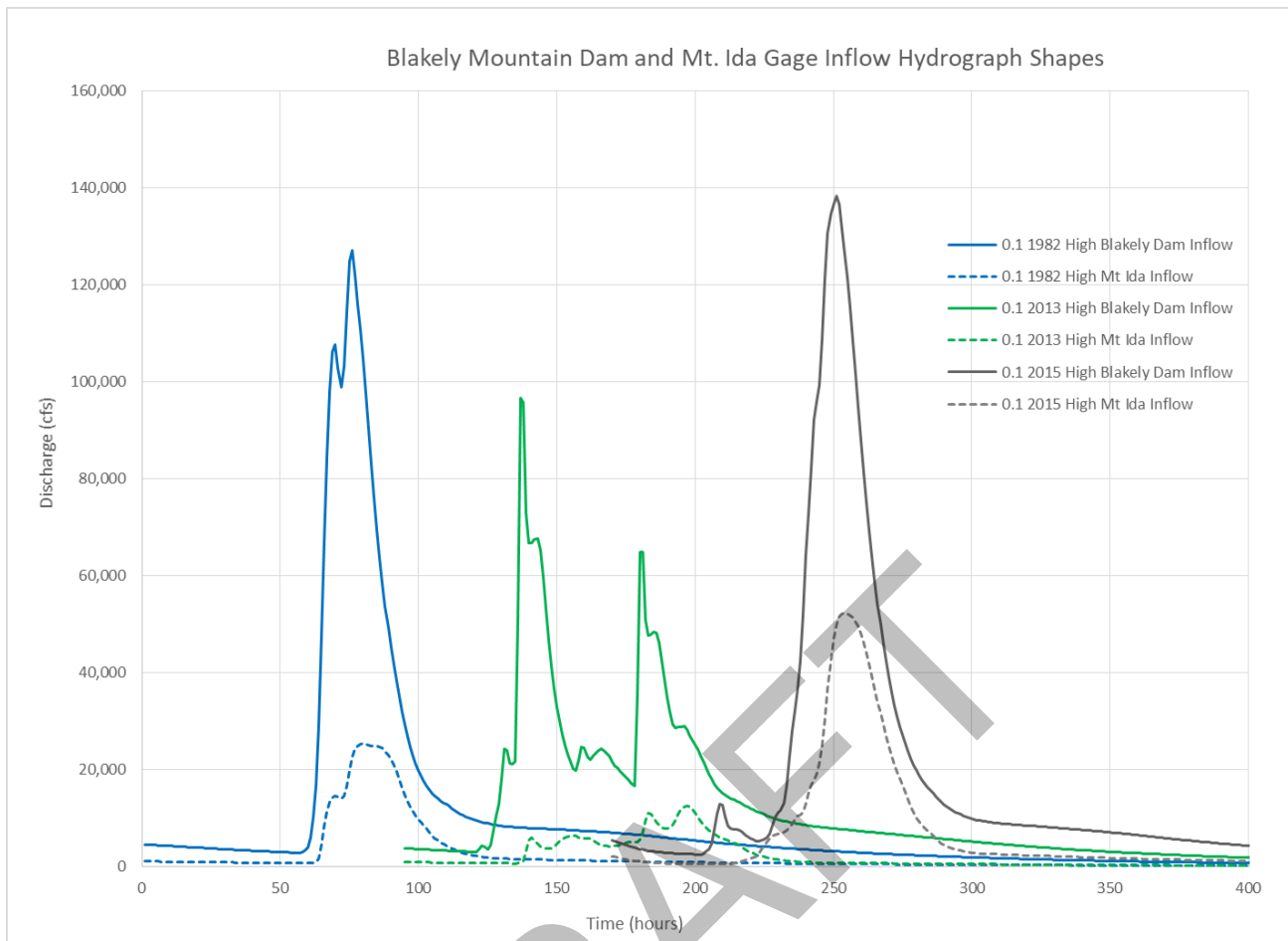


Figure 4-26. Blakely Mountain and Mt. Ida inflow hydrograph shapes.

3-day inflow volumes developed from each simulation set of inflow hydrographs were used to inform the Bayesian volume-frequency analysis. High and Low inflow estimates were developed using different constant loss assumptions (e.g., High: 0.03 in/hr and Low: 0.07 in/hr, where lower loss rates create higher runoff volume). 3-day inflow volumes at the Mt. Ida gage and Blakely Mountain Dam are listed in Table 4-9 and Table 4-10.

Table 4-9. Blakely Mountain 3-day high inflow estimates.

High Flow Estimates (Low Constant Loss Rate 0.03 in/hr)			
		Mt. Ida Gage	Blakely Mountain Dam
AEP	Storm Event Template	Peak 3-Day Inflow (cfs)	Peak 3-Day Inflow (cfs)
0.1	1982	10,790	42,240
0.1	2013	6,701	35,395
0.1	2015	20,968	51,206
0.02	1982	18,966	64,872
0.02	2013	13,660	55,186
0.02	2015	30,238	72,985
1E-03	1982	37,006	114,371
1E-03	2013	27,859	98,561
1E-03	2015	50,415	120,480
1E-04	1982	53,824	160,780
1E-04	2013	41,589	140,399
1E-04	2015	69,382	164,958
1E-05	1982	73,862	216,017
1E-05	2013	57,802	190,167
1E-05	2015	91,931	217,773

Table 4-10. Blakely Mountain 3-day low flow estimates.

Low Flow Estimates (High Constant Loss Rate 0.07 in/hr)			
		Mt. Ida Gage	Blakely Mountain Dam
AEP	Storm Event Template	Peak 3-Day Inflow (cfs)	Peak 3-Day Inflow (cfs)
0.1	1982	8,570	31,571
0.1	2013	5,105	26,803
0.1	2015	16,894	35,383
0.02	1982	16,176	52,461
0.02	2013	11,074	42,412
0.02	2015	25,863	55,657
1E-03	1982	33,718	99,754
1E-03	2013	23,046	78,625
1E-03	2015	45,688	101,500
1E-04	1982	50,023	144,661
1E-04	2013	35,602	115,189
1E-04	2015	64,485	145,253
1E-05	1982	69,775	198,969
1E-05	2013	50,471	159,618
1E-05	2015	86,918	197,623

4.3.4.5 Frequency Analysis with Causal Information

Figure 4-24 shows the 3-day volume-frequency curve based on the paleoflood data with regional skew data compared to the expected 3-day basin-average regional precipitation-frequency curve (converted to 3-day flow), shown in blue. As described in the previous sections, rainfall-frequency events were routed through the calibrated HEC-HMS model by scaling three historical storm events: December 1982, May 2013, and December 2015. The rainfall-runoff results are shown as colored diamonds in Figure 4-27. As can be seen, the rainfall-runoff results predicted an AEP for the SDF of approximately $1\text{E-}5$ or more frequent. The rainfall-runoff results are reasonably consistent with the historical and paleoflood data, but slightly more frequent than the most likely curve from the Bayesian fit.

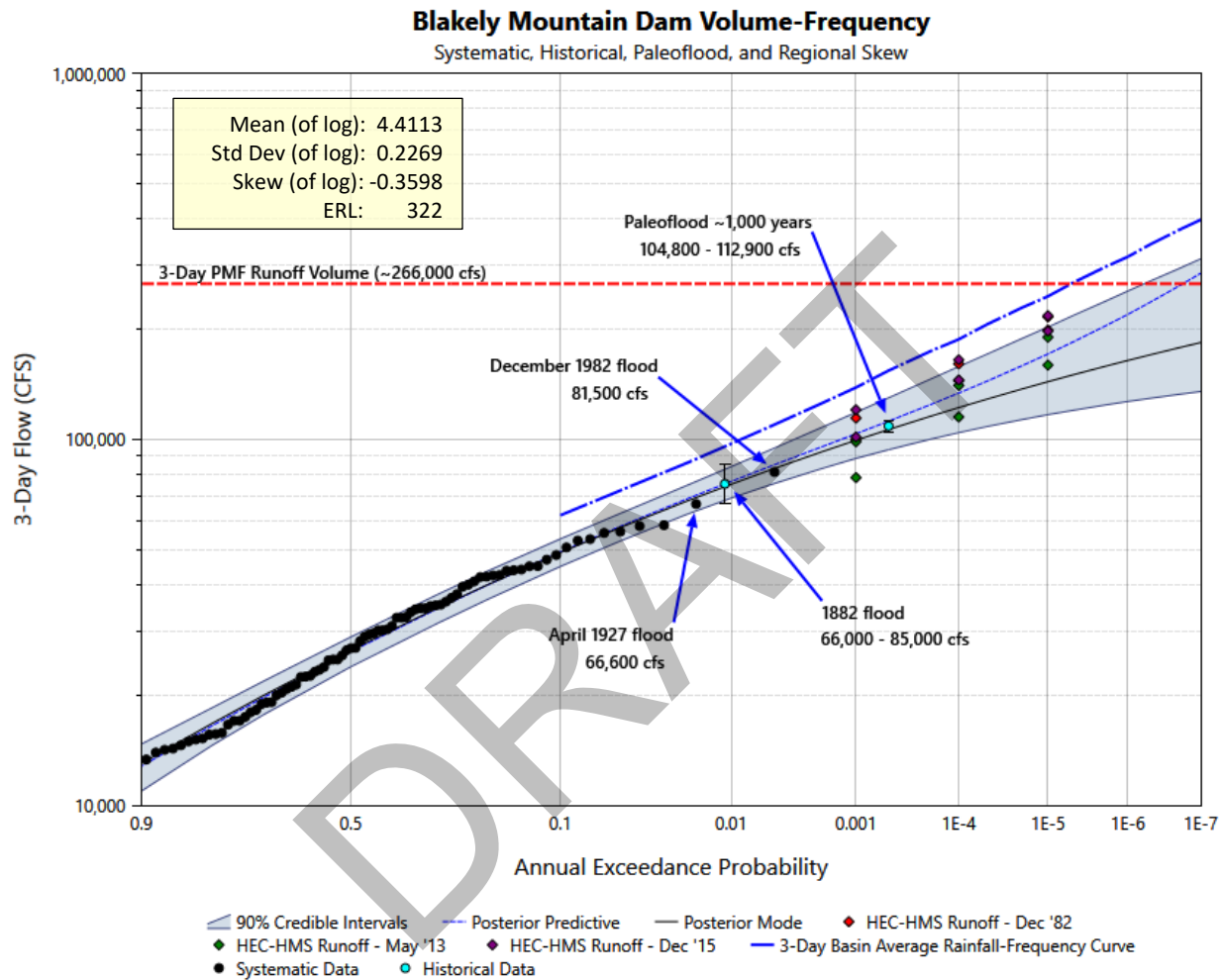


Figure 4-27. The expected 3-day basin-average regional precipitation-frequency curve (shown in blue dash dot) and rainfall-runoff results from HEC-HMS (shown as colored diamonds) compared with the Bayesian fit of the LPIII distribution using systematic, historical, paleoflood, and regional skew data.

The modelled runoff results using regional rainfall-frequency can be incorporated into the Bayesian analysis by estimating the prior distribution of discharge for a specified exceedance probability [(Coles & Tawn, 1996); (Viglione, Merz, Salinas, & Blöschl, 2013)]. For this study, a distribution of discharge was estimated for the $1\text{E-}4$ AEP. This AEP was selected because in order to add information to the fit, it needed to be rarer than the paleoflood event, while not being so rare as to be overly influential. The choice of quantile for the causal information has a large influence on the resulting fit and credible bounds. If a more frequent quantile is chosen, such as $1\text{E-}2$, more weight would be given to the historical and paleoflood data and the causal prior would have little influence on the fit. Whereas, the choice of the $1\text{E-}4$ quantile gives significant weight to the causal prior, reflecting the high confidence the experts had in the regional

precipitation-frequency analysis and modelled rainfall-runoff results. In general, the rarer the chosen quantile for the causal information, the more influence it will have in the Bayesian analysis.

The distribution of rainfall at the 1E-4 AEP was determined to be approximately normally distributed with a mean of 155,000 cfs and a standard deviation of 22,000 cfs. The HEC-HMS rainfall-runoff results at all of the sites indicate that rainfall losses can vary significantly depending on the antecedent conditions and the spatial and temporal temperature pattern of the event.

Final results are shown in Figure 4-28. The ERL is 532 years, demonstrating that the combination of temporal, spatial and causal information expansion significantly increases the ERL beyond that of the temporal expansion alone. The most likely and expected curves are much closer together, indicating a substantial reduction in uncertainty. Consequently, the results shown in Figure 4-28 are considered to be the best estimate for the 3-day inflow volume-frequency relationship at Blakely Mountain Dam.

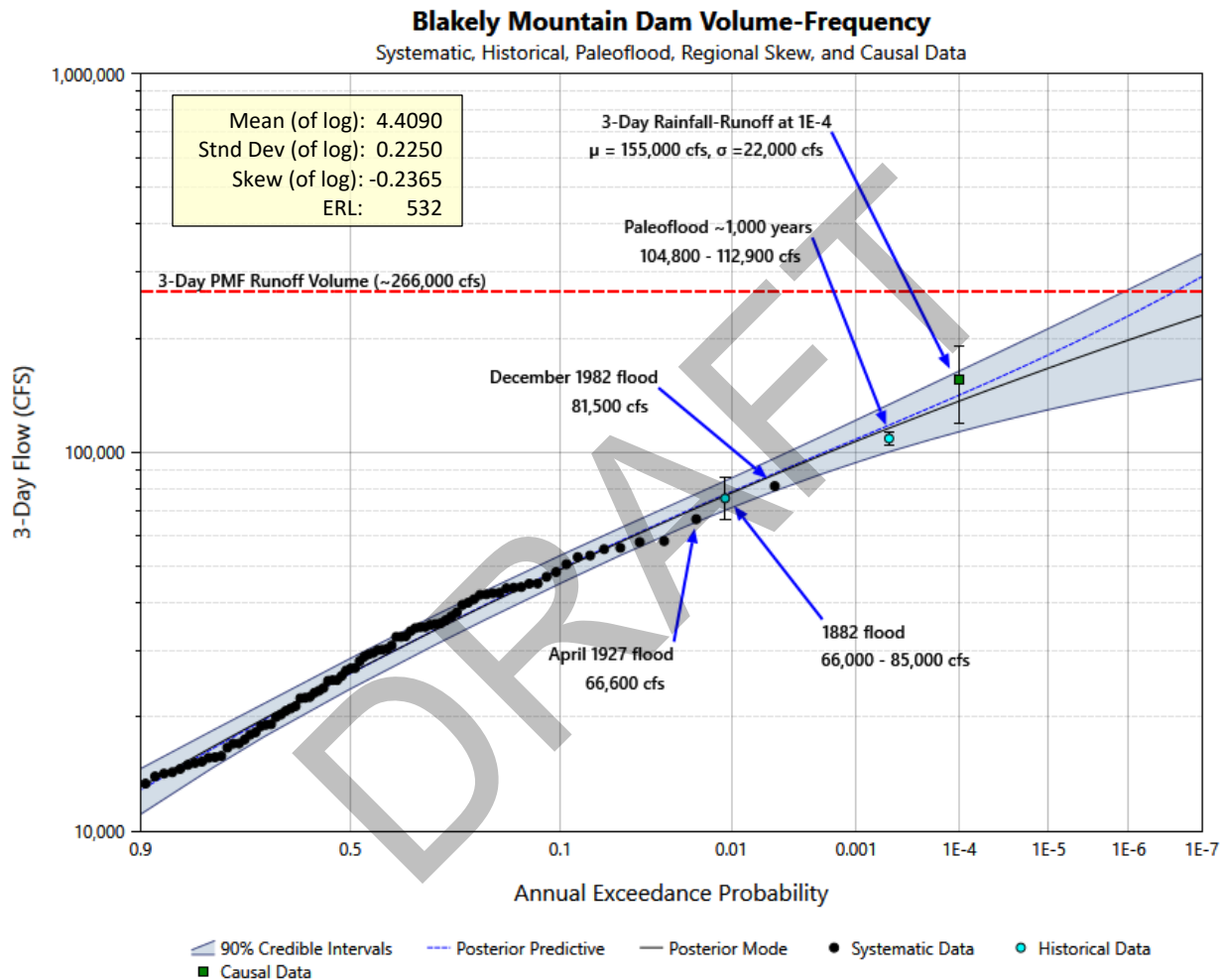


Figure 4-28. Bayesian results for the unregulated annual maximum 3-day inflow volume-frequency curve and 90 percent credible intervals based on systematic, historical, paleoflood, regional skew, and causal rainfall-runoff data.

4.4 Summary and Conclusions

Summary statistics for the 3-day volume frequency analysis scenarios using RMC-BestFit are provided in Table 4-11. As this chapter demonstrates, EMA and Bayesian estimation methods produce similar flood frequency curves for systematic, historical, and paleoflood data. There are minor trade-offs between mean, standard deviation, and skew between the two estimation methods, and the confidence intervals can be slightly different. However, the differences are typically inconsequential, and the computed curves for each method often lead to similar statistical inferences.

One of the primary differences between the methods is that EMA is not capable of incorporating the causal rainfall-runoff information in a formal, probabilistic manner, whereas the Bayesian estimation method is capable of incorporating and accounting for uncertainty in all available sources of hydrologic information, such as systematic flood records, historical floods, paleofloods, and regional rainfall-runoff results. Consequently, the Bayesian method is considered to provide higher confidence in the fitted volume frequency curves and the resulting estimates for reservoir stage-frequency.

Table 4-11. Summary statistics for 3-day volume-frequency analysis scenarios using RMC-BestFit

Analysis	Mean (of log)	Standard Deviation (of log)	Skew (of log)	ERL (years)
Systematic	4.4114	0.2265	-0.4291	91
Systematic, Historical, & Paleoflood	4.4106	0.2266	-0.3535	307
Systematic, Historical, Paleoflood, & Regional Skew	4.4113	0.2269	-0.3598	322
Systematic, Historical, Paleoflood, Regional Skew, & Causal Data	4.4090	0.2250	-0.2365	532

5 Reservoir Stage-Frequency Analysis

The reservoir stage-frequency curves developed for the Blakely Mountain Dam used an inflow volume-frequency based approach performed by RMC – Reservoir Frequency Analysis (RMC-RFA) v.1.1.0 (Smith C. H., 2017). RMC-RFA is a modeling tool that utilizes a nested Monte Carlo approach to capture both the knowledge (epistemic) uncertainty and natural (aleatory) variability associated with a deterministic flood routing model (Smith & England, 2017). This chapter discusses the RMC-RFA methodology, development of the model input parameters, and the final hazard curves for Blakely Mountain Dam.

5.1 Overview of the RMC-RFA Methodology

RMC-RFA utilizes a deterministic flood routing model to produce an inflow volume-based stage-frequency curve with uncertainty bounds. Uncertainty is handled by treating the seasonal occurrence of inflow events, the antecedent reservoir stage, the inflow volume, and the inflow hydrograph shape as variable rather than fixed inputs. To quantify the knowledge uncertainty and natural variability in stage-frequency estimates RMC-RFA employs a two-looped, nested Monte Carlo methodology. The *inner loop*, also referred to as a *realization*, simulates the natural variability of the reservoir stage prior to the flood event. Each inner loop simulation represents a single realization. The *outer loop* simulates the knowledge uncertainty associated with the inflow volume-frequency distribution. Many thousands of realizations are simulated within the outer loop. The basic construct of the simulation procedure employed by RMC-RFA is illustrated in Figure 5-1.

The framework of the RMC-RFA model is based on discrete inflow events, which is similar to the framework used in the Stochastic Event Flood Model (SEFM) (MGS Engineering Consultants, Inc., 2005) and the Flood Risk Analysis (FRA) Compute Option in HEC-WAT (U.S. Army Corps of Engineers, 2017). This framework was deemed appropriate for the Blakely Mountain Dam because the precipitation events that drive the upper tail of the hazard curve are usually individual events, lasting 3 to 6 days. Additionally, storms of longer durations, which can be seen in the watershed, are accounted for due to the random sampling of starting reservoir stages within RMC-RFA.

Table 5-1 summarizes the model input parameters that are treated as random variables in the RMC-RFA model. Development of model inputs is aided by tools within the program that allow the user to estimate inputs, such as flood seasonality or starting pool duration curves, in a consistent and automated manner. Other inputs, such as the volume-frequency curve or reservoir operations, are developed by the user independently. For a detailed description of RMC-RFA model framework and approach, refer to RMC-RFA User's Manual (Smith C. H., 2017).

By varying flood frequency parameters across realizations, the uncertainty in the probability of an event, such as overtopping the dam, can be assessed. Each realization will produce an estimate of the probability of overtopping the dam based on the parameters used to drive the realization. Percentiles (such as the 5th and 95th percentiles) of these probabilities produce a confidence interval for the probability of overtopping. The expected probability curve is the best estimate of the probability for a given stage because it represents the relative likelihood of all probabilities of a stage exceeding a certain predefined stage, rather than the point where 50% of the exceedance probabilities lie either above or below the median (Smith & England, 2017).

The model selects a starting reservoir stage, routes a randomly generated annual maximum inflow event through a model simulating reservoir storage and release, and calculates the resulting peak stage. In order to account for consecutive inflow events leading to peak reservoir stage, the RMC-RFA model selects a starting stage from starting stage duration curves that account for consecutive inflow events prior to simulating a peak reservoir stage from a single inflow event.

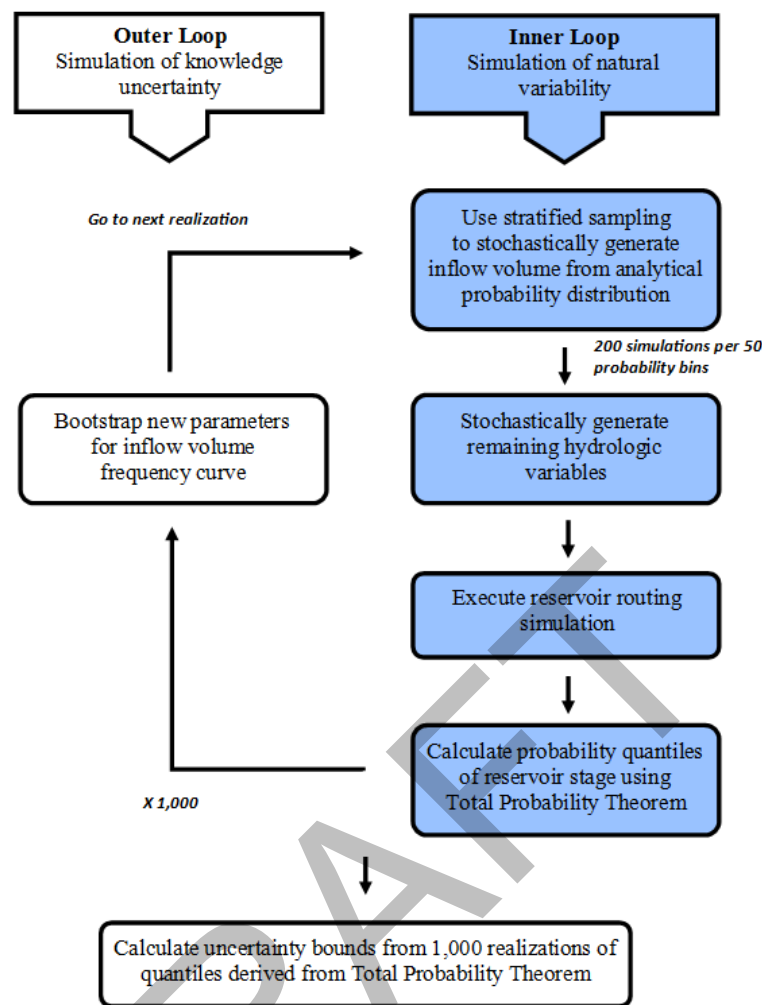


Figure 5-1. Flowchart of the steps involved in the RMC-RFA simulation (Smith C. H., 2017).

Table 5-1. Model parameters treated as random variables in the RMC-RFA simulation.

Input Parameter	Dependency	Statistical Distribution	Sampling Approach
Inflow Volume	Independent	Analytical	Importance/Stratified
Inflow Hydrograph Shape	Independent	Empirical	Monte Carlo
Flood Seasonality	Independent	Empirical	Monte Carlo
Reservoir Starting Stage	Flood Season	Empirical	Monte Carlo

5.2 Inflow Hydrograph Shapes

Discrete inflow hydrograph shapes are used in simulations as representative watershed responses to a precipitation event. The hydrograph patterns are scaled up or down based on the stochastically sampled inflow volume to create a suite of flood simulations (Smith C. H., 2017). Historical documents, systematic flow data, and past studies were used to develop four inflow hydrograph shapes that are characteristic of the different types of responses of the Blakely Mountain Dam watershed. Three hydrographs represent historical runoff events – April 1927, December 1982, and

December 2015 – and the fourth represents the design PMF inflow hydrograph (Figure 5-2). Hydrograph shapes were tested in a sensitivity analysis to assess the impact of hydrograph shape on the stage-frequency curve (section 5.7).

The four hydrographs are approximately 3 days in duration, consistent with the 3-day critical duration that was determined from systematic data. The hydrograph shapes were equally weighted such that each hydrograph shape was sampled in the Monte Carlo simulations with equal probability. The four hydrograph shapes were each normalized by their respective peak 3-day volumes and compared in Figure 5-3. Each ordinate was scaled by the maximum 3-day flow volume, which preserves the timing and critical duration of the inflow hydrograph shape when scaled to a new volume. This represents the process by which RMC-RFA normalizes the inflow hydrograph shapes to the critical duration volume, then scales each ordinate to match the 3-day volume randomly sampled from the inflow volume-frequency curve for each realization.

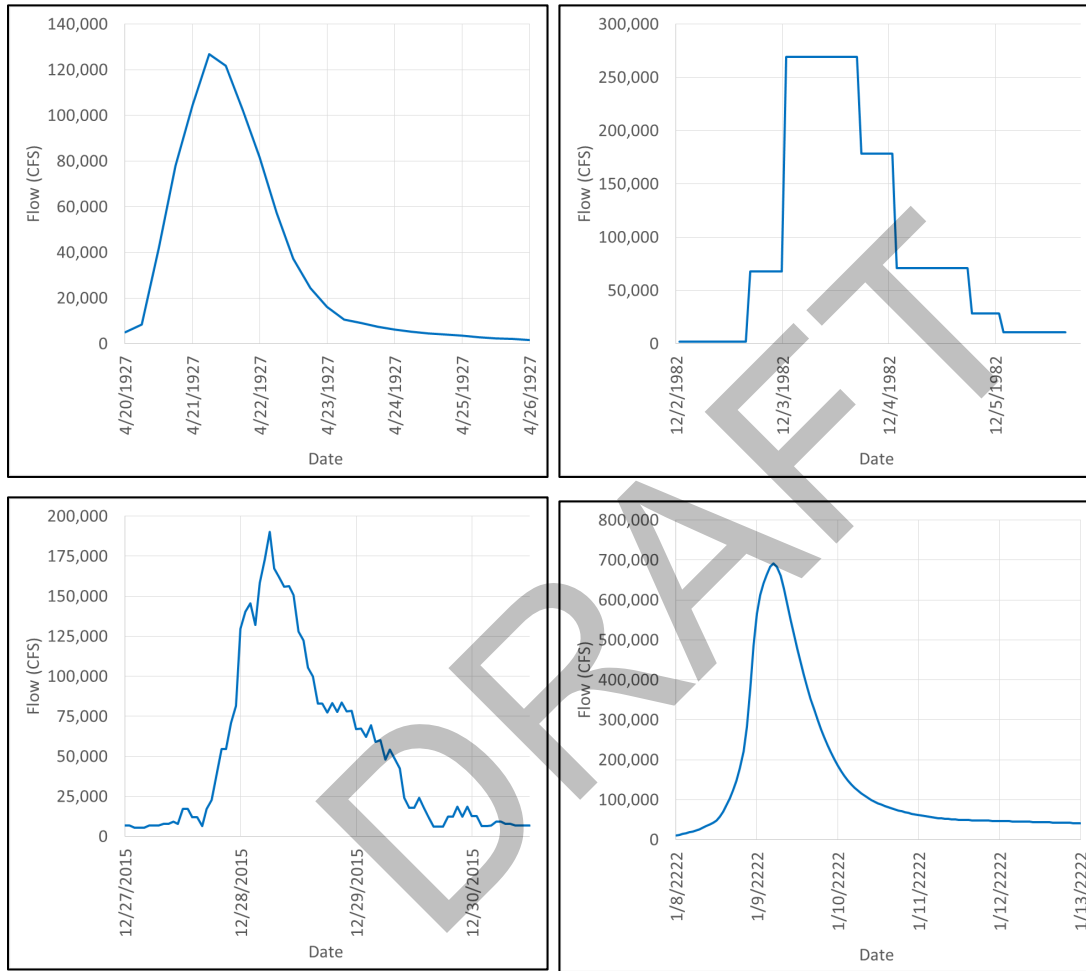


Figure 5-2. Variation in inflow hydrograph shapes for Blakely Mountain Dam.

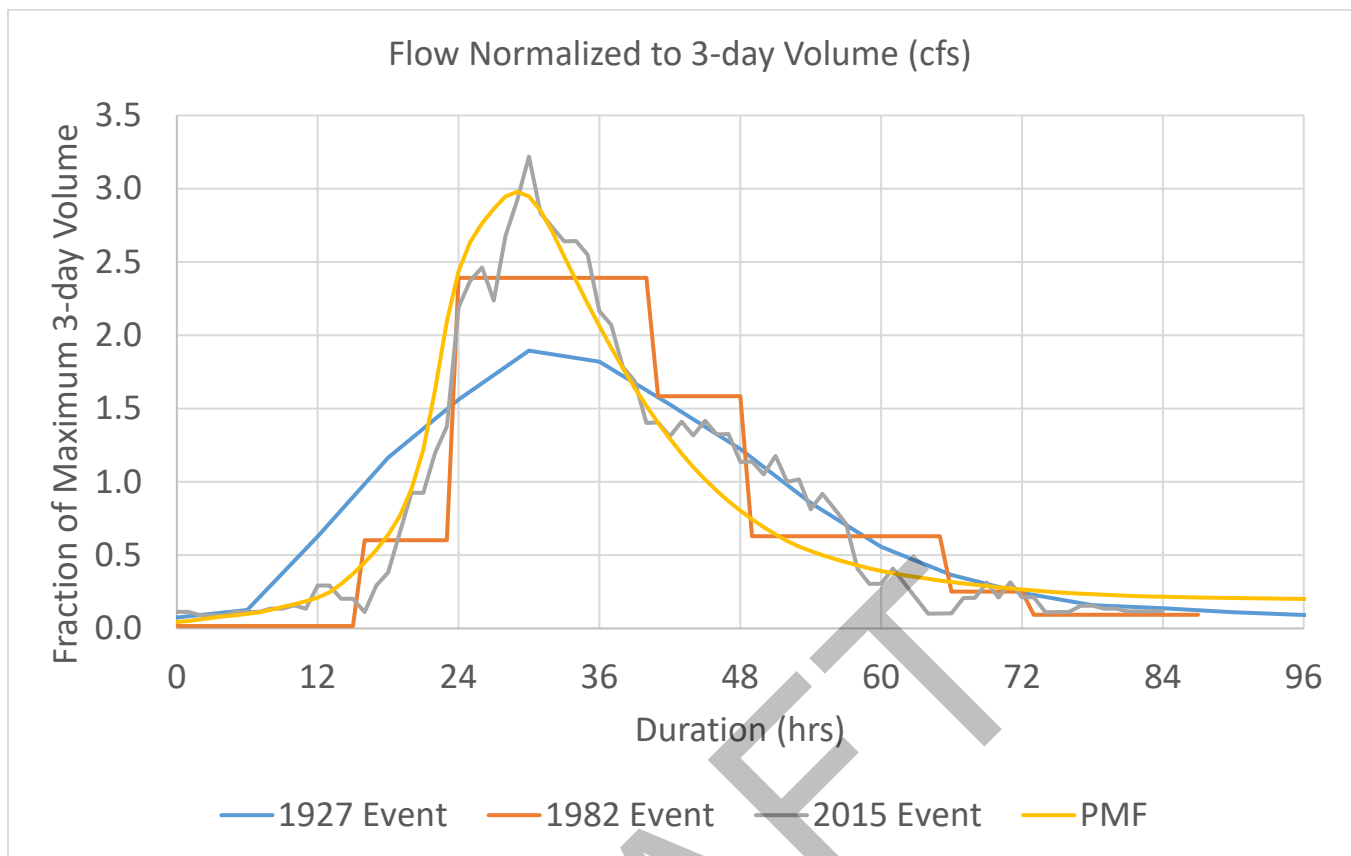


Figure 5-3. Inflow hydrograph shape comparison normalized to 3-day volume.

5.3 Flood Seasonality

In the RMC-RFA modeling process, flood seasonality is intended to describe the frequency of occurrence of floods on a seasonal (monthly) basis. Flood seasonality is used to sample a starting reservoir stage that is representative of the time of year when rare floods are most likely to occur. A rare flood is defined as any event where the flow exceeds a user-specified threshold for a specified flow duration (Smith C. H., 2017). The flood seasonality for Blakely Mountain Dam was determined by identifying the month in which the annual maximum stage occurred, termed *stage seasonality*.

Figure 5-4 shows a histogram of flood seasonality that was developed for Blakely Mountain Dam using the stage seasonality method for annual maximum stages for the period of record of available daily pool data (1970-2018). The period 1961-1969 was omitted from the seasonality analysis because there appears to have been a change in unofficial operations prior to 1970, which may be due to operator changes or due to the initial filling of the reservoir. The period of data 1970-2018 represents current operations. The data were converted into an annual maximum series (AMS), then ranked and plotted using Weibull plotting positions. Most precipitation resulting in peak annual stages occurs in May and the bordering late-spring months. An additional increase in frequency occurs in December.

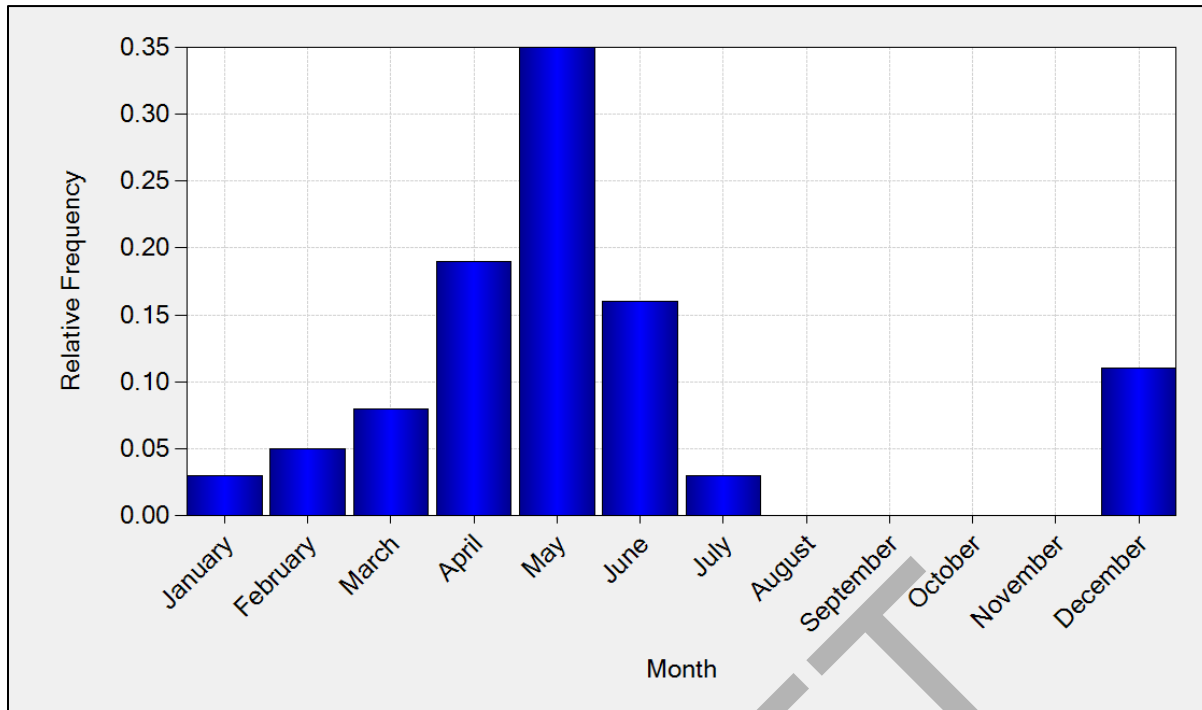


Figure 5-4. Histogram of RMC-RFA flood seasonality model input for Blakely Mountain Dam using stage seasonality method.

5.4 Starting Stage Duration

Reservoir starting stage duration curves represent the percent of time during which an antecedent reservoir stage is exceeded and used to represent multiple or stacked events in the watershed (U.S. Army Corps of Engineers, 2018). A monthly pool stage-duration plot is shown in Figure 5-5. The curves were developed from the available daily pool elevation record from 1970-2018. The period 1961-1969 was omitted from the starting stage analysis because there appears to have been a change in unofficial operations prior to 1970, which may be due to operator changes or due to the initial filling of the reservoir. A pool change threshold of 0 feet and a typical high pool duration of 28 days were used to influence the starting stage-duration curves, as determined through calibration of the model.

The upper tail of the stage-duration curves (below 10 percent of time exceeded) spreads approximately 5 to 10 feet of reservoir stage, suggesting the reservoir is not highly sensitive to the starting stage. A sensitivity analysis was run on reservoir starting stage to assess the impact on the stage-frequency curve (see section 5.7).

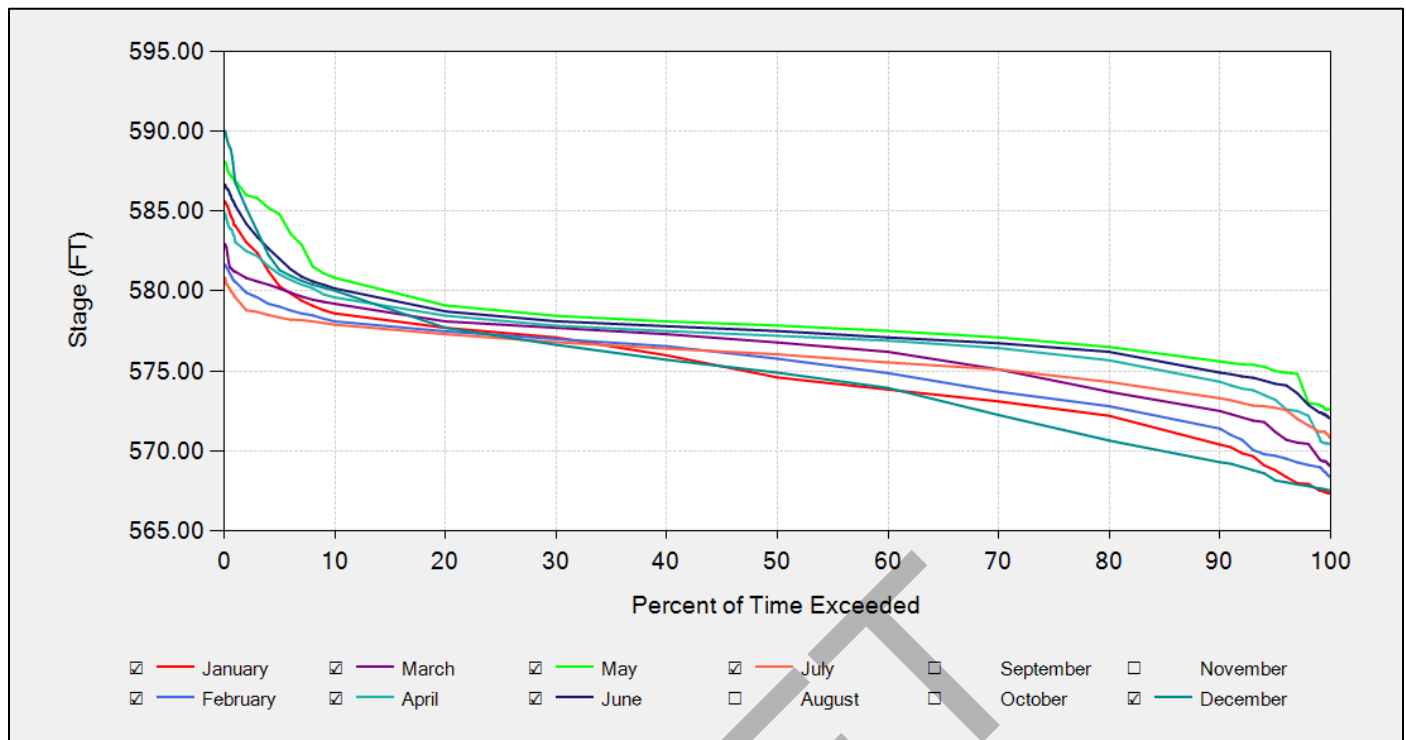


Figure 5-5. Monthly stage-duration curves.

5.5 Reservoir Model

Reservoir routing in RMC-RFA is based on the Modified Puls routing method, also known as storage routing or level-pool routing (Smith C. H., 2017). Reservoir routing with RMC-RFA requires the following data to describe the reservoir storage and hydraulic characteristics:

1. Elevation-Storage curve to describe the size and shape of the reservoir.
2. Elevation-Discharge curve to describe the hydraulic behavior of the reservoir outlet structure(s), spillway, and overtopping conditions.

5.5.1 Reservoir Geometry

The stage-storage curve for Blakely Mountain Dam was defined from a combination of bathymetric data and a LiDAR-based stage-storage curve. Pool bathymetric data were sourced from the Definite Project Report. The LiDAR-based curve was computed from 1-D HEC-River Analysis System (HEC-RAS) model. The stage-storage table is reported in increments ranging from 0.2 to 2.0 feet using the NAVD88 datum. The Top of Dam is at a stage of 615.7 ft, and the Spillway is at 591.7 ft. The resulting stage-storage rating curve is tabulated in Table 5-2 and shown in Figure 5-6.

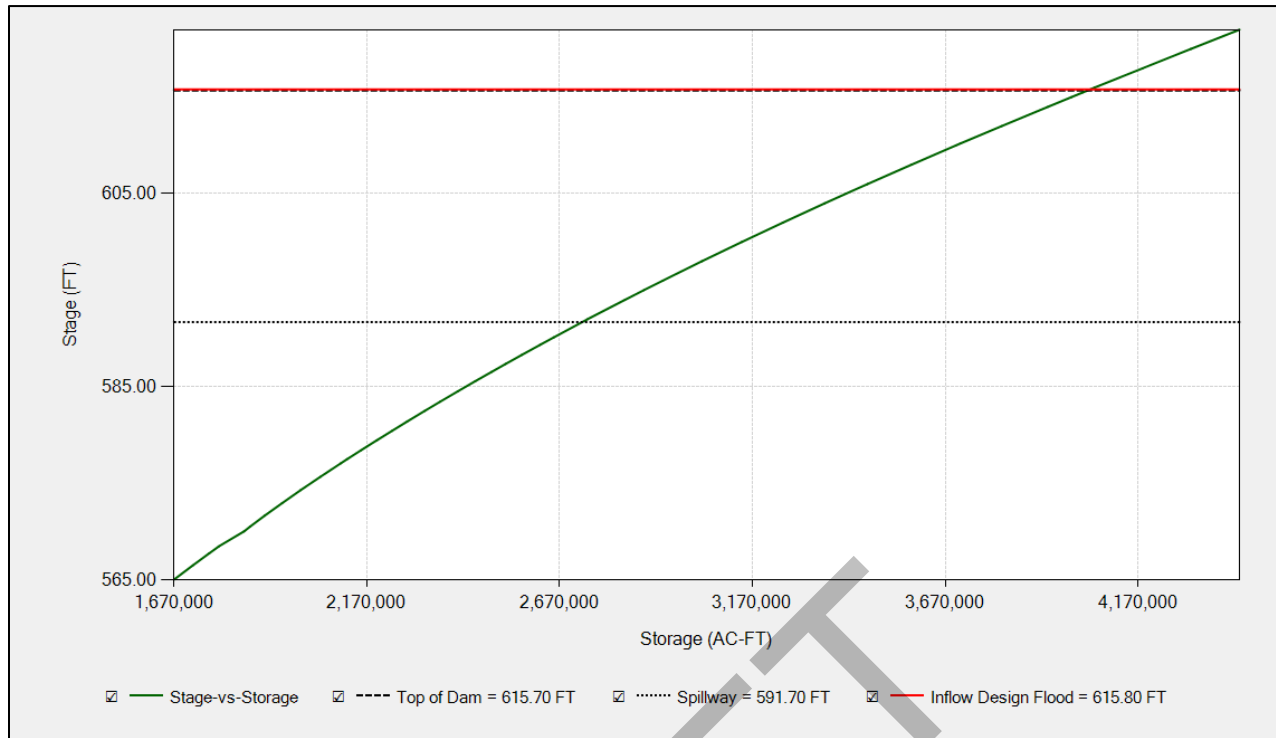


Figure 5-6. Blakely Mountain Dam stage-storage rating curve for RMC-RFA simulations.

5.5.2 Reservoir Operations

RMC-RFA reservoir operations use a simplified elevation-discharge curve to simulate actual operational rules. For the purposes of the RFA model, it was assumed that no releases would occur below the spillway crest (i.e., no outlet works) as Blakely Mountain Dam is an uncontrolled spillway. In reality, the reservoir is operated by the USACE Vicksburg District for flood control, with operational rules specified in the WCM, and by Entergy for power generation. This is a conservative assumption that is backed by current operational procedures, where flood water is typically held in the reservoir during precipitation events until downstream channel capacity becomes available. A sensitivity analysis on reservoir operations was run to test the assumption of zero release below the spillway crest (see section 5.7).

The stage-discharge rating curve was developed using 1-D and 2-D HEC-RAS modeling and LiDAR terrain data. The outflow capacity was updated from prior analyses, which were derived from WCM operations and an understanding of downstream controls, and adjusted to mimic large scale event operations. For the current analysis, overtopping discharge was determined from the 1-D and 2-D HEC-RAS models, which simulated spillway and dam overtopping routing and tested various different Manning's n roughness coefficients to recommend a final stage-discharge rating curve. As stated previously, discharge was assumed to be zero below the spillway elevation, and discharge above the spillway was extrapolated using RAS model results. The resulting stage-discharge curve is tabulated in Table 5-2 and plotted in Figure 5-7.

Table 5-2. Stage-Storage-Discharge rating table for Blakely Mountain Dam for RMC-RFA simulations.

Elevation (FT-NAVD88)	Storage (AC-FT)	Discharge (CFS)	Elevation (FT-NAVD88)	Storage (AC-FT)	Discharge (CFS)
565.00	1,670,000	0	589.33	2,618,900	0
566.48	1,719,000	0	590.10	2,654,300	0
567.73	1,761,500	0	592.00	2,742,482	0
568.44	1,786,000	0	594.00	2,839,200	1,000
570.00	1,851,600	0	595.00	2,888,100	3,000
570.49	1,867,200	0	596.00	2,937,700	4,000
571.63	1,905,100	0	598.00	3,038,700	6,875
572.92	1,950,100	0	600.00	3,142,300	11,180
574.27	1,998,300	0	602.00	3,248,300	17,090
575.64	2,049,000	0	604.00	3,356,900	24,110
577.05	2,102,400	0	605.00	3,412,100	27,960
577.48	2,118,700	0	606.00	3,467,900	32,170
578.39	2,154,200	0	608.00	3,581,700	41,270
579.44	2,195,900	0	610.00	3,697,000	51,800
580.43	2,235,700	0	610.20	3,708,495	52,840
581.38	2,274,200	0	612.00	3,815,400	63,520
582.49	2,320,100	0	614.00	3,935,600	76,440
583.43	2,359,600	0	615.00	3,996,035	83,550
584.51	2,405,500	0	615.50	4,026,371	87,220
585.57	2,451,200	0	616.00	4,056,619	91,080
586.52	2,492,900	0	618.00	4,181,300	114,600
587.35	2,529,600	0	620.00	4,306,100	148,000
588.29	2,571,700	0	622.00	4,432,300	188,100

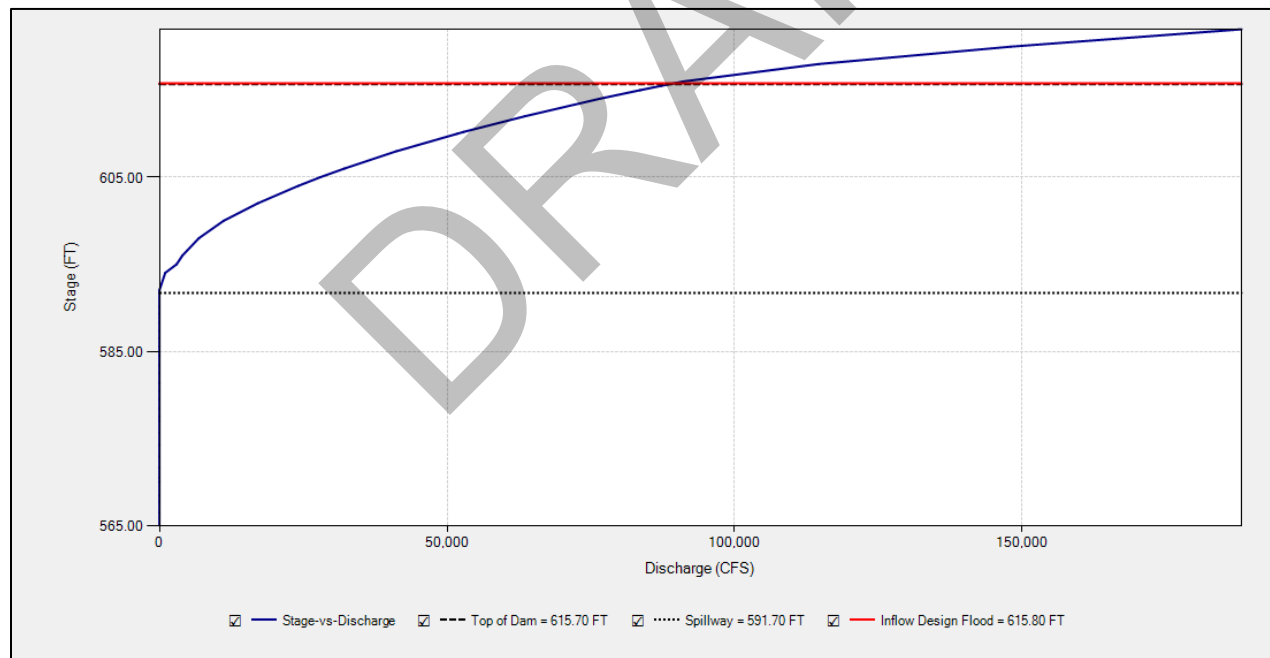


Figure 5-7. Blakely Mountain Dam stage-discharge rating curve for RMC-RFA simulations.

5.6 Empirical Stage-Frequency

Empirical stage frequencies were used to estimate reservoir stage probabilities and to evaluate the performance of the stage-frequency curve developed from the RMC-RFA modeling. The empirical frequency curve was based on the annual maximum series from the POR of stage data from 1970-2018, which reflects current operations. The peak reservoir stage AMS by water year was assigned a frequency using Weibull plotting positions. The empirical frequency curve for Blakely Mountain Dam is shown in Figure 5-8.

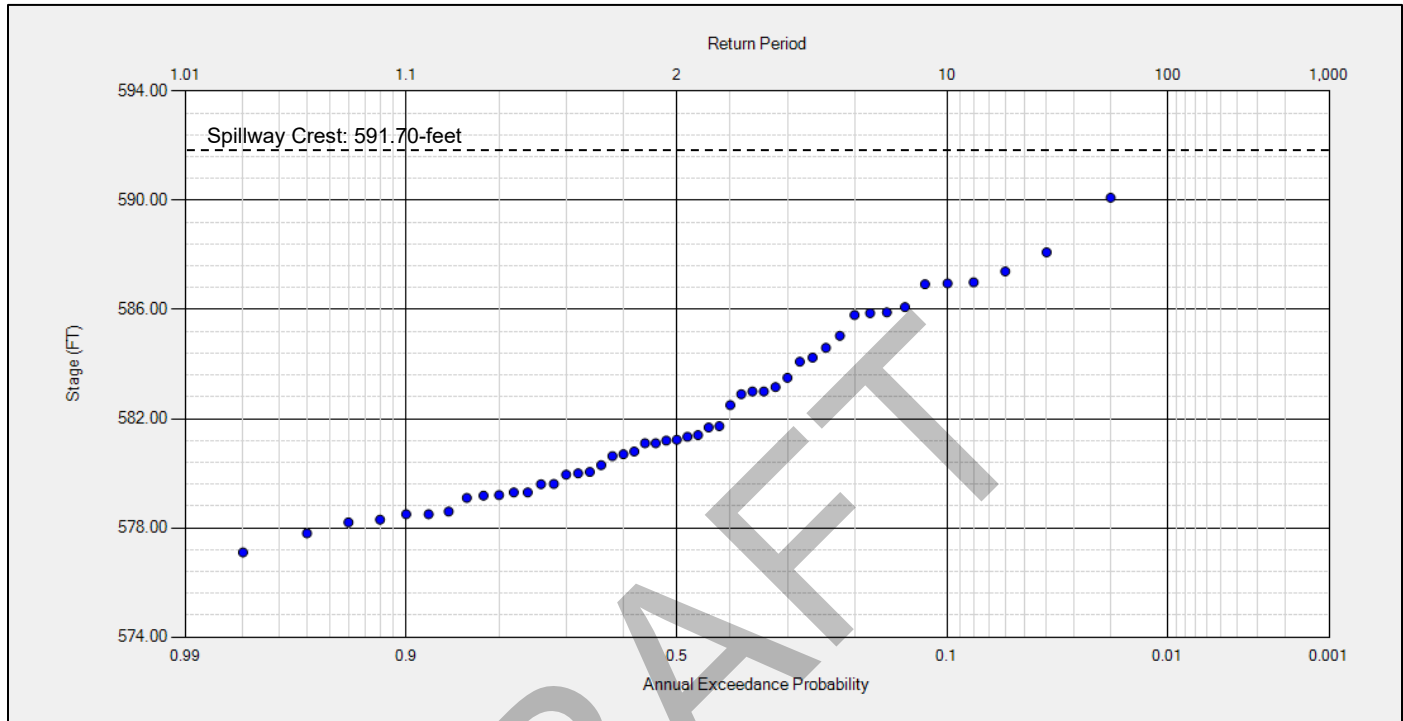


Figure 5-8. Empirical stage-frequency curve for Blakely Mountain Dam for 1970-2018.

5.7 Stage-Frequency Sensitivity Analysis

Several sensitivity analyses were conducted to develop an understanding of the uncertainty of certain key inputs to the hazard curve computation. Four sensitivity analyses were run in RMC-RFA to test the impacts of: hydrograph shape, inflow volume-frequency using RMC-BestFit, inflow volume-frequency using B17C, and reservoir operations. All sensitivity analysis simulations were run in RMC-RFA and results were plotted on a normal probability scale. The simulations were specified to produce the expected frequency curve only (without confidence limits) to reduce computation time, therefore the data will not align exactly with the final loading curve results. Additionally, a routing time window of 6 days and a 6-hour routing time step was used for efficiency, as the difference between a 6-hour and a 1-hour time step was negligible.

The results of the sensitivity analyses were ranked by their impact on the AEP of the intersection of the loading curve with the top of dam. Three categories of risk impacts were identified: Primary, Secondary, or Tertiary. Primary risk impacts are those that have the potential to alter the probability of overtopping the dam by an order of magnitude or greater. Secondary risk impacts alter the probability by half an order of magnitude or greater. Tertiary risk impacts alter the probability by less than one-half an order of magnitude. The five sensitivity analyses with each resulting risk impact level is summarized in Table 5-3.

Table 5-3. Summary of sensitivity analyses and assigned risk impact levels.

Sensitivity Analysis	Number of Alternatives	Risk Impact Level
Hydrograph Shape	4	Tertiary
BestFit Inflow Volume-Frequency	3	Secondary
B17C Inflow Volume-Frequency	2	Tertiary
Reservoir Operations	4	Secondary

5.7.1 Hydrograph Shape Sensitivity

Each of the four hydrograph patterns were individually run in a RMC-RFA simulation to test the sensitivity of the final loading curve to hydrograph shape. Hydrograph shape had minimal effect on the probability of overtopping. This is the expected result, as the four hydrograph patterns are similar in shape, duration, and volume, as seen in the comparison plot in Figure 5-3. As the loading curve probabilities changed by less than half an order of magnitude, hydrograph shape is assigned a **tertiary risk impact** level. The expected frequency curves are shown in Figure 5-9.

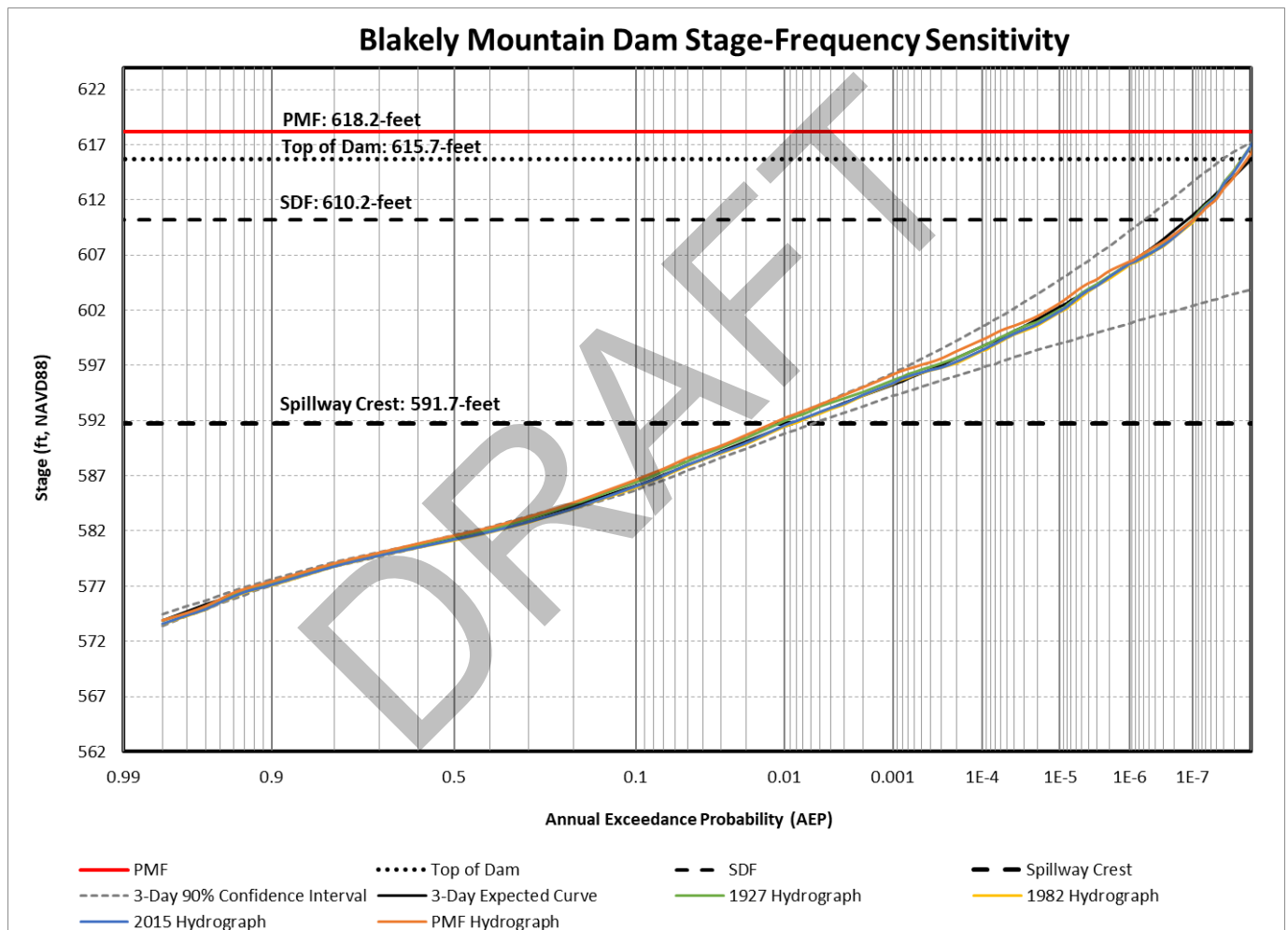


Figure 5-9. Stage-frequency sensitivity analysis of hydrograph shape.

5.7.2 BestFit Inflow Volume-Frequency Sensitivity

A sensitivity analysis was run to assess the impacts of different inputs to the RMC-BestFit determination of inflow volume-frequency. As discussed in Chapter 4, inflow volume-frequency curves were developed using RMC-BestFit using a Bayesian framework. Various data inputs were included to build the final recommended curve. Table 4-11 in section 4.4 summarizes the statistics for each of the curves: (1) systematic data only, (2) systematic, historical, and

paleoflood data, (3) systematic, historical, paleoflood, and regional skew, and (4) all previous inputs combined with causal rainfall-runoff information. The final recommended BestFit inflow volume-frequency curve, using all available information, was compared in a sensitivity analysis to the three curves developed incrementally using the sources of data listed previously.

This sensitivity analyses reflects the impact to the curve when including additional sources of information. The inclusion of additional data serve to extend the ERL, which can be defined as the “value” of the data, and to reduce the uncertainty bounds. The curve developed using systematic data only results in a probability of overtopping of approximately one order of magnitude more frequent. This is primarily due to the limited length of information available, with an ERL of 91 years. The inclusion of historical and paleoflood data decrease the probability of overtopping as the ERL is increased and the skew is decreased. The inclusion of regional skew has minimal impact on the curve, as discussed in section 4.3.3. The BestFit inflow volume-frequency curve is assigned a **secondary risk impact** level. The expected frequency curves are shown in Figure 5-10.

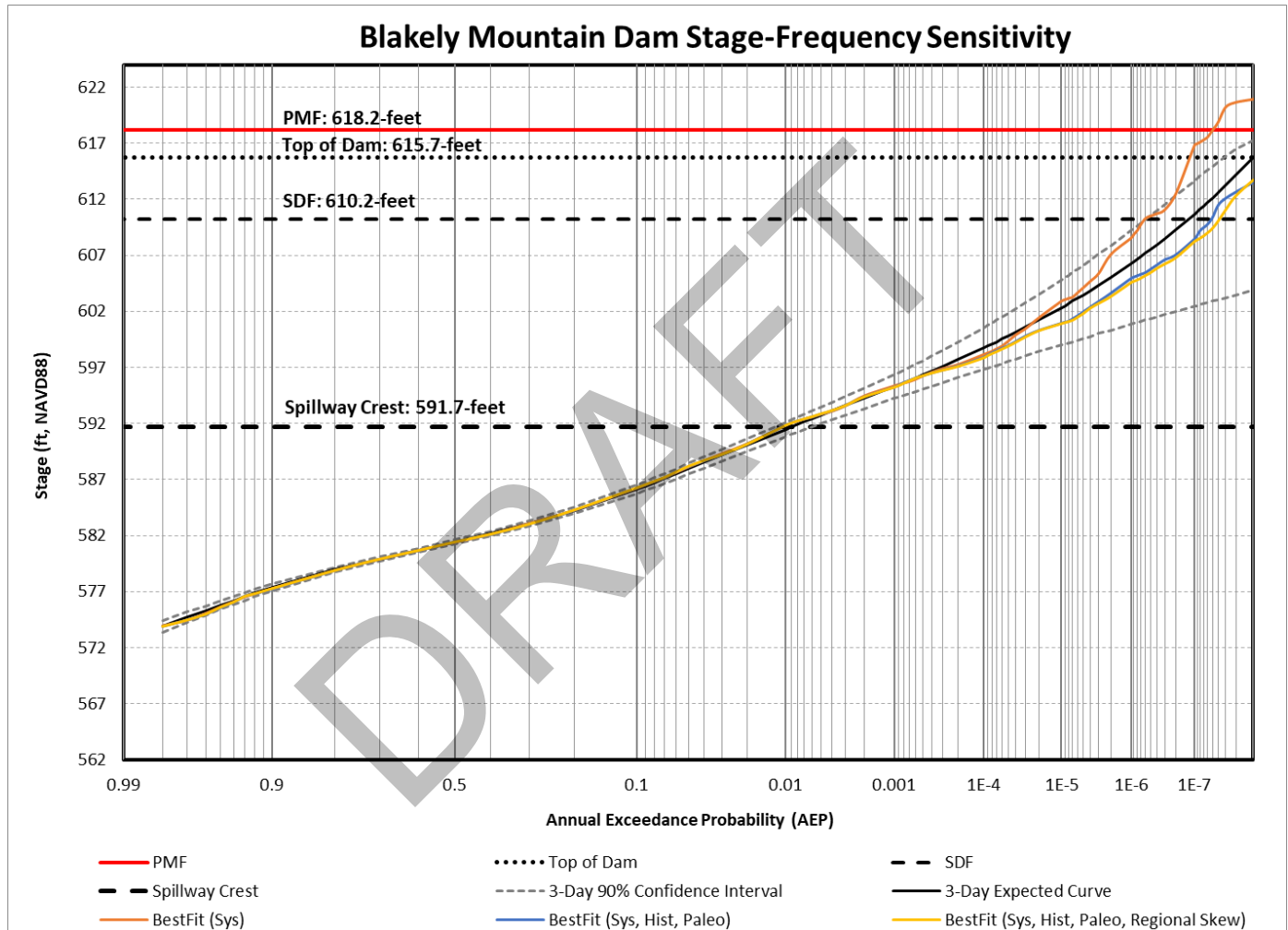


Figure 5-10. Stage-frequency sensitivity of data inputs used to develop RMC-BestFit inflow volume-frequency curves.

5.7.3 B17C Inflow Volume-Frequency Sensitivity

A sensitivity analysis was run to assess the impacts of inflow volume-frequency curves developed using B17C methodology. As discussed in Chapter 4, inflow volume-frequency curves were developed both using B17C with EMA methodology and using Bayesian methodology. The Bayesian results were used as the final recommended curve for development of the hydrologic loading curve. These results were compared to loading curves developed using the B17C methodology. Two different computations of skew were simulated in the sensitivity analysis: weighted skew using B17B methods, and weighted skew using EMA methods.

Both B17C volume-frequency methods resulted in similar stage-frequency curves that differ by less than one-half order of magnitude for the probability of overtopping. This is the expected result, as the two methods (B17C and BestFit) were

used to develop inflow volume-frequency using near-identical data inputs. As the loading curve probabilities changed by less than one-half an order of magnitude, the inflow volume-frequency curve is assigned a **tertiary risk impact** level. The expected frequency curves are shown in Figure 5-11.

The difference in results can be primarily attributed to ERL and skew. The recommended Bayesian inflow volume-frequency curve (using RMC-BestFit) incorporated additional data beyond that of the B17C analyses, including causal rainfall-runoff information, which allowed for a longer ERL. This decrease in uncertainty is seen through a reduced interval of confidence limits. The BestFit analysis also resulted in a curve with higher (less negative) skew, which causes a reduction in the probability of overtopping. The BestFit analysis is considered to provide higher confidence in the fitted volume frequency curves and the resulting estimates for reservoir stage-frequency.

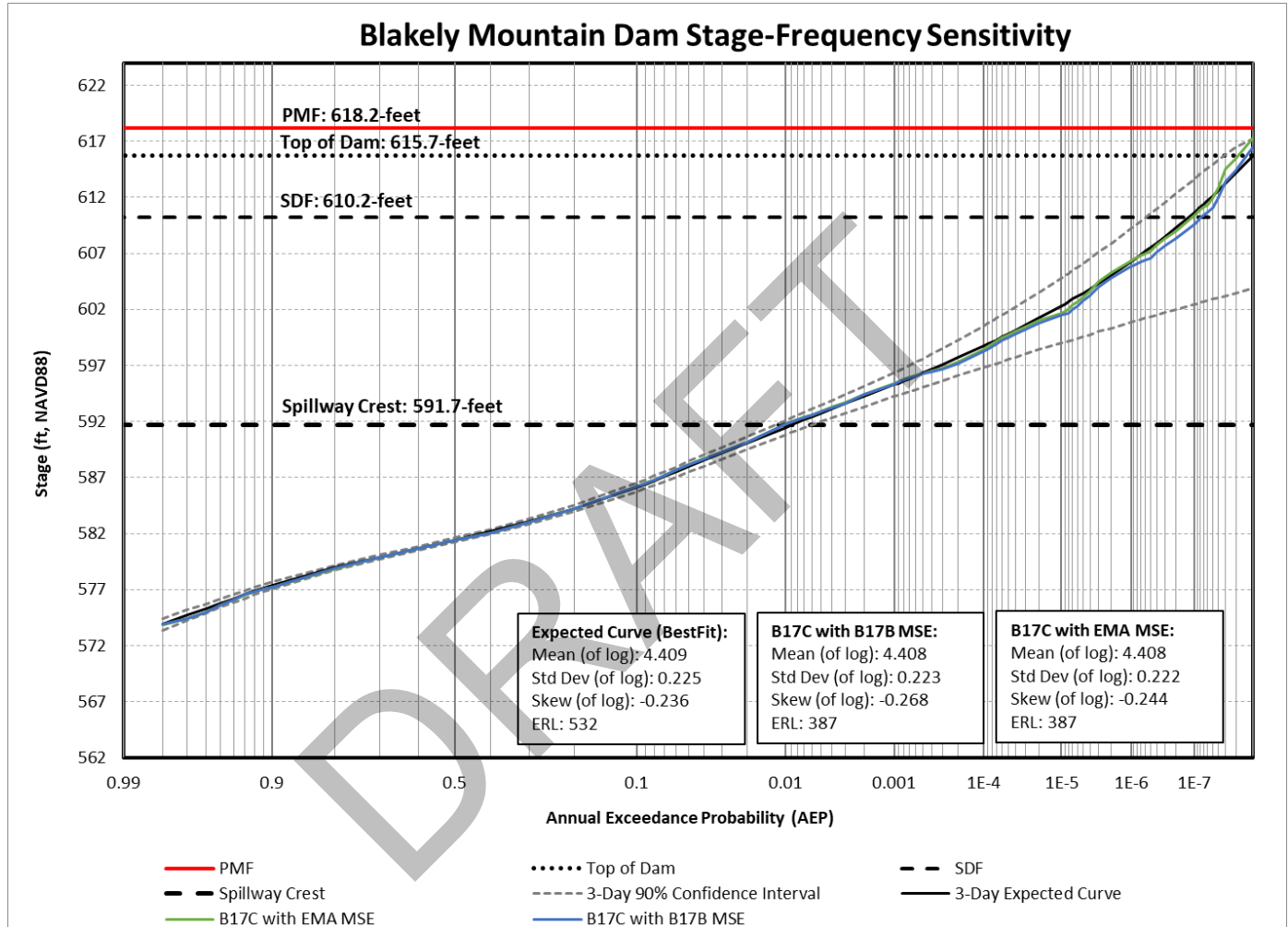


Figure 5-11. Stage-frequency sensitivity of inflow volume-frequency curve and weighted skew methodology.

5.7.4 Reservoir Operations Sensitivity

The reservoir model in RFA was built with the assumption of zero releases below the spillway elevation. In reality, some releases are made for power generation, and releases may be made according to the WCM and depending on downstream channel capacity. Four sensitivity analyses were run to assess the impacts of utilizing outlet release capacity at different stages to the probability of overtopping. The following reservoir operation alternatives were simulated:

1. Constant outlet release of 3,000 cfs below the spillway crest (592 ft)
2. Maximum outlet release (outlet capacity) beginning at spillway crest (592 ft)
3. Maximum outlet release (outlet capacity) beginning at 591 ft

4. Maximum outlet release (outlet capacity) beginning at 586 ft

Stage-frequency results differed by half- to under one-order of magnitude, particularly in the range of elevations near the spillway crest (586 ft to 610 ft), and is assigned a ***secondary risk impact*** level. The expected frequency curves are shown in Figure 5-12 and Figure 5-13. The additional release provided by the outlet works, if utilized, has the potential to reduce the overall flood risk. For example, the more aggressive alternative of utilizing full outlet capacity beginning at a stage of 586 ft reduces the AEP of overtopping the spillway from 8.6E-03 (1:115 years) to 2.2E-03 (1:450 years). However, this utilization of the reservoir outlets is dependent on downstream channel capacity, and has the potential to exacerbate downstream flooding.

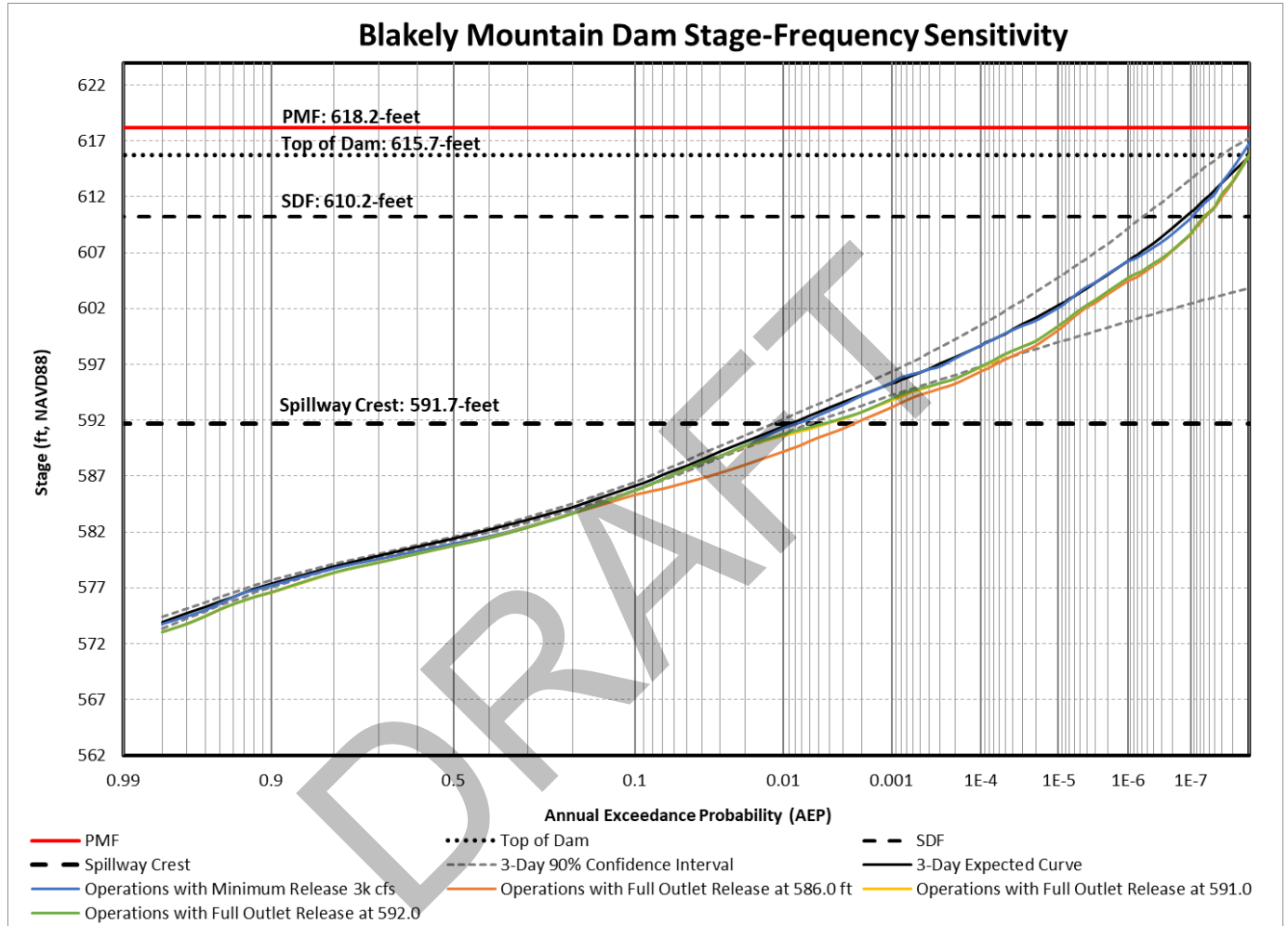


Figure 5-12. Stage-frequency sensitivity of reservoir model operations.

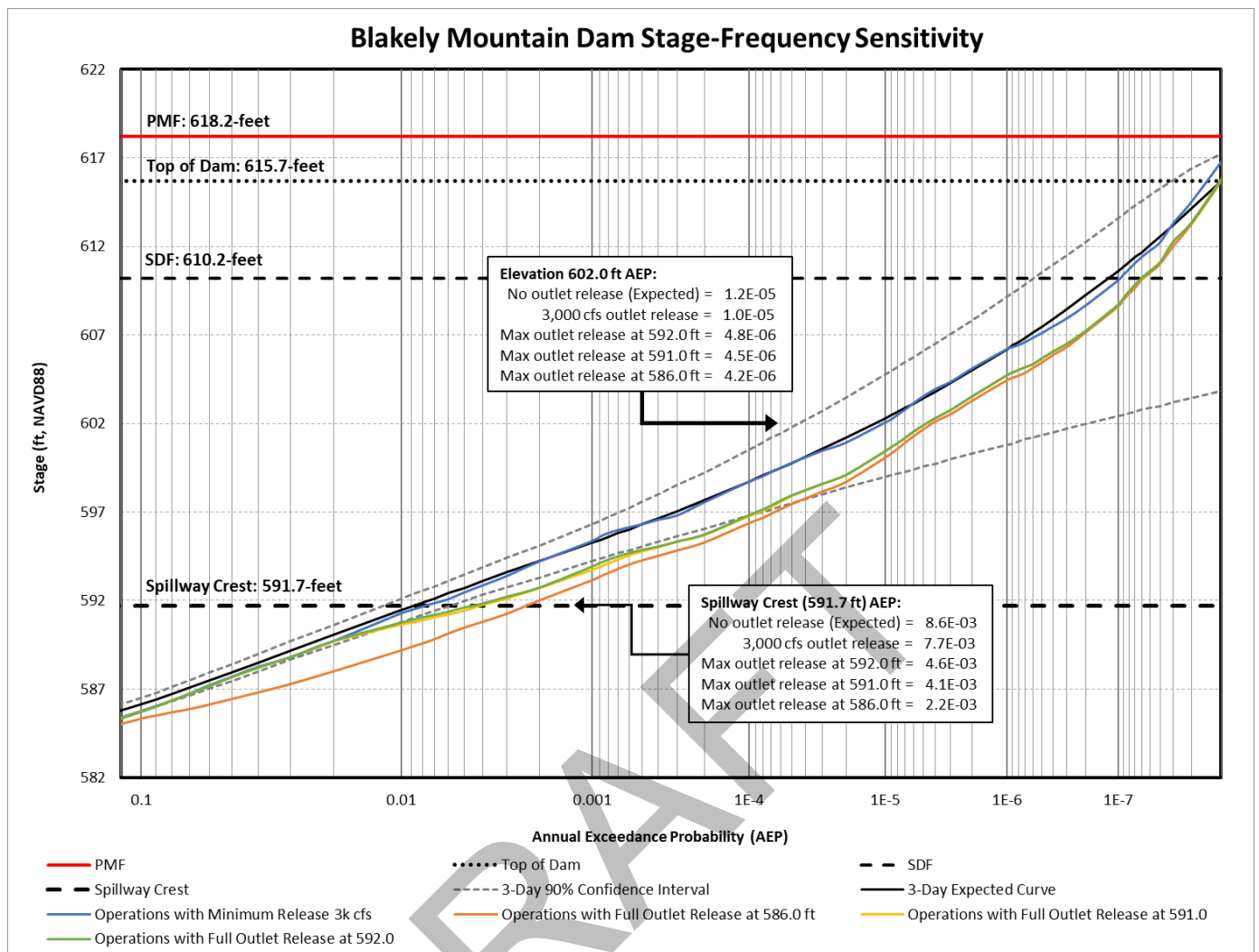


Figure 5-13. Stage-frequency sensitivity of reservoir operations (focused on elevations between the spillway crest, 592 ft, and the top of dam, 616 ft).

5.8 Stage Frequency Results

The final RMC-RFA model simulation was run with full uncertainty, which calculated a median curve, an expected curve, and uncertainty bounds representing a 90% confidence interval. The final curves were developed based on 5,000 simulated events. A routing time window of 6 days at a 1-hour time step was used to capture the annual maximum peak reservoir stage. The simulations included the volume-frequency curve of the critical inflow duration of 3-days, the flood seasonality analysis, starting stage duration analysis, reservoir routing model, and inflow hydrograph shapes. The four inflow hydrograph shapes were equally weighted, giving an equal chance a specific shape is selected at random.

5.8.1 Final Stage-Frequency Curve Results

The final best estimate (unregulated) stage-frequency loading curve for Blakely Mountain Dam is plotted in Figure 5-14 and tabulated in Table 5-5. The 0.95 and 0.05 percent exceedance confidence limits (90% confidence interval) are shown. The uncertainty bounds (dashed lines) represent the uncertainty in stage-frequency due to knowledge uncertainty. The expected curve represents the combined uncertainty due to both natural variability and knowledge uncertainty.

The simulated hazard curve is also plotted against empirical stage-frequency, shown as orange points, which is based on the POR of reservoir stage data for 1970-2018. The model matches the observed stage elevations well at elevations above 579 feet. The ability of the model to fit well to observed stage elevations gives more confidence in the extrapolated AEP for stages beyond the observed events (i.e., above the spillway crest elevation). However, the loading curve suggests a lower (rarer) frequency of surpassing the SDF or the PMF than is indicated by the inflow volume-frequency curve.

The expected curve is recommended to represent the stage-frequency for use in the IES hazard analysis. Table 5-4 summarizes the pertinent elevations for Blakely Mountain Dam and the annual exceedance probability determined from the stage-frequency curve.

Table 5-4. RMC-RFA expected probability with uncertainty for pertinent reservoir pool elevations for Blakely Mountain Dam.

Critical Reservoir Elevation	Elevation (ft-NAVD88)	AEP	Return Period (years)
PMF (2019)	618.2	<1E-08	>100,000,000
Top of Dam (2018 survey)	615.7	<1E-08	>100,000,000
SDF (1950)	610.2	1.2E-07	8,000,000
Spillway Crest (2018 survey)	591.7	8.6E-03	115

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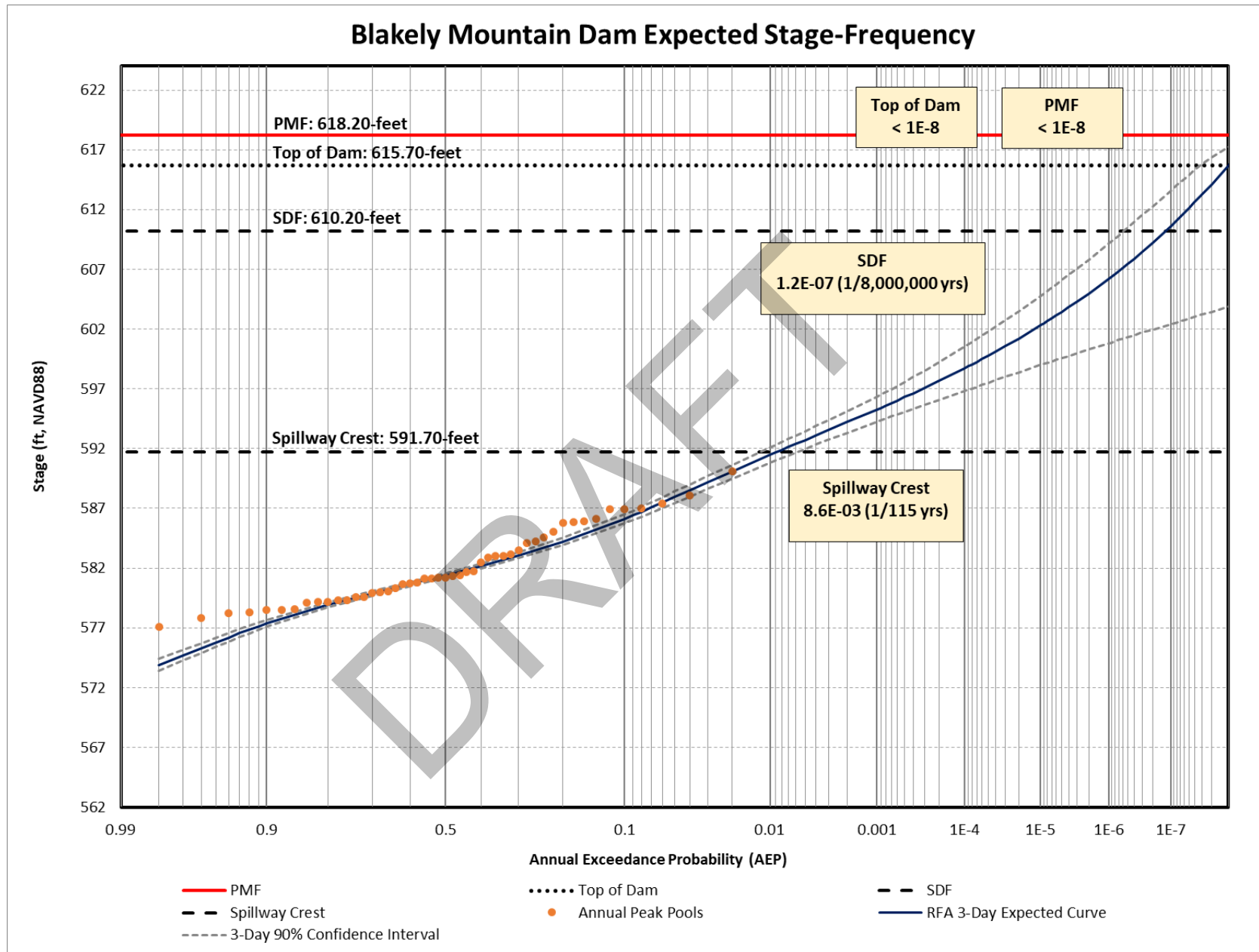


Figure 5-14. RMC-RFA reservoir stage-frequency loading curve for Blakely Mountain Dam.

Table 5-5. RMC-RFA reservoir stage-frequency loading curve tabulated for Blakely Mountain Dam.

AEP	Expected Curve (ft-NAVD88)	Median Curve (ft-NAVD88)	95% Uncertainty Bound (ft-NAVD88)	5% Uncertainty Bound (ft-NAVD88)	AEP	Expected Curve (ft-NAVD88)	Median Curve (ft-NAVD88)	95% Uncertainty Bound (ft-NAVD88)	5% Uncertainty Bound (ft-NAVD88)
9.80E-01	573.92	573.93	574.44	573.39	8.00E-05	599.10	598.67	600.97	597.04
9.70E-01	574.72	574.73	575.17	574.27	7.00E-05	599.29	598.86	601.22	597.19
9.60E-01	575.32	575.33	575.73	574.91	6.00E-05	599.55	599.07	601.49	597.36
9.50E-01	575.80	575.81	576.18	575.43	5.00E-05	599.80	599.29	601.83	597.54
9.40E-01	576.21	576.21	576.57	575.85	4.00E-05	600.17	599.60	602.24	597.76
9.30E-01	576.57	576.57	576.90	576.21	3.00E-05	600.61	599.97	602.76	598.03
9.20E-01	576.87	576.87	577.18	576.55	2.00E-05	601.24	600.49	603.51	598.42
9.10E-01	577.14	577.15	577.44	576.83	1.00E-05	602.34	601.34	604.81	599.04
9.00E-01	577.38	577.39	577.67	577.09	9.01E-06	602.49	601.47	605.02	599.12
8.00E-01	578.95	578.95	579.16	578.75	8.01E-06	602.70	601.64	605.23	599.19
7.00E-01	579.89	579.90	580.08	579.72	7.01E-06	602.91	601.80	605.50	599.29
6.00E-01	580.67	580.67	580.86	580.49	6.01E-06	603.16	601.97	605.77	599.45
5.00E-01	581.42	581.42	581.61	581.23	5.01E-06	603.46	602.22	606.12	599.61
4.00E-01	582.20	582.20	582.40	581.99	4.01E-06	603.84	602.46	606.54	599.76
3.00E-01	583.09	583.08	583.32	582.86	3.00E-06	604.33	602.82	607.09	600.03
2.00E-01	584.24	584.24	584.52	583.97	2.00E-06	605.03	603.30	607.85	600.32
1.00E-01	586.13	586.13	586.52	585.75	1.00E-06	606.24	604.10	609.20	600.84
9.00E-02	586.42	586.41	586.81	586.03	9.02E-07	606.44	604.23	609.40	600.91
8.00E-02	586.73	586.73	587.15	586.32	8.02E-07	606.65	604.38	609.63	601.02
7.00E-02	587.10	587.09	587.53	586.66	7.02E-07	606.88	604.53	609.90	601.13
6.00E-02	587.50	587.50	587.95	587.04	6.01E-07	607.18	604.68	610.21	601.24
5.00E-02	587.97	587.96	588.44	587.49	5.01E-07	607.52	604.92	610.56	601.35
4.00E-02	588.53	588.52	589.01	588.02	4.01E-07	607.92	605.15	611.00	601.52
3.00E-02	589.21	589.19	589.72	588.67	3.01E-07	608.48	605.49	611.54	601.74
2.00E-02	590.08	590.06	590.64	589.50	2.01E-07	609.28	605.93	612.32	601.99
9.99E-03	591.49	591.46	592.12	590.83	1.00E-07	610.67	606.66	613.68	602.45
8.99E-03	591.68	591.65	592.34	591.01	9.03E-08	610.90	606.75	613.87	602.50
7.99E-03	591.91	591.88	592.57	591.21	8.03E-08	611.12	606.86	614.10	602.60
6.99E-03	592.15	592.12	592.83	591.45	7.02E-08	611.42	607.02	614.33	602.70
5.99E-03	592.43	592.40	593.14	591.69	6.02E-08	611.73	607.19	614.62	602.81
4.99E-03	592.73	592.70	593.49	592.00	5.02E-08	612.13	607.35	614.93	602.91
4.00E-03	593.12	593.08	593.90	592.35	4.02E-08	612.61	607.60	615.33	603.01
3.00E-03	593.61	593.56	594.43	592.75	3.01E-08	613.24	607.86	615.80	603.22
2.00E-03	594.26	594.19	595.13	593.33	2.01E-08	614.15	608.27	616.42	603.47
9.99E-04	595.29	595.19	596.36	594.27	1.00E-08	615.74	608.90	617.30	603.89
8.99E-04	595.43	595.34	596.53	594.38	9.04E-09	615.97	608.98	617.42	603.95
7.99E-04	595.62	595.49	596.76	594.54	8.04E-09	616.25	609.11	617.54	604.01
6.99E-04	595.83	595.70	596.99	594.70	7.03E-09	616.54	609.25	617.66	604.07
6.00E-04	596.05	595.91	597.28	594.86	6.03E-09	616.90	609.40	617.85	604.13
5.00E-04	596.34	596.17	597.60	595.08	5.02E-09	617.30	609.54	618.04	604.27
4.00E-04	596.66	596.48	598.03	595.34	4.02E-09	617.80	609.79	618.23	604.41
3.00E-04	597.08	596.89	598.56	595.67	3.02E-09	618.45	610.03	618.52	604.56
2.00E-04	597.71	597.45	599.27	596.10	2.01E-09	619.33	610.43	618.86	604.74
1.00E-04	598.75	598.40	600.57	596.84	1.01E-09	620.87	611.03	619.53	605.08

9.00E-05	598.93	598.53	600.76	596.94					
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5.8.2 Comparison to Previous Studies

The final stage-frequency curve was plotted against loading curves developed in past studies: the 2012 IES study, and the 2015 SQRA study. The 2012 study was based on a volume-duration-frequency analysis based on empirical inflow data and fitting to unofficial PMF routing results with an assigned AEP. The 2015 study expanded on 2012 results by incorporating NOAA Atlas 14 frequency rainfall routed through a rainfall-runoff model and using B17B methodology and regional skew maps. The graphical representation of the comparison to previous studies is shown in Figure 5-15.

The hydrologic hazard curve developed for this IES results in a probability of overtopping approximately 3 orders of magnitude less frequent than those found in the 2015 SQRA report. The 2015 loading was developed using simple B17B analysis on an abbreviated period of record with a modest level of effort to investigate sensitivity to skew. Additional routings were made for ratios of the PMF and balanced hydrographs, and for Atlas 14 precipitation frequency events adjusted for basin area and storm duration. These routings provide sparse independent data for development of the less frequent portion of the loading curve. This IES H&H study utilized additional historical data points, including the results of the Paleoflood study and a regional precipitation-frequency analysis, to extend the ERL beyond that used in the 2015 study. Refinements to methodology since 2015, including the B17C use of thresholds and flow intervals for treatment of historical data and the utilization of a Bayesian framework, resulted in a broader data portfolio for use in more robust statistical tools. The greatest impact on the loading curve resulted from a more detailed analysis of inflow volume-frequency, regional precipitation-frequency, and greater sensitivity testing for skew.

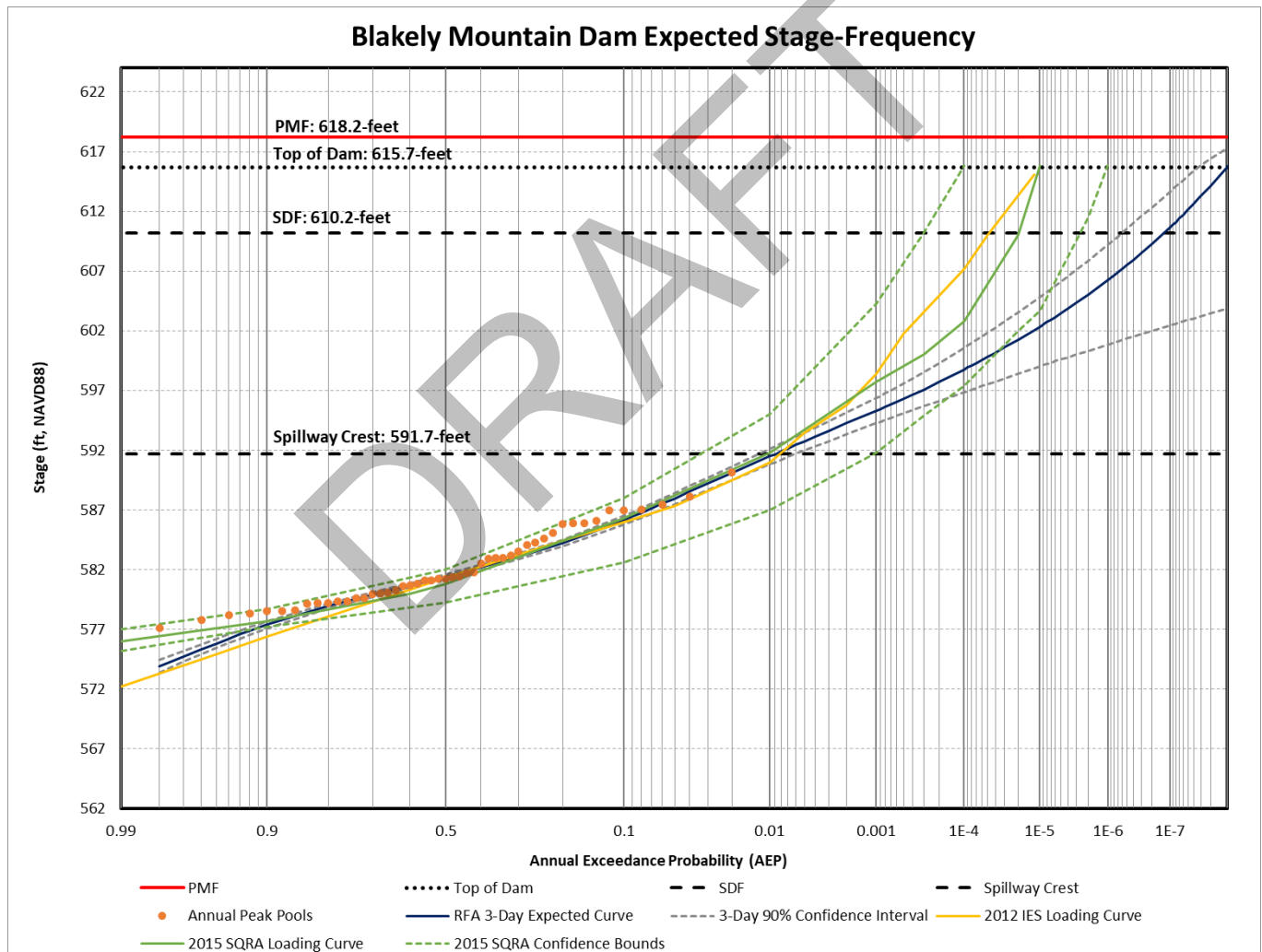


Figure 5-15. Comparison of expected stage-frequency curve to past studies of Blakely Mountain Dam.

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